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Memory, Storage, and Software Terms

Digital Electronics Terms

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Component Names and Pin Labels

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Software, Network, File, and Test Abbreviations

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Z80 ROMless SBC - WIP Engineering & Build Specification

Repository: github.com/gloveboxes/Z80ROMlessSBC **PDF edition:** [README.pdf](#)

Overview

This document describes a theoretical Z80 single-board computer design that has not yet been built, wired, or validated in hardware. Treat it as an engineering proposal and bring-up plan, not as a proven reference design. Every electrical assumption, timing margin, and firmware interaction still needs bench validation with the staged tests in Section 8 before the design should be considered reliable.

The proposed system is a ROMless Z80 computer supervised by a Raspberry Pi Pico 2 W. The Pico supplies the Z80 clock, holds and releases reset, loads a boot image directly into SRAM by taking the Z80 bus, and then lets the Z80 execute from RAM. A 64 KB SRAM provides the Z80 memory space; level shifters and bus transceivers isolate the Pico's 3.3 V GPIO from the 5 V Z80/SRAM bus; an MCP23S17 expands the Pico's address-drive capability during DMA and trapped I/O cycles.

The key design idea is that the Pico acts as both supervisor and virtual peripheral controller. It can stop the fully static Z80 clock on an I/O cycle while the GAL asserts WAIT#, inspect the requested port, exchange one data byte, release WAIT# only after the selected data path is ready, and then resume execution. Terminal I/O is intended to be one of those virtual peripherals: Z80 IN and OUT instructions feed nonblocking queues on the Pico, while a WebSocket terminal server runs on the Pico's other core so Wi-Fi and browser traffic do not disturb Z80 timing.

Z80 software lives in a reserved region of the Pico's own onboard flash rather than on removable media or compiled into the running firmware image. The Pico reads boot binaries, monitor images, and CP/M disk images directly from that memory-mapped flash region, then uses the already-validated DMA path to populate SRAM before the Z80 is released. Once the Z80 is running, the same I/O trap mechanism can expose sector-oriented virtual disk ports backed by that same flash partition (Section 6.3).

CP/M Boot and Disk Flow

The complete software and storage path is:

1. The build assembles the Z80-optimized CCP/BDOS and board BIOS, then places them in a reset-ready 64 KiB Z80 memory image.
2. The builder wraps that image as `z80boot.pkg` with a manifest and CRC, and inserts the same optimized resident system into Drive A's reserved system tracks. Drives B-D are built as separate native 320 KiB CP/M images.
3. The boot package and four disk images are provisioned into fixed regions of the Pico 2 W's 4 MiB onboard flash. They may be combined with the Pico firmware as `z80romless-flash.bin`.
4. On cold boot, the Pico validates the package and journal, holds the Z80 in reset, takes ownership of the bus, copies the 64 KiB image from flash into SRAM, verifies the copy, installs the CP/M page-zero vectors, and releases reset.
5. The Z80 starts from SRAM, enters the BIOS, and reaches the CP/M `A>` prompt. The BIOS converts CP/M's disk requests into 128-byte virtual-I/O transfers; the Pico services those transfers from the flash disk regions.
6. The Pico caches ordinary writes and journals directory or other important writes. On a CP/M warm boot, the BIOS reloads the resident CCP/BDOS records from flash-backed Drive A into SRAM before returning to the prompt.

This gives the project one simple separation: **flash provides persistent storage, SRAM provides the Z80's executing memory, and the BIOS plus Pico connect the two**. The [CP/M boot-image README](#), [disk-media README](#), [Section 6.3](#), and [Appendix D](#) provide the construction details, addresses, protocols, and validation evidence.

The build plan is deliberately phased. Early phases prove power, translation, SPI expansion, bus isolation, SRAM DMA, and clock control before the Z80 is installed. Later phases prove virtual-ROM boot, synchronous I/O trapping, WebSocket terminal service, and finally the maximum qualified clock rate. Passing an earlier phase is a prerequisite for trusting the assumptions used by the next one.

Project Documentation

The repository also includes focused documentation for specific implementation areas:

- [Stage 0: Power and passive wiring](#) - safe initial power, continuity, resistance, rail-voltage, and diode-OR checks.
- [Native CP/M boot image](#) - CP/M memory layout, Z80 optimization, boot-package construction, BIOS behavior, and host testing.
- [DCC debug I/O adapter](#) - complete project-hosted Altair-compatible port drivers, SBC native disks, interrupt services, private environment configuration, and ANSI terminal input.
- [CP/M disk media](#) - source disk formats, conversion, native geometry, generated images, and flash provisioning.
- [PDF edition](#) - printable version of this specification.

0. Project Inventory

The quantities below build one complete three-breadboard prototype. Purchase quantities include a small allowance for breadboard spares.

0.1 Semiconductors

Required	Component	Package	Function
1	Raspberry Pi Pico 2 W	Module with two 20-pin headers	Supervisor, clock, DMA, virtual I/O, and Wi-Fi terminal
1	Z84C0020PEC	40-pin PDIP	CMOS Z80 CPU
1	AS6C1008-55PCN	32-pin PDIP	SRAM; lower 64 KB used
1	MCP23S17-E/SP	28-pin SPDIP	SPI-to-16-bit address interface
1	ATF22V10B-15PC or ATF22V10C-15PU	24-pin PDIP	SRAM arbitration, data interlock, MCP reset, and I/O WAIT# control
1	SN74AHCT244N	20-pin PDIP	Eight-channel 5 V output buffer for clock, BUSREQ#, SPI, and SRAM controls
1	SN74AHCT245N	20-pin PDIP	Fixed Pico-to-5 V data-bus path; A side faces Pico, B side faces bus
1	SN74LVC245AN	20-pin PDIP	Fixed 5 V-to-Pico data-bus path with 5 V-tolerant inputs and Ioff protection
1	SN74LVC244AN	20-pin PDIP	5 V-to-3.3 V buffering for Z80 status/control and MCP SO
1	2N3904	TO-92	Pulls MCP23S17 RESET# LOW whenever GAL address access is disabled
1	1N5819	Axial diode	Schottky power OR from external +5 V to Pico VSYS

Nine-package all-through-hole design: The ATF22V10 computes all three SRAM controls from RESET#, BUSACK#, the Pico DMA controls, and the Z80 MREQ#/RD#/WR# outputs (Section 1.2). The ATF22V10 guarantees only a TTL-level 2.4 V HIGH, so its SRAM outputs pass through the SN74AHCT244N rather than driving the 5 V SRAM directly. The same AHCT244 translates Pico CLK, BUSREQ#, and the three MCP23S17 SPI inputs to guaranteed 5 V levels. Pico GP3 drives the Z80 RESET# input directly because the Z84C00 specifies a 2.2 V HIGH threshold for non-clock inputs; the Z80 clock uses the AHCT244 path to satisfy its stricter threshold. The data bus uses separate fixed-direction SN74AHCT245N and SN74LVC245AN devices because no single PDIP part provides a dual-supply, bidirectional, power-off-safe function. Two spare GAL inputs accept the existing DIR and master-enable controls, and two spare GAL outputs generate mutually exclusive active-low OEs. A third spare GAL output and one 2N3904 hold the MCP23S17 in reset outside DMA/trap access, allowing its reset-default input ports to connect directly to pulled-up A0-A15 and safely share the address bus. The design uses 9 active packages with deterministic isolation and safe power sequencing. The SN74LVC244AN buffers RD#/WR# because Pico 2 GP27 and GP28 are standard ADC-capable pads, not 5 V-tolerant FT pads. It also buffers BUSACK#, IORQ#, and MCP SO: although GP0, GP1, and GP20 are FT pads, their 5.5 V tolerance requires RP2350 IOVDD to be present at 3.3 V. Using the spare LVC244 channels preserves safe power sequencing and powered-off isolation (Section 5.3).

GAL pin 13 now accepts raw IORQ# and pin 20 drives WAIT#, adding the deterministic I/O interlock without another package.

AHCT244 sourcing: Specify the exact SN74AHCT244N PDIP-20 part. TI lists this N package as active and in production. It is available from authorized distributors and in breadboard-friendly DIP form on AliExpress (for example, selectable-part listing 1005005865735693). Do not substitute SN74AHC244N: at 5 V its CMOS input threshold does not guarantee recognition of Pico/GAL TTL HIGHS.

0.2 Sockets and Headers

Quantity	Item	Used for	Width / form factor
1	40-pin DIP socket	Z84C0020PEC Z80	Wide: 0.6-inch (15.24 mm) row spacing
1	32-pin DIP socket	AS6C1008-55PCN SRAM	Wide: 0.6-inch (15.24 mm) row spacing
1	28-pin DIP socket	MCP23S17-E/SP	Narrow: 0.3-inch (7.62 mm) row spacing
1	24-pin DIP socket	ATF22V10B-15PC or ATF22V10C-15PU	Narrow: 0.3-inch (7.62 mm) row spacing
4	20-pin DIP sockets	SN74AHCT244N, SN74AHCT245N, SN74LVC245AN, and SN74LVC244AN	Narrow: 0.3-inch (7.62 mm) row spacing
2	20-pin 0.1-inch male headers	Pico 2 W, if headers are not fitted	Single-row header strips
2	20-position 0.1-inch socket strips	Removable Pico 2 W mounting; do not substitute DIP IC sockets	Single-row socket strips

0.3 Capacitors and Resistors

Fitted	Purchase	Item	Purpose
8	11	100 nF X7R ceramic capacitors, at least 10 V	One at every DIP logic supply pair
3	5	22 uF capacitors, at least 10 V	One per breadboard
1	2	100 uF electrolytic capacitor, at least 10 V	5 V supply-entry bulk capacitance
29	35	10 kOhm, 1/4 W resistors	Defined startup levels and temporary test pulls
1	3	4.7 kOhm, 1/4 W resistor	GAL-to-2N3904 base current limiting
1	3	47 kOhm, 1/4 W resistor	2N3904 base-emitter pull-down
3	4	8x10 kOhm bussed SIP resistor networks, 9-pin	Two A0-A15 pull-ups (common to +5 V) and one Pico D0-D7 pull-down (common to GND)
Reused for tests	10	1 kOhm, 1/4 W resistors	First-drive current limiting and manual input tests

Place each 100 nF capacitor directly across its IC supply pins with the shortest practical leads. If scope captures show more than 250 mV rail excursion at the farthest board, add 47-100 uF there and repeat the capture; bulk capacitance does not replace local 100 nF capacitors. On the 5 V side, fit 10 kOhm pull-ups to BUSREQ#, BUSACK#, MREQ#, IORQ#, RD#, WR#, MCP SO, SRAM CE#, SRAM OE#, SRAM WE#, WAIT#, INT#, and NMI#. RESET# is the direct GP3 node and must not have a 5 V pull-up. Also pull SN74AHCT244 2A2, 2A3, and 2A4 up to 5 V so its SRAM-control outputs remain inactive if the GAL is physically absent. Fit 10 kOhm pull-ups on A0-A15 using two 9-pin bussed SIP networks. On the Pico side, use 10 kOhm pull-ups to 3.3 V on GP4 (BUSREQ#), GP5 (SRAM CE#), GP21 (SPI CS#), GP22 (SRAM WE#), and GP26 (SRAM OE#). Use 10 kOhm pull-downs to GND on GP2 (CLK), GP3 (RESET#), GP6 (DATA_DIR), GP7 (DATA_ENABLE), GP9 (ADDR_ENABLE), and GP18/GP19 (SPI SCK/SI). Leave GP8 unconnected. Connect GP7 and GP6 directly to ATF22V10 pins 9 and 11; its TTL-compatible inputs accept 3.3 V. Fit the third 8x10 kOhm bussed SIP network from GP10-GP17 to GND so the SN74AHCT245N A inputs stay defined while Pico GPIOs are inputs or the Pico is absent. It loads each active HIGH by only 0.33 mA at 3.3 V. Once the Pico 3.3 V rail is valid, these external resistors establish safe levels before the firmware configures SIO. During a cold power ramp the Pico-side pull-ups cannot hold active-low controls HIGH while their 3.3 V rail is still at 0 V; RESET# remains asserted and SRAM contents are therefore treated as indeterminate until the boot image is loaded and verified. The AHCT244-input pull-ups cover an absent GAL, and the final SRAM-node pull-ups cover an absent AHCT244. They do not provide power-off isolation for an installed source IC. Tie every unused CMOS input to a defined level.

0.4 Construction and Power

Quantity	Item	Requirement
3	830-tie-point solderless breadboards (BusBoard/X-ON BB830, ABS body; 63 terminal rows plus two 100-point power-distribution strips per board)	Memory, CPU core, and peripheral zones
1	Regulated 5 V supply	At least 1 A; adjustable current limiting preferred
1	USB data cable	Pico programming and serial diagnostics
1 set	22 AWG solid-core hookup wire	Multiple colors for signal groups
1 set	Short male-to-male jumpers	Inter-board and temporary connections
1 set	Short ground and power-distribution links	Connects all three boards; includes +5 V, 3.3 V, and multiple GND links
1	In-line power switch or removable supply jumper	Emergency power removal; rated at least 1.5 A continuous
As required	Labels or wire markers	Bus bits, controls, and socket pin 1

Feed the breadboard's +5 V logic rail directly from the regulated supply. Feed Pico VSYS from that rail only through the 1N5819: anode to external +5 V, banded cathode to VSYS. Pico 2 already has a Schottky diode from USB VBUS to VSYS, so this second diode safely ORs USB and external power without back-powering either source. Never link the external +5 V rail directly to Pico VBUS or VSYS. Feed the SN74LVC245AN and SN74LVC244AN from the Pico 3.3 V rail. Connect the SN74AHCT245N and all other logic to the regulated 5 V rail. No 5 V output connects directly to a Pico GPIO; both LVC devices provide power-off isolation. The GAL inputs do not source 5 V into GP6/GP7, and the SN74LVC244AN's I_{off} protection keeps the five monitored inputs isolated while its 3.3 V supply is absent or ramping. Tie the Pico's AGND pin (pin 33) to the common digital ground: this design does not use the ADC function on GP26-GP29, so no separate analogue ground plane is needed.

All installed 5 V logic devices must be powered from the same 5 V rail whenever that rail is energized. In particular, do not apply 5 V with an installed ATF22V10 or SN74AHCT244 unpowered: downstream pull-ups can otherwise raise an output above its $V_{CC} + 0.75\text{ V}$ GAL limit or the corresponding logic-family absolute maximum. The absent-device pull-up behavior is valid only when that device is physically removed from its socket. Verify 5 V continuity at every installed IC before power-up; a missing VCC socket contact is a fault, not a supported operating mode.

The plug-in supply shown on the middle breadboard may provide the 5 V entry point only after a load test proves 4.75 V to 5.25 V at the board under at least 500 mA load without regulator overheating. Do not connect that module's 3.3 V output to the Pico 3V3 pin or the system 3.3 V rail; the Pico must remain the only 3.3 V source. Reserve terminal rows 1-3 under the module body even if its pins enter only the distribution strip. If the actual module obscures more than three terminal rows, move it off-board or to the Memory Board rather than compressing the Core Board placement.

0.5 Bring-Up Equipment

- Digital multimeter with resistance, continuity, and DC voltage modes.
- Oscilloscope: RIGOL DH0814, four analog channels, 100 MHz bandwidth, 1.25 GSa/s maximum real-time sample rate, 25 Mpts maximum memory depth, and 12-bit vertical resolution. Use its four supplied 150 MHz passive probes for the analog signal-quality and timing checks below.
- Logic analyzer with at least 16 channels; 24 or more is preferred.
- Current-limited bench supply or an in-line 5 V current meter.
- Fine probe hooks or test clips suitable for DIP pins.
- A programmer explicitly supporting the selected ATF22V10B or ATF22V10C device and generic 22V10 JEDEC files.

RIGOL DHO814 Setup and Capture Plan

The DHO814 validates actual voltage, edge shape, ringing, pulse width, and timing margin at the device pins. Its four channels allow one stable trigger and three related nodes to be captured in the same acquisition. It does not replace the logic analyzer: four analog channels cannot prove the state or ordering of the complete A0-A15 and D0-D7 buses.

Probe and Instrument Preparation

1. Connect each probe to its channel, set the physical probe switch to **10X**, and open that channel's **Vertical > Probe > Probe Ratio** menu. Set the scope ratio to **10X** as well. A mismatched ratio makes every displayed voltage incorrect.
2. Before each bring-up session, connect each probe in turn to the front panel compensation output and its adjacent ground terminal. Adjust the probe until the square wave has flat tops and square corners, matching the manual's "Perfectly compensated" example. Repeat for CH1-CH4.
3. Fit a ground spring or a ground lead shorter than 20 mm to every probe. Connect all probe grounds only to nearby points on the common circuit GND. The DHO814 BNC shells and probe grounds are common; never connect a ground clip to +3.3 V, +5 V, or any signal node.
4. Recall the default setup, then configure every active channel as **1 MOhm input, DC coupling, 10X probe, Invert OFF, Delay 0 s**, and **Bandwidth Limit OFF**. Use the full 100 MHz bandwidth for timing, edge, overshoot, and ringing captures. A bandwidth limit may be used only for a separately labelled supply-noise capture.
5. Set **Horizontal > Acquisition = Normal**. The DHO800 manual identifies Normal as the best mode for most waveforms and confirms that sampling is real-time only. Do not use Average to qualify glitches or transitions; averaging can hide non-repetitive faults. Peak mode may be used as a second search capture, but the pass/fail evidence must include a Normal acquisition.
6. Start with **Memory Depth = Auto** for repetitive clock checks. Select **25 Mpts** for single-shot RESET#, BUSREQ#/BUSACK#, DMA, I/O-trap, or intermittent-fault captures, then verify the sample-rate label remains at least 500 MSa/s. Reduce the time span or memory depth if necessary to restore that sample rate. Place the trigger at about 40% of the screen so both pre-trigger cause and post-trigger response are visible.
7. For 3.3 V nodes start at **1 V/div**; for 5 V nodes start at **2 V/div**. Position ground markers near the bottom of each lane without overlapping traces. Adjust one step finer when useful, but keep ground and both logic levels visible. Start the horizontal scale at 200 us/div for 1 kHz, 2 us/div for 100 kHz, 200 ns/div for 1 MHz, and 50 ns/div for 2-8 MHz. Tighten the scale around an edge when measuring rise time or ringing.
8. Open **Trigger**, choose **Type = Edge, Source = CH1, Coupling = DC**, and choose the edge stated in the table below. Set the level to **1.65 V** when CH1 is a 3.3 V node or **2.5 V** when it is a 5 V node. Use **Sweep = Normal** for repetitive captures and **Sweep = Single** for ownership, reset, trap, and other one-off events; arm Single before issuing the firmware command. Do not use Auto sweep as pass evidence because it can force an acquisition without the requested event.
9. Add automatic measurements appropriate to the capture: **Frequency, Period, +Duty, -Duty, Rise Time, Fall Time, +Width, -Width, Vmax, Vmin, Vpp, Overshoot**, and the applicable **Delay** measurement between CH1 and another channel. Check timing with cursors as well; automatic measurements are invalid if the relevant edges or levels are not fully visible.
10. Stop the acquisition before moving a live probe. Save a screen image and waveform for every pass gate, naming it with the phase, clock rate, stimulus, and probed signals. Record the displayed sample rate, memory depth, probe ratio, bandwidth limit, and measured minima/maxima.

Four-Channel Connections and Expected Results

Connect the channels exactly as listed, using the physical IC pin as the probe point. CH1 is the normal trigger source unless the row explicitly names another source. Where a row lists alternatives, repeat the capture for every alternative while leaving the other channels in place where possible.

Purpose	CH1 / trigger	CH2	CH3	CH4	Trigger and expected outcome
Clock translation	Pico GP2, header pin 4	AHCT244 pin 18	Z80 CLK pin 6	Z80 RESET# pin 26	CH1 rising. At 1 kHz, 100 kHz, 1 MHz, and each qualification rate, CH1 is about 0-3.3 V and CH2/CH3 about 0-5 V. CH2 and CH3 match frequency and duty cycle, contain no extra edges, and differ only by interconnect delay. RESET# stays HIGH while running.

Purpose	CH1 / trigger	CH2	CH3	CH4	Trigger and expected outcome
Reset sequence	Z80 RESET# pin 26	Z80 CLK pin 6	Z80 M1# pin 27	Z80 MREQ# pin 19	CH1 rising, Single. RESET# remains LOW for at least three complete clocks. After release, M1# and MREQ# produce valid active-LOW opcode-fetch activity with no runt RESET# or CLK pulse.
DMA ownership	Z80 BUSREQ# pin 25	Z80 BUSACK# pin 23	Z80 RESET# pin 26	Z80 CLK pin 6	CH1 falling, Single. BUSACK# subsequently falls and remains LOW for DMA; CLK is deliberately stopped only by the firmware sequence. On release, BUSREQ# rises before BUSACK# returns HIGH and normal clocks resume without a runt edge.
SRAM write control	Z80 CLK pin 6	Z80 MREQ# pin 19	Z80 WR# pin 22	SRAM WE# pin 29	CH1 rising. During a CPU write, MREQ# and WR# assert LOW and SRAM WE# follows the selected write control through GAL/AHCT244 propagation. WE# has no extra pulse and its LOW width is at least 45 ns.
SRAM read timing	Z80 CLK pin 6	Z80 MREQ# pin 19	SRAM OE# pin 24	One SRAM data pin D0-D7: pins 13-15, 17-21	CH1 rising. Repeat CH4 for all eight bits and the 00/FF/55/AA patterns. MREQ# and OE# assert LOW once per read; CH4 reaches the expected 0 V or 5 V state and is stable before the Z80 sampling edge. Use the logic analyzer to prove the complete byte and setup margin.
Address integrity	Z80 CLK pin 6	SRAM A0 pin 12 or A7 pin 5	SRAM A8 pin 27	SRAM A15 pin 31	CH1 rising. Repeat with A0 and A7 on CH2 at 1, 2, 3, and 4 MHz, then at every Section 8.12 rate. For 0000/FFFF/5555/AAAA and walking patterns, each observed line matches the commanded bit, reaches valid 0/5 V levels, is stable during the active memory control interval, and has no double edge or excessive ringing. The logic analyzer proves A0-A15 together.
GAL ownership mux	RESET# at Z80 pin 26 or BUSACK# at Z80 pin 23	Selected Z80 control: WR# pin 22, RD# pin 21, or MREQ# pin 19	Matching GAL output: pin 14, 15, or 16	Matching SRAM control: WE# pin 29, OE# pin 24, or CE# pin 22	Trigger on the ownership input transition, Single; repeat rising/falling and all three paths. GAL and SRAM outputs remain inactive HIGH while ownership changes when both candidate controls are HIGH. Under CPU ownership they follow the Z80 control; under RESET#/BUSACK# DMA ownership they follow the Pico control. No active-LOW glitch is permitted.
Data-transceiver interlock	DATA_ENABLE at Pico GP7, header pin 10	DATA_DIR at Pico GP6, header pin 9	GAL pin 17 / AHCT245 OE# pin 19	GAL pin 18 / LVC245 OE# pin 19	CH1 rising and falling, Single. With DATA_ENABLE LOW both OE# outputs remain HIGH. With DATA_ENABLE HIGH and DATA_DIR HIGH, CH3 is LOW and CH4 HIGH; with DATA_DIR LOW, CH3 is HIGH and CH4 LOW. CH3 and CH4 must never be LOW simultaneously, including during transitions.
I/O trap and WAIT#	Z80 IORQ# pin 20	Z80 WAIT# pin 24	Z80 CLK pin 6	DATA_ENABLE at Pico GP7, header pin 10	CH1 falling, Single. WAIT# falls from GAL hardware before the Z80 sampling edge; the clock stops at a complete edge. WAIT# rises only after DATA_ENABLE and direction are valid, then clocking resumes. WAIT# does not reassert before IORQ# and RD#/WR# are inactive. Use a second capture with CH4 on RD# pin 21, then WR# pin 22.
Supply integrity	+5 V logic-rail entry at the 100 uF bulk capacitor	Farthest-board +5 V rail	Pico 3V3 header pin 36	Z80 CLK pin 6	Trigger on CH4 rising for repetitive operation; use Single on the relevant command for transients. Repeat at idle, DMA patterns, Z80 memory loop, disk write, and Wi-Fi traffic. CH1/CH2 remain 4.75-5.25 V and CH3 remains within the Pico 3.3 V rail specification; no capture may show more than 250 mV rail droop or a reset/clock disturbance. For ripple detail, AC coupling or a bandwidth limit is allowed only in an additional labelled capture; retain the DC-coupled full-bandwidth capture as pass evidence.

For logic nodes, a measured LOW must satisfy the receiving device's LOW limit and a measured HIGH must satisfy its HIGH limit; use the device-specific thresholds and margins stated in the relevant phase rather than treating the trigger level as a pass threshold. Investigate overshoot below GND or above the node's supply, non-monotonic threshold crossings, ringing that creates a second crossing, or a rise/fall time near the DHO814's own approximately 3.5 ns 100 MHz front-end limit. The scope validates analog quality on the listed nodes; the logic analyzer remains mandatory for simultaneous bus-wide ordering and the final frequency claim.

0.6 Optional Items

- LEDs with 1 kOhm series resistors for low-speed diagnostics only; do not permanently load high-speed buses with LEDs.
- No card socket or extra storage hardware is needed for boot images or CP/M disks: they live in a reserved region of the Pico's own onboard flash (Section 6.3), provisioned once with `picotool` and read back over the existing memory-mapped XIP bus. Reflash that region with `picotool` to change a disk image; no wiring changes.
- Spare 100 nF capacitors, resistors, jumper wire, and an IC extractor.

1. Reference Pin Mapping & Logic Domain Verification

This architecture leverages the fully static nature of the Z84C0020PEC CMOS CPU. The Pico controls the clock and uses the Z80 BUSREQ#/ BUSACK# handshake for DMA ownership. During RESET, the address and data buses float and control outputs are inactive. During BUSACK#, A0-A15, D0-D7, MREQ#, IORQ#, RD#, and WR# float while BUSACK# remains actively driven LOW; M1#, RFSH#, and HALT# are not part of the floated bus set.

1.0 Raspberry Pi Pico 2 Header Pin Map

GPIO numbers in the firmware are not physical header numbers. Use this table for construction; the ranges preserve ascending bit order.

Pico function	GPIO	Physical header pin	Connection
BUSACK# monitor	GP0	1	SN74LVC244AN 1Y1
IORQ# trap input	GP1	2	SN74LVC244AN 1Y2
Z80 CLK	GP2	4	SN74AHCT244 1A1 pin 2
Z80 RESET#	GP3	5	Z80 pin 26 and ATF22V10 pin 1, direct 3.3 V logic
Z80 BUSREQ#	GP4	6	SN74AHCT244 1A2 pin 4
SRAM CE#	GP5	7	ATF22V10 pin 7
Data DIR / ENABLE	GP6 / GP7	9 / 10	ATF22V10 pins 11 / 9; GAL pins 17 / 18 generate mutually exclusive data OE# signals
Unused	GP8	11	Leave open; no address transceivers are fitted
Address enable	GP9	12	ATF22V10 pin 10; HIGH releases MCP RESET# through GAL pin 19 and Q1
D0-D3	GP10- GP13	14-17	SN74AHCT245N and SN74LVC245AN A1-A4 pins 2-5
D4-D7	GP14- GP17	19-22	SN74AHCT245N and SN74LVC245AN A5-A8 pins 6-9
SPI SCK / MOSI / MISO / CS#	GP18- GP21	24-27	AHCT244 1A4 pin 8 / 2A1 pin 11, LVC244 2Y1, AHCT244 1A3 pin 6
SRAM WE#	GP22	29	ATF22V10 pin 3
SRAM OE#	GP26	31	ATF22V10 pin 5
Z80 RD# monitor	GP27	32	SN74LVC244AN 1Y3
Z80 WR# monitor	GP28	34	SN74LVC244AN 1Y4
3.3 V output	-	36	SN74LVC245AN VCC, LVC244 VCC, and 3.3 V pull-ups
VSYS	-	39	1N5819 banded cathode; anode to external +5 V

Connect Pico GND pins 3, 8, 13, 18, 23, 28, and 38, plus AGND pin 33, to common ground with multiple short links. Leave RUN pin 30, ADC_VREF pin 35, and 3V3_EN pin 37 open. Do not connect the external 5 V rail to VBUS pin 40; USB supplies that node internally.

Component	Signal Name / Group	Hardware Pin Numbers	Electrical Constraint / Logic Domain
Z84C0020PEC (CPU)	CLK	Pin 6	Input. Driven by level-shifted 5V CMOS square wave. Clock can be safely frozen indefinitely in either a HIGH or LOW state.
	IORQ#	Pin 20	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A2/1Y2 and monitored by Pico 2 GP1 to trigger clock-stop trapping. The raw 5 V node also feeds ATF22V10 pin 13 for hardware WAIT generation.
	MREQ#	Pin 19	Output. Active Low, three-state. Connects to ATF22V10 pin 8 for CPU-owned SRAM cycles; pulled HIGH when the CPU is absent.
	RD#	Pin 21	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A3/1Y3 and sampled by Pico 2 GP27 to resolve cycle intent.
	WR#	Pin 22	Output. Active Low. Buffered to 3.3 V through SN74LVC244AN channel 1A4/1Y4 and sampled by Pico 2 GP28 to resolve cycle intent alongside RD#.
	BUSACK#	Pin 23	Output. Active Low. Feeds ATF22V10 pin 2 for SRAM-source arbitration (Section 1.2) and is buffered to 3.3 V through SN74LVC244AN channel 1A1/1Y1 for monitoring at Pico 2 GP0.
	BUSREQ#	Pin 25	Input. Active Low. Level-shifted up to 5V to request DMA ownership.
	RESET#	Pin 26	Input. Active Low. Driven directly by Pico GP3 at TTL-compatible 3.3 V and also connected to ATF22V10 pin 1 for SRAM-source arbitration. Hold LOW for at least three full clock cycles.
	M1#	Pin 27	Output. Active Low. Probe point for the Phase 7 opcode-fetch test; not connected to Pico logic.
	A0 – A15 (Address Bus)	Pins 30 – 40, 1 – 5	5V Tri-state Bus. Driven by CPU during active run, floats during DMA/RESET.
	D0 – D7 (Data Bus)	Pins 7 – 10, 12 – 15	5V Bi-directional Tri-state Bus. Connected directly to SRAM data bus pins.
	WAIT#	Pin 24	Input. Active Low. Driven by ATF22V10 pin 20 during I/O trapping and pulled to +5 V through 10 kOhm so it remains inactive if the GAL is physically absent.
	INT#	Pin 16	Input. Active Low. Unused; pull to +5V through 10 kOhm to prevent a spurious interrupt from a floating input.

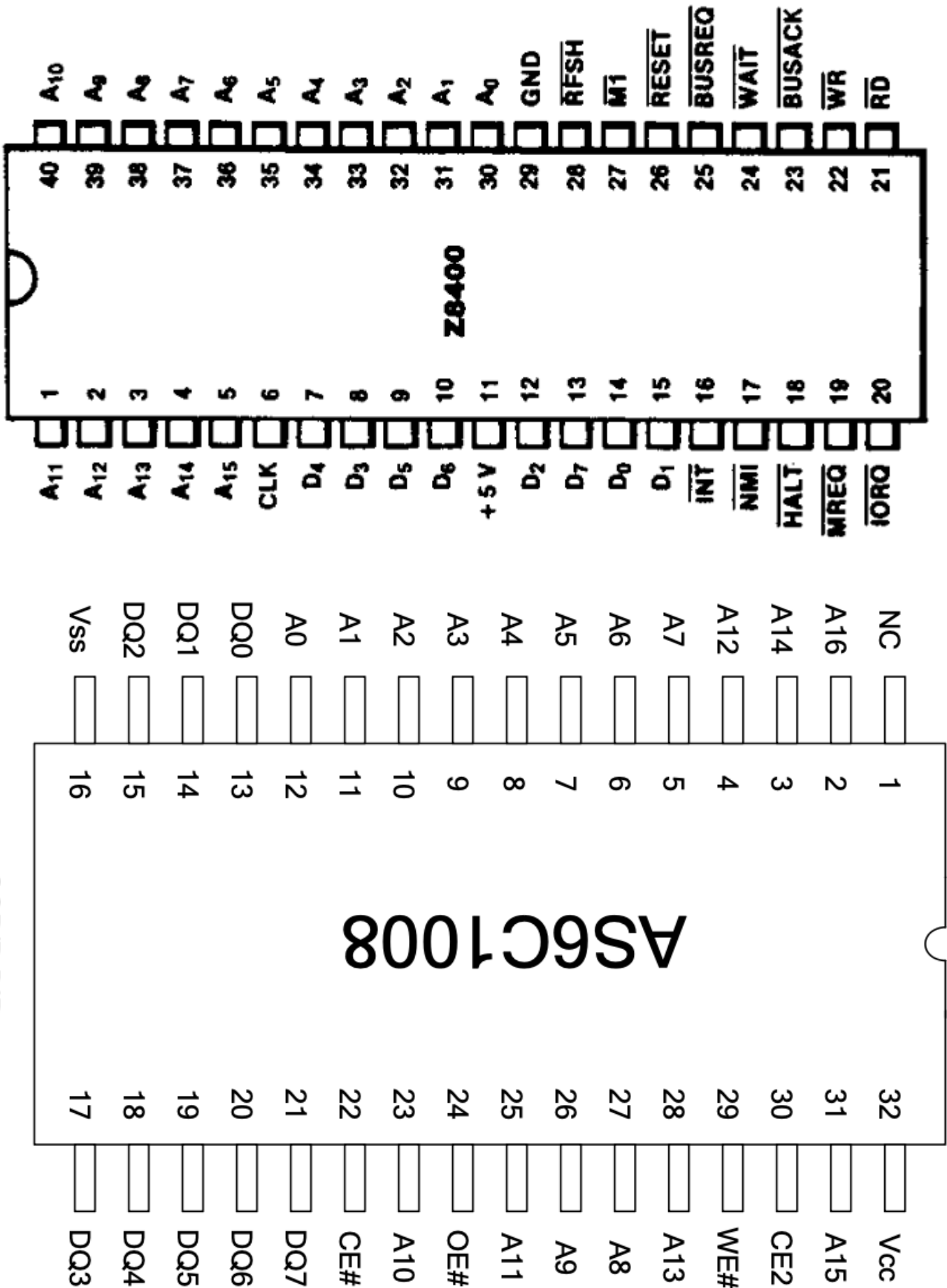
Component	Signal Name / Group	Hardware Pin Numbers	Electrical Constraint / Logic Domain
	NMI#	Pin 17	Input. Active Low, edge-sensed. Unused; pull to +5V through 10 kOhm to prevent a spurious non-maskable interrupt from a floating input.
AS6C1008-55PCN (SRAM)	A0 – A15 (Address lines)	Pins 12-10, 9-7, 6-5, 27-26, 23, 25, 4, 28, 3, 31	5V Input. Connected directly to the 5V Z80 address bus. Driven by MCP23S17 during DMA.
	A16	Pin 2	5V Input. Tie to GND to select the lower 64KB bank; do not leave floating.
	D0 – D7 (Data lines)	Pins 13 – 15, 17 – 21	5V Bi-directional Bus. Connected directly to the Z80 data bus.
	WE#	Pin 29	Input. Active Low. Driven by SN74AHCT244 2Y2 pin 7 from the ATF22V10 arbitration output (Section 1.2): Z80 WR# during CPU ownership, Pico DMA write signal during DMA ownership.
	OE#	Pin 24	Input. Active Low. Driven by SN74AHCT244 2Y3 pin 5 from the ATF22V10 arbitration output (Section 1.2): Z80 RD# during CPU ownership, Pico DMA read signal during DMA ownership.
	CE#	Pin 22	Input. Active Low. Driven by SN74AHCT244 2Y4 pin 3 from the ATF22V10 arbitration output (Section 1.2): Z80 MREQ# during CPU ownership so I/O cycles cannot access SRAM, Pico DMA chip-enable signal during DMA ownership.
	CE2	Pin 30	Input. Active High. Tie permanently to VCC (+5V) to enable the device through the active-low CE# input.

1.1 Z84C0020PEC-to-AS6C1008-55PCN Wiring

The following is the direct run-mode connection verified against the manufacturers' 40-pin PDIP and 32-pin PDIP pin assignments. The AS6C1008 is a 128K x 8 device, so A16 must be tied LOW to expose its lower 64KB as the Z80's address space. CE2 must be tied HIGH, and pin 1 is not connected.

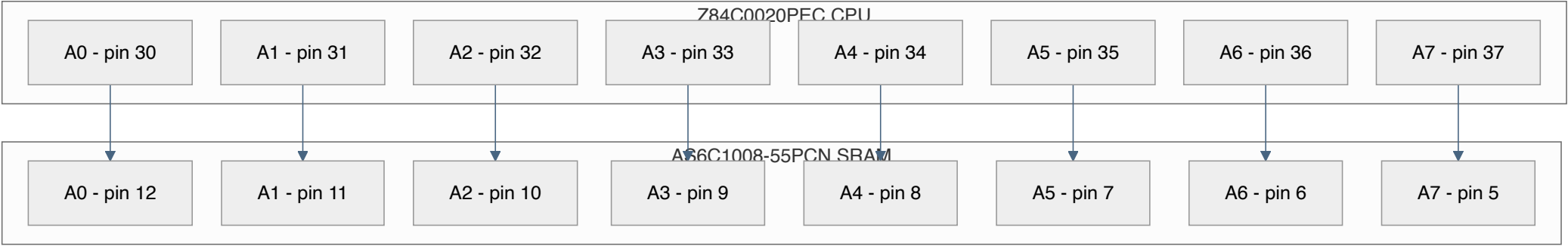
Manufacturer datasheets:

- [Zilog Z84C00 CMOS Z80 CPU Product Specification](#)
- [Alliance Memory AS6C1008 128K x 8 Low Power CMOS SRAM Datasheet](#)



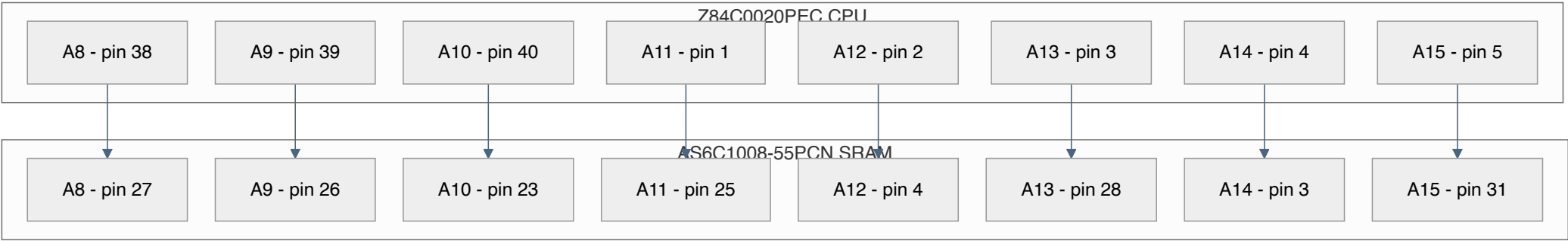
Address Bus A0-A7

Address Bus A0-A7



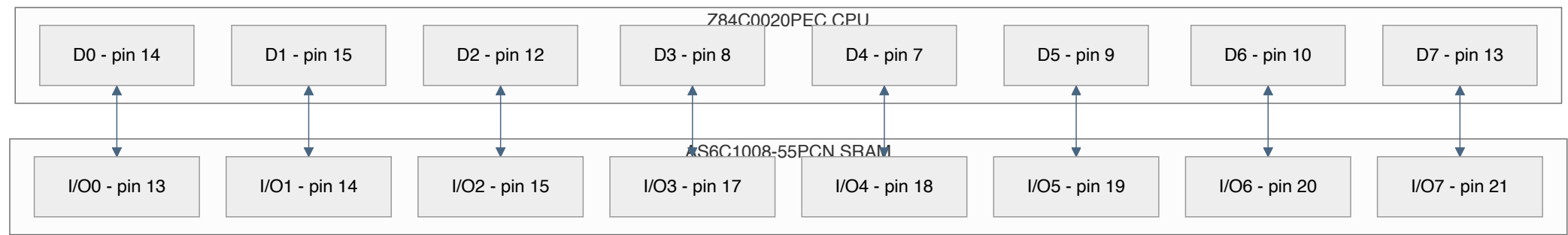
Address Bus A8-A15

Address Bus A8-A15

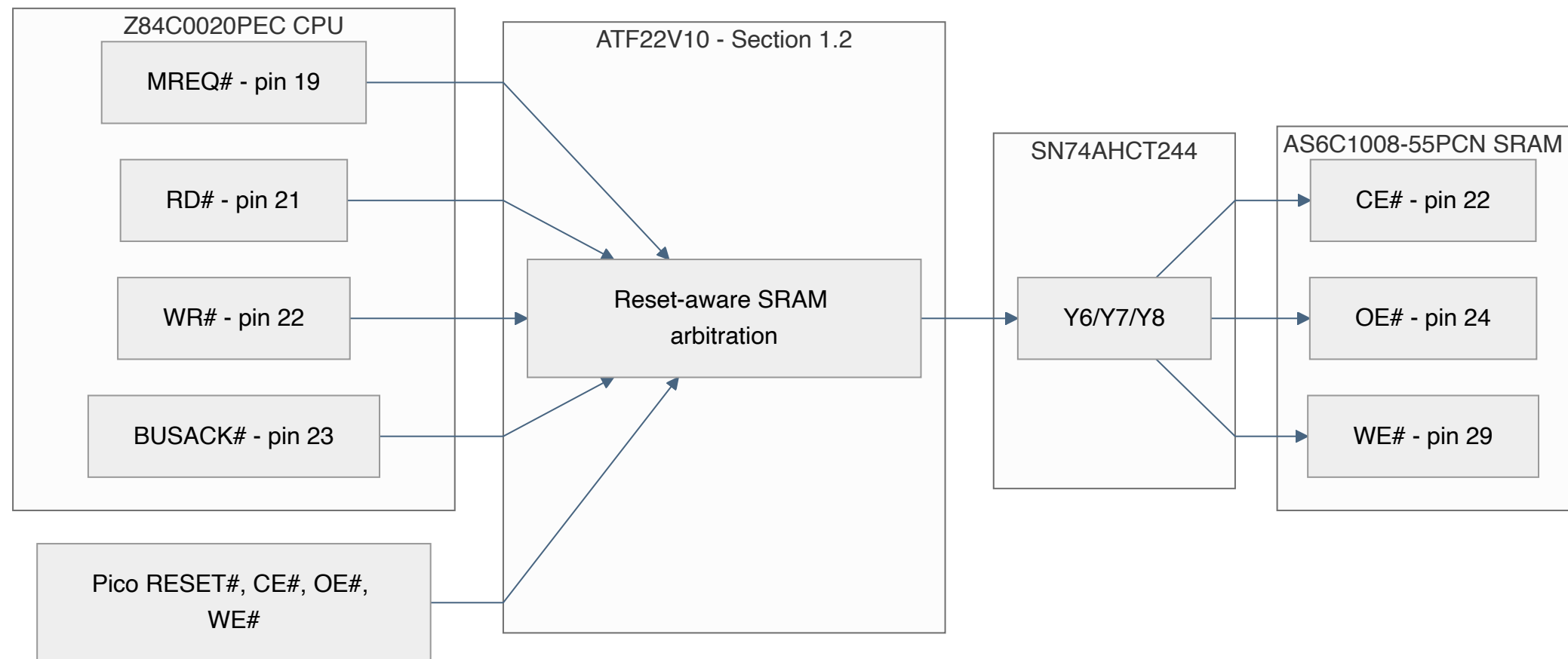


Data Bus D0-D7

Data Bus D0-D7

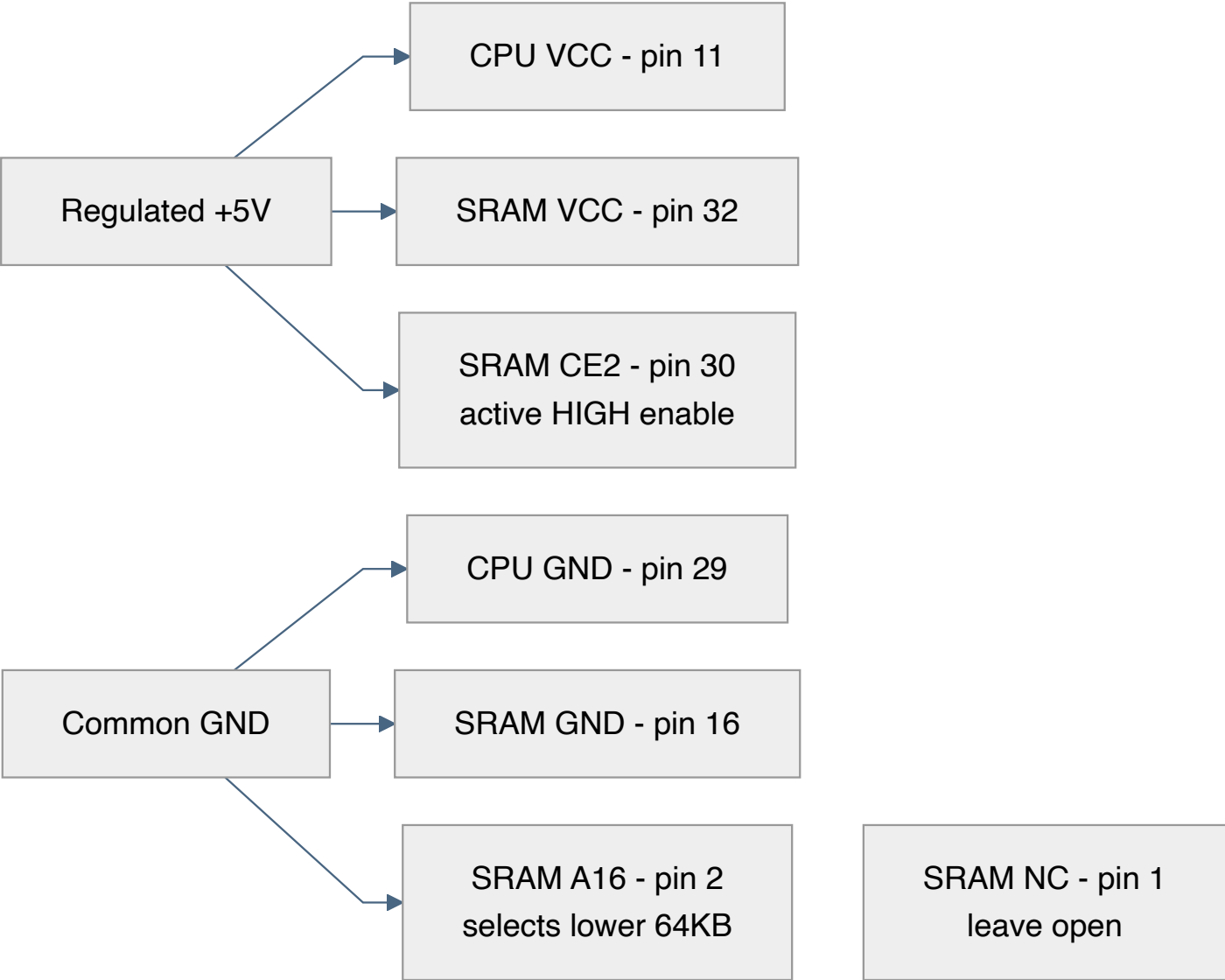


Memory Control



Power and Fixed Pins

Power and Fixed Pins



SRAM control-source arbitration: MREQ#/RD#/WR# above are the CPU-owned run-mode inputs of the ATF22V10 described in Section 1.2, never directly to the SRAM. The programmed equations select either these inputs or the Pico DMA controls; the three results then pass through dedicated SN74AHCT244 channels to the SRAM.

1.2 SRAM Control-Source Arbitration: ATF22V10B/C

The Z80 and Pico must never drive SRAM CE#, OE#, or WE# simultaneously. The ATF22V10 implements three combinational 2:1 selectors with a shared ownership condition:

```
CPU_OWNS_SRAM = RESET# AND BUSACK#
```

```
SRAM_WE_PRE# = CPU_OWNS_SRAM ? Z80_WR# : PICO_WE#
```

```
SRAM_OE_PRE# = CPU_OWNS_SRAM ? Z80_RD# : PICO_OE#
```

```
SRAM_CE_PRE# = CPU_OWNS_SRAM ? Z80_MREQ# : PICO_CE#
```

RESET#	BUSACK#	Selected source	Required state
0	X	Pico	Held-reset DMA or Z80 absent
1	0	Pico	BUSREQ#/BUSACK# DMA ownership
1	1	Z80	Normal CPU execution

This covers the two cases that BUSACK# alone cannot distinguish. During RESET the Z80 drives BUSACK# inactive HIGH, and with the Z80 physically absent its pull-up also reads HIGH. RESET# LOW therefore selects the Pico in both cases. Only an out-of-reset CPU that has not granted the bus can select MREQ#/RD#/WR#.

The PLD source adds the logically redundant consensus term Z80_CONTROL# & PICO_CONTROL# to each output. It does not change this truth table, but it holds an inactive HIGH continuously while RESET# or BUSACK# transfers ownership with both candidate controls HIGH. Without that term, unequal GAL product-term delays can create a static-1 hazard, seen externally as a short active-low SRAM control pulse. Each hardened equation uses four product terms, within every ATF22V10 macrocell's capacity.

Program the selected ATF22V10B or ATF22V10C with [src/pld/sram_control.pld](#). Use the exact device algorithm listed by the programmer, write the generated JEDEC file, read it back, and require a verify pass before installation. Do not enable the optional power-down mode or security fuse.

ATF22V10 pin	Signal	Connection
1	RESET#	Pico GP3 and Z80 pin 26 direct node
2	BUSACK#	Z80 pin 23
3	PICO_WE#	Pico GP22
4	Z80_WR#	Z80 pin 22
5	PICO_OE#	Pico GP26
6	Z80_RD#	Z80 pin 21
7	PICO_CE#	Pico GP5
8	Z80_MREQ#	Z80 pin 19
9	DATA_ENABLE	Pico GP7; 10 kOhm pull-down to GND
10	ADDR_ENABLE	Pico GP9; 10 kOhm pull-down to GND
11	DATA_DIR	Pico GP6; 10 kOhm pull-down to GND
12	GND	Common ground
13	Z80_IORQ#	Z80 pin 20 direct 5 V node; existing 10 kOhm pull-up
14	SRAM_WE_PRE#	SN74AHCT244 2A2 pin 13

ATF22V10 pin	Signal	Connection
15	SRAM_OE_PRE#	SN74AHCT244 2A3 pin 15
16	SRAM_CE_PRE#	SN74AHCT244 2A4 pin 17
17	DATA_UP_OE#	SN74AHCT245N OE# pin 19
18	DATA_DOWN_OE#	SN74LVC245AN OE# pin 19
19	MCP_RESET_DRIVE	4.7 kOhm to Q1 base; Q1 collector drives MCP RESET#
20	WAIT#	Z80 pin 24 direct node; existing 10 kOhm pull-up
21-23	Unused outputs	Program constant LOW; leave open
24	VCC	Regulated +5 V

The ATF22V10's guaranteed HIGH output is 2.4 V, which satisfies the AHCT244's 2.0 V input threshold but not the AS6C1008's 5 V CMOS input threshold. Never bypass the AHCT244 on these three paths. Fit 10 kOhm pull-ups at AHCT244 2A2-2A4 and at the final SRAM CE#/OE#/WE# nodes. The input pull-ups keep the permanently enabled AHCT244 deterministic if the GAL is missing; the output pull-ups keep the SRAM inactive if the AHCT244 is physically missing. All installed source devices must be powered as required by Section 0.4.

WAIT# may connect directly from GAL pin 20 to Z80 pin 24 because the Z84C00 ordinary-input HIGH minimum is 2.2 V, below the GAL's guaranteed 2.4 V HIGH. The existing 10 kOhm pull-up defines WAIT# HIGH if the GAL is physically absent; it is not a substitute for powering an installed GAL.

The same 2.4 V guaranteed HIGH exceeds the 2.0 V input-HIGH minimum of both data-transceiver OE# inputs. Two additional equations use two product terms each:

```
DATA_UP_OE#   = NOT DATA_ENABLE OR NOT DATA_DIR
DATA_DOWN_OE# = NOT DATA_ENABLE OR DATA_DIR
MCP_RESET_DRIVE = NOT ADDR_ENABLE
WAIT_N        = Z80_IORQ_N OR DATA_ENABLE
```

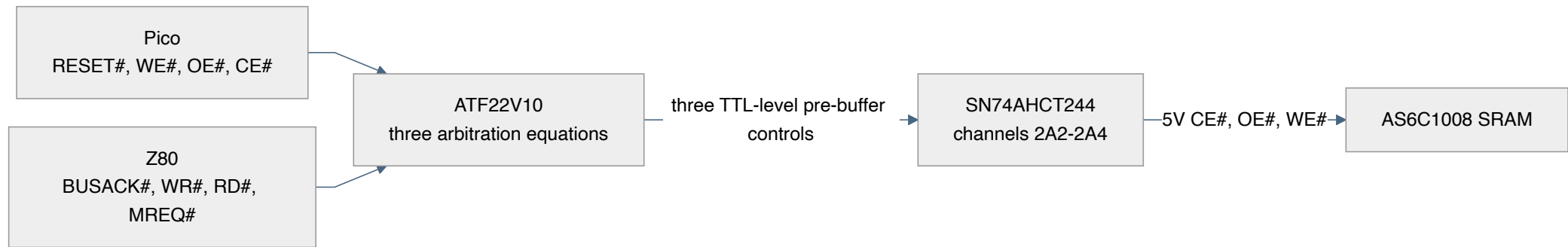
Both paths are disabled whenever DATA_ENABLE is LOW. When it is HIGH, exactly one path is enabled. Firmware changes DATA_DIR only while DATA_ENABLE is LOW, so the shared NOT DATA_ENABLE product term holds both OE# outputs inactive throughout the transition. SN74LVC245A I_{off} permits its OE# pin to remain driven from the 5 V GAL while the LVC device's 3.3 V supply is absent.

While IORQ# is inactive HIGH, WAIT# is always HIGH. When IORQ# falls, WAIT# goes LOW because DATA_ENABLE is still LOW. After the trap configures the correct direction and stable read/write data, raising DATA_ENABLE both enables exactly one data path and releases WAIT#. Firmware keeps DATA_ENABLE HIGH until IORQ# and RD#/WR# are inactive; lowering it early would reassert WAIT# in the same cycle.

For address isolation, connect GAL pin 19 through 4.7 kOhm to the base of Q1 (2N3904), add 47 kOhm base-to-emitter, ground the emitter, pull MCP RESET# up to 5 V through 10 kOhm, and connect the collector to RESET#. ADDR_ENABLE LOW makes GAL pin 19 HIGH and Q1 asserts reset, which forces all MCP GPIO registers back to input mode. ADDR_ENABLE HIGH releases reset; firmware then configures IODIR before using the address bus.

During Phase 6 with the Z80 absent, hold GP3 RESET# LOW. The programmed logic then selects the Pico controls regardless of the pulled-up, undriven BUSACK# input.

1.2 SRAM Control-Source Arbitration: ATF22V10B/C



2. 16-Bit Address Expansion Interface: MCP23S17-E/SP

The MCP23S17 acts as the dedicated 16-bit register shifter interfacing the Pico 2's SPI bus with the shared 5V address bus. During DMA block injection, it drives the target SRAM locations. During an active I/O trap, its GPIO ports remain inputs, allowing the MCP23S17 to monitor the address bus states driven by the frozen Z80 CPU.

MCP23S17 Pin Designation	Target Connection	System Logic Role
GPA0 – GPA7 (Port A)	Shared A0-A7 directly	Lower address byte drive/sample
GPB0 – GPB7 (Port B)	Shared A8-A15 directly	Upper address byte drive/sample
CS# (Pin 11)	SN74AHCT244 1Y3 pin 14, from Pico GP21 via 1A3 pin 6	SPI Hardware Chip Select (Active Low), 5 V translated
CLK / SCK (Pin 12)	SN74AHCT244 1Y4 pin 12, from Pico GP18 via 1A4 pin 8	SPI Master Clock Train Input, 5 V translated
SI (Pin 13)	SN74AHCT244 2Y1 pin 9, from Pico GP19 via 2A1 pin 11	SPI Master-Out-Slave-In (MOSI Path), 5 V translated
SO (Pin 14)	SN74LVC244AN channel 2A1/2Y1 to Pico 2 GP20	SPI Master-In-Slave-Out (MISO Path), buffered from 5 V to 3.3 V; fit the Section 0.3 pull-up because SO is high-impedance while CS# is HIGH
A0, A1, A2 (Pins 15-17)	Tied to GND	Hardware address = 000; matches the fixed 0x40/0x41 opcode used in firmware regardless of the IOCON.HAEN state, and prevents floating address-select inputs
RESET# (Pin 18)	10 kOhm pull-up to 5 V and Q1 collector	LOW outside address access; reset default makes all GPIO inputs

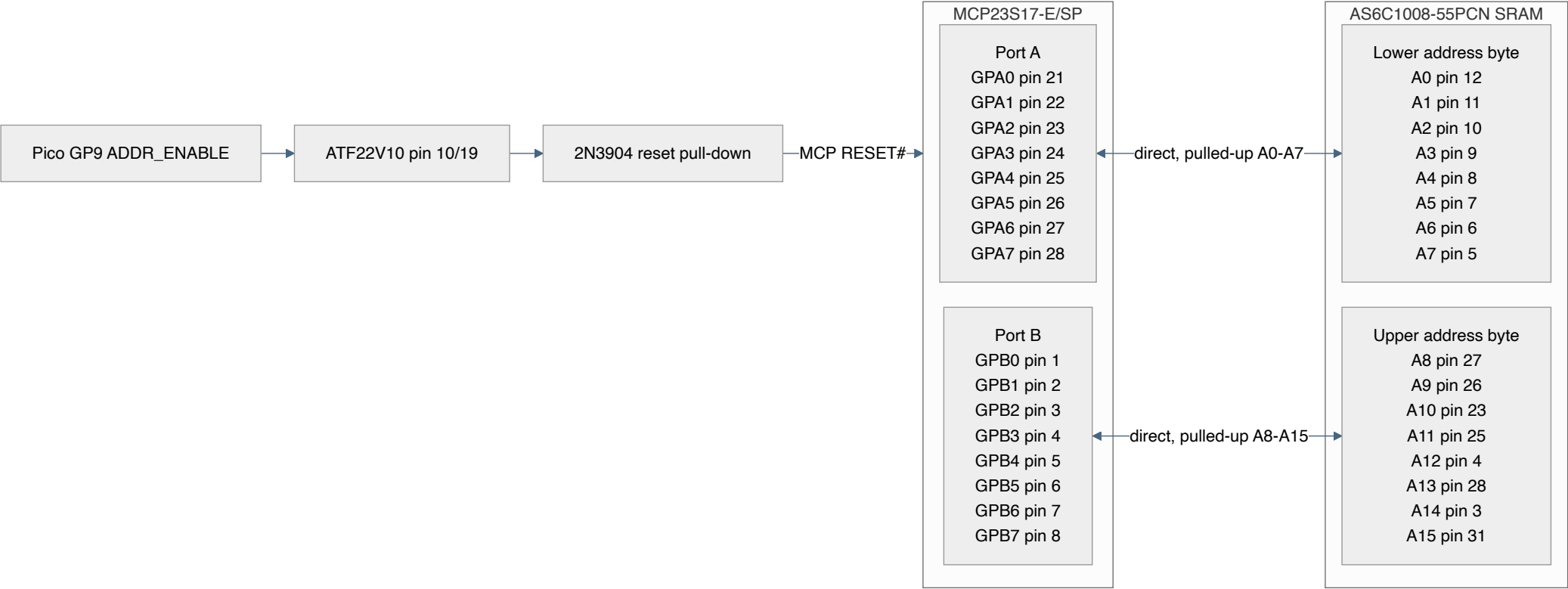
[Microchip MCP23017/MCP23S17 16-Bit I/O Expander with Serial Interface Datasheet](#)

2.1 MCP23S17-to-SRAM Address Wiring

The MCP23S17 connects directly to the shared address bus: GPA0-GPA7 to A0-A7 and GPB0-GPB7 to A8-A15. Fit one 8x10 kOhm bussed pull-up network per byte, with each common pin at 5 V. The Z84C00 guarantees 4.2 V at its light-load VOH2 point against the MCP's 4.0 V input minimum; the external pull-ups improve the static HIGH level and add at most about 0.5 mA load per LOW bit. MCP GPIO outputs guarantee at least 4.3 V at 5 V, exceeding the SRAM's 3.5 V input minimum.

Hardware isolation comes from RESET#, not a bus transceiver. GP9 ADDR_ENABLE feeds GAL pin 10. When LOW, GAL pin 19 drives Q1 and holds MCP RESET# LOW, forcing IODIRA/IODIRB to their all-input reset values. When HIGH, Q1 releases RESET#; firmware waits, then programs OLAT before changing IODIR to outputs. It reasserts reset before releasing the Z80. GP9 has a 10 kOhm pull-down, so Pico reset or power loss fails closed.

2.1 MCP23S17-to-SRAM Address Wiring



3. Physical Partitioning & Breadboard Topology

The layout enforces a strict three-zone model across three 830-point breadboards to minimize cross-talk and propagation delay across the distinct 3.3V and 5V power domains. Each zone below lists the specific chips to place on that board and where to seat them.

Place the three BB830s side by side with their long edges parallel: Memory on the left, Core in the center, and Peripheral on the right. Rows 1-63 run in the same direction on all three boards, so equal row numbers align across both board boundaries. The Core Board's central position is deliberate: the dominant Memory/Core paths are the shared 16-bit address and 8-bit data buses plus the MCP23S17 address/SPI interface, while the dominant Core/Peripheral paths are the Pico's 8-bit data interface and supervisor controls. Seven Pico-to-GAL signals span both board boundaries without a buffer: PICO_CE#, PICO_OE#, PICO_WE#, RESET#, DATA_ENABLE, DATA_DIR, and ADDR_ENABLE. RESET# is a three-way node also tapped by the Z80 on the Core Board; the others simply cross the Core Board without connecting to a Core component. MCP SO also crosses both boundaries to the Peripheral Board's LVC244. Keep these low-activity paths away from CLK. Putting either outer cluster in the center would shorten these four wires only by forcing one of the much wider address/data interfaces to span two board widths.

- **Memory Board (Left Zone):** AS6C1008-55PCN SRAM, the programmed ATF22V10, and MCP23S17. The GAL's five CPU-side inputs (BUSACK#, MREQ#, IORQ#, RD#, WR#) cross from the Core Board, and its three pre-buffer outputs cross back to the Core Board's SN74AHCT244 before the buffered result returns here as SRAM CE#/OE#/WE#. This deliberate round trip keeps the AHCT244's clock channel local to the Z80 instead (see below), which matters far more than SRAM control length since SRAM CE#/OE#/WE# only toggle at the Z80 bus-cycle rate, the same class as MREQ#/RD#/WR#. The MCP23S17 now sits here so its 16 port lines join the pulled-up shared address bus at the Memory/Core boundary. Q1 beside it provides reset-based hardware isolation. AHCT244 SPI outputs also cross only the Memory/Core boundary. Its SO output is the one MCP signal that continues across Core to the LVC244. Row budget: 16 (SRAM) + 12 (ATF22V10) + 14 (MCP23S17) = 42 of 63 rows.
- **Core Board (Center Zone):** Z84C0020PEC CPU and the SN74AHCT244N output buffer. Keep the AHCT244 beside the Z80 so its Y1 clock output never crosses a board boundary, the same placement rule the discrete design used for its dedicated clock buffer. The Z80 address pins join the SRAM/MCP address trunk directly. The photographed plug-in supply reserves rows 1-3, leaving 60 usable terminal rows. The chip budget is 20 (Z80) + 10 (SN74AHCT244N) = 30 of those 60 rows, leaving 30 rows for socket-body clearance, decoupling, and wiring. *No separate wait-state latch or flip-flop is used; the GAL drives WAIT# combinatorially.*
- **Peripheral Board (Right Zone):** Raspberry Pi Pico 2, SN74LVC244AN monitor buffer, SN74AHCT245N upward data path, and SN74LVC245AN downward data path. The ATF22V10 on Memory provides the mutually exclusive enables. Rotate both data transceivers so their B-port pin rows face the Core Board and their A-port pin rows face the Pico; tie AHCT DIR HIGH and LVC DIR LOW. Orient the LVC244 so its 1A input pin row faces the Core Board and its 1Y output pin row faces inward toward the Pico; its fifth input, MCP SO, arrives from Memory across Core. Row budget: 20 (Pico) + 10 (LVC244) + 10 (AHCT245) + 10 (LVC245) = 50 of 63 rows.

The following row-aligned schedule was selected by evaluating every valid per-board package ordering and gap distribution against grouped signal-count weights, iterating the comparison across all three boards. This is a routing aid, not an instruction to equalize individual wire lengths:

Board	Terminal-row schedule	Unallocated rows
Memory	Unallocated 1-4; GAL 5-16; gap 17; SRAM 18-33; gap 34; MCP23S17 35-48	1-4 and 49-63 (19)
Core / middle	Supply clearance 1-3; unallocated 4-7; AHCT244 8-17; gap 18; Z80 19-38	4-7 and 39-63 (29)
Peripheral	Pico 1-20; gap 21; LVC244 22-31; gap 32; AHCT245 33-42; gap 43; LVC245 44-53	54-63 (10)

3.1 Package Orientation and Pin 1

Use this convention for both the table and the placement image: view each BB830 from above and rotate it 90 degrees counter-clockwise from the manufacturer's landscape drawing, so **row 1 is at the top, row 63 is at the bottom, A-E are on the left, and F-J are on the right**. Memory, Core, and Peripheral then sit left-to-right. Do not mirror any board.

Seat every DIP socket across the center ravine before wiring it. A 0.3-inch DIP uses the E/F holes immediately beside the ravine. The Z80 and SRAM are 0.6-inch-wide DIPs: dry-fit their specified 0.6-inch sockets in two terminal columns matching the actual 15.24 mm lead-row span; do not force or bend them into E/F. Both pin rows must remain on opposite, electrically isolated sides of the ravine. Install the IC only after marking the socket's pin-1 corner and matching the IC notch or dot to the socket.

Board / device	Occupied rows	Body orientation	Pin 1 location (top view)	Opposite corner check
Memory / ATF22V10	5-16	Notch toward row 1	E5	Pin 24 at F5
Memory / AS6C1008 SRAM	18-33	Notch toward row 1	A-E pin-row side at row 18	Pin 32 on F-J side at row 18
Memory / MCP23S17	35-48	Notch toward row 63	F48	Pin 28 at E48
Core / SN74AHCT244	8-17	Notch toward row 63	F17	Pin 20 at E17; 1Y1 pin 18 is then close to Z80 CLK
Core / Z84C0020	19-38	Notch toward row 1	A-E pin-row side at row 19	Pin 40 on F-J side at row 19
Peripheral / Pico 2 W	1-20	USB connector toward row 1	Header pin 1 on A-E side at row 1	Header pin 40 on F-J side at row 1
Peripheral / SN74LVC244	22-31	Notch toward row 1	E22	Pin 20 at F22; 1A inputs face Core
Peripheral / SN74AHCT245	33-42	Notch toward row 63	F42	Pin 20 at E42; B1-B8 face Core
Peripheral / SN74LVC245	44-53	Notch toward row 63	F53	Pin 20 at E53; B1-B8 face Core

For the non-DIP keyed parts:

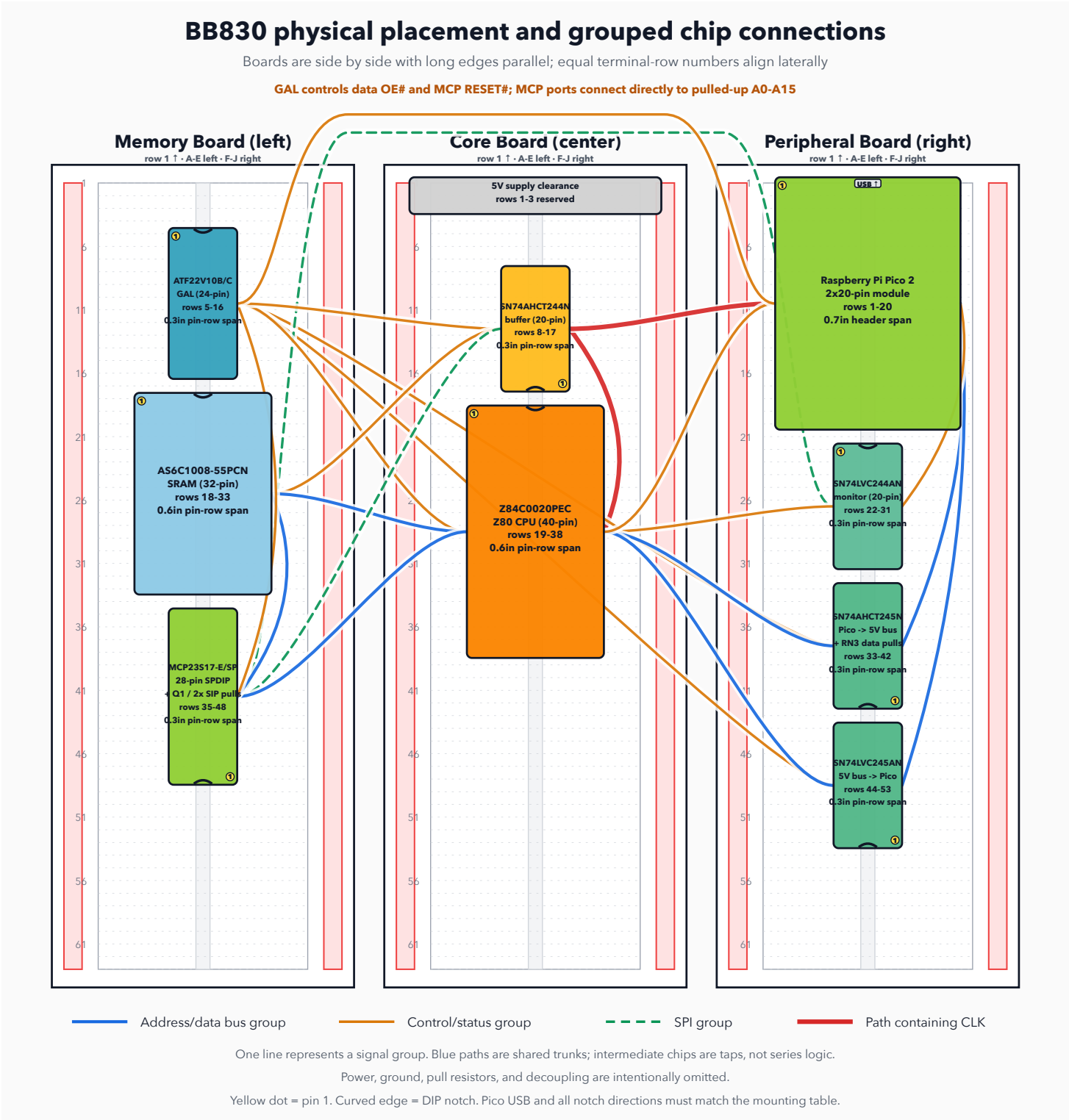
- **Q1 (2N3904):** place beside the MCP, not across the ravine. TO-92 lead order can vary by manufacturer; use the purchased part's datasheet to identify emitter/base/collector and mark E-B-C beside its holes. Do not rely on flat-face orientation alone.
- **RN1/RN2:** place as single-row SIPs near SRAM/MCP, parallel to the ravine, with the dot/common pin toward row 1 and wired to +5 V.
- **RN3:** place beside the AHCT/LVC A-port node with its dot/common pin toward row 1 and wired to GND.
- **1N5819 and electrolytics:** the diode band faces Pico VSYS; every electrolytic + lead goes to its positive rail. Mark polarity on the breadboard before insertion.
- **Plug-in supply module:** orient it only from its printed +, -, input, and output labels, then confirm every rail with a meter while unloaded. There is no generic module orientation; never infer polarity from USB-jack position or board color.

Before applying power, inspect every socket from above and verify the pin-1 location against this table. Then use continuity mode to prove that opposite-side pins at the same row are not shorted through a terminal strip.

This placement uses row alignment to reduce diagonal jumper length:

- **Memory/Core:** SRAM rows 18-33 overlap the Z80's rows 19-38 for the full A0-A15/D0-D7 trunk. GAL rows 5-16 overlap the AHCT244, while MCP23S17 rows 35-48 meet the lower end of the direct Z80 address trunk.
- **Core/Peripheral:** Pico rows 1-20 overlap AHCT244 and the top of the Z80. LVC244 rows 22-31 overlap the Z80 monitor sources. Both data transceivers overlap the lower Z80 region while remaining directly adjacent to the Pico on the same board.
- **Peripheral/Memory:** Pico rows 1-20 overlap GAL rows 5-16, reducing the vertical component of the seven Pico-to-GAL paths and two OE# returns. MCP SO remains one grouped low-activity crossing to the LVC244.

This schedule includes one empty row between every socket or module and still fits all three boards. Mark the actual supply-module overhang and socket outlines on the empty boards before wiring; the schedule is not valid if the supply consumes more than the assumed three rows.



3.2 KiCad Electrical Schematic

The native KiCad 10 schematic is the electrical source of truth for pin numbers, named nets, explicit no-connect markers, and ERC. It uses project-local symbols so the exact ATF22V10, Z80, SRAM, Pico, and translator pin definitions travel with the repository. RESET# is drawn as an explicit three-way Pico/Z80/GAL wire junction. Address, data, SPI, monitor, and control paths are shown as unfolded KiCad vector or group buses with every member named; matching labels on the chip-pin stubs provide the exact electrical connections without implying that signals pass through an intermediate chip as series logic.

Artifact	Purpose
KiCad project and native schematic	Editable KiCad 10 source
Project symbol library	Exact local pin names, numbers, and ERC electrical types
SVG export and PDF export	Zoomable full schematic renderings
KiCad netlist and independent net manifest	Machine-readable connectivity
ERC report	KiCad 10.0.5 result: zero violations with errors, warnings, and exclusions included

The schematic contains 59 physical components, including 31 discrete resistors, three SIP networks, and 12 fitted capacitors, plus one nonphysical #FLG01 power marker used only by ERC. KiCad's exported netlist matches the independently generated manifest exactly: 79 real nets and 349 component pin endpoints. The ERC-only power marker and KiCad's synthetic no-connect nets are excluded from that comparison.

To regenerate and validate the native source, exports, strict ERC report, and independent netlist comparison:

```
python3 -m venv .venv-kicad
.venv-kicad/bin/pip install -r scripts/requirements-kicad.txt
PYTHON=.venv-kicad/bin/python npm run kicad
```

The command runs [build-kicad-schematic.py](#), KiCad CLI upgrade/export/ERC, and [check-kicad-netlist.py](#). Any ERC violation or net/endpoint mismatch fails the build.

3.3 High-Speed Interconnect Routing

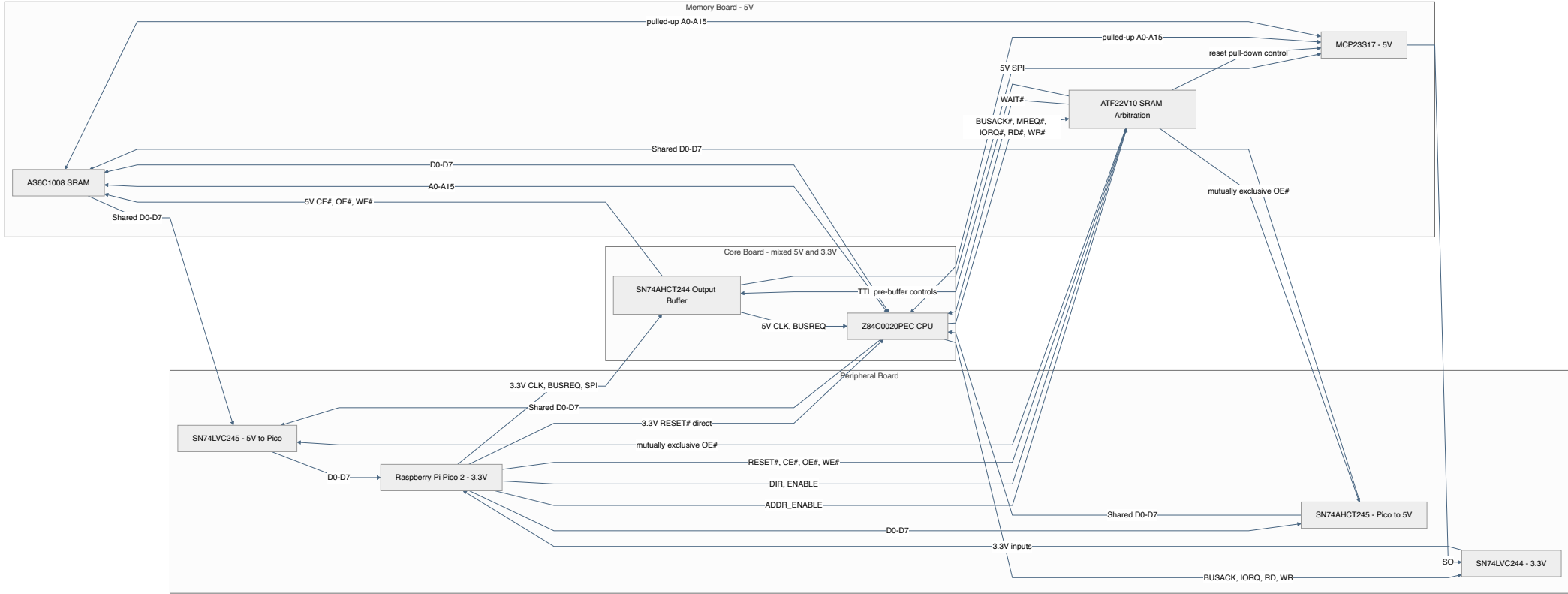
At the qualified 1-6 MHz clock rates, propagation skew from a few millimetres of wire-length difference is negligible compared with the Z80 timing budget. Solderless-breadboard reliability is instead dominated by total wire length, stubs, loop area, contact resistance, and fast-edge ringing. Route each bus as a short grouped trunk with roughly similar paths, but do not add serpentine wire merely to make lengths equal:

- **A0-A15 (hard construction rule):** keep the Z80, SRAM, MCP23S17, and both SIP pull-up networks on one short common trunk. Do not build this as a star and do not leave long branches to any device. A layout that cannot satisfy this rule fails the placement review and must be rearranged before wiring the remaining signals.
- **D0-D7:** join both Peripheral data-transceiver B ports to one short 5 V trunk crossing to the Core Board, then continue that shared trunk through the Z80 to the adjacent SRAM. Keep the two A-port taps to the Pico short and parallel; never route one translator through the other.
- **SN74AHCT244 2Y2-2Y4 to SRAM WE#/OE#/CE#:** these cross from the Core Board's AHCT244 to the Memory Board's SRAM; route them as one grouped trunk at the board boundary. Exact length matching is unnecessary.
- **CLK (SN74AHCT244 1Y1 pin 18 to Z80 pin 6):** route this as the single shortest, most direct jumper, kept away from the address bus. Do not lengthen it to match other nets; clock is the most edge-rate-sensitive signal in the design. This is why the AHCT244 is seated on the Core Board rather than beside the GAL.

Route a ground jumper alongside every inter-board signal group and add ground probe points near CLK, IORQ#, MREQ#, RD#, WR#, SRAM CE#/OE#/WE#, and each bus transceiver. Keep all jumpers as short as the placement allows.

3.4 Major Chip Interconnection Overview

3.4 Major Chip Interconnection Overview



4. Output Buffer Mapping: SN74AHCT244N (DIP-20)

The 5 V-powered SN74AHCT244 provides all eight required high-level outputs. Its TTL-compatible inputs accept both Pico 3.3 V signals and the ATF22V10's guaranteed 2.4 V HIGH. At the light CMOS loads used here, its 5 V outputs satisfy the Z80 clock's strict V_{IHC} threshold, the MCP23S17's $0.8V_{DD}$ SPI threshold, and the SRAM's CMOS control-input threshold. Its current TI datasheet specifies a worst-case 9.5 ns A-to-Y delay at 5 V with a 50 pF load over -40°C to 85°C. Combined with the ATF22V10C-15's 15 ns combinational delay and the SRAM's 55 ns chip-enable access, these datasheet maxima give a conservative 79.5 ns component-delay sum for the control-to-data path. This is not complete timing closure: breadboard interconnect and Z80 setup allowance are additional. This is why 1-6 MHz is measured rather than assumed, and why this design is not a 20 MHz system despite using a 20 MHz-rated CPU.

Channel	AHCT244 input	AHCT244 output	Function
1A1	Pin 2 from Pico GP2	1Y1 pin 18 to Z80 CLK pin 6	Master clock
1A2	Pin 4 from Pico GP4	1Y2 pin 16 to Z80 BUSREQ# pin 25	DMA request
1A3	Pin 6 from Pico GP21	1Y3 pin 14 to MCP23S17 CS# pin 11	SPI chip select
1A4	Pin 8 from Pico GP18	1Y4 pin 12 to MCP23S17 SCK pin 12	SPI clock
2A1	Pin 11 from Pico GP19	2Y1 pin 9 to MCP23S17 SI pin 13	SPI MOSI
2A2	Pin 13 from ATF22V10 pin 14	2Y2 pin 7 to SRAM WE# pin 29	Arbitrated write enable
2A3	Pin 15 from ATF22V10 pin 15	2Y3 pin 5 to SRAM OE# pin 24	Arbitrated output enable
2A4	Pin 17 from ATF22V10 pin 16	2Y4 pin 3 to SRAM CE# pin 22	Arbitrated chip enable

Tie both active-low output enables, OE1# pin 1 and OE2# pin 19, to GND. Connect VCC pin 20 to regulated +5 V and GND pin 10 to common ground. The buffer is permanently enabled; the Pico-side defaults and GAL equations therefore establish safe inactive outputs before firmware enables its GPIO drivers.

Pico GP3 RESET# bypasses the AHCT244 and connects directly to Z80 pin 26 and ATF22V10 pin 1. Its 3.3 V HIGH exceeds the Z84C00's 2.2 V and the ATF22V10's 2.0 V input-HIGH minima. Keep its 10 kOhm pull-down to GND and never fit a 5 V pull-up on this node.

5. Transceiver Operating Modes & Isolation Tables

The transceivers isolate the supervisor elements (Pico 2 and MCP23S17) from the main bus during standard execution, preventing bus contention.

5.1 All-PDIP Data-Bus Translation and Interlock

No single PDIP part provides an SN74LVC8T245-equivalent combination of dual supplies, bidirectional translation, deterministic direction, three-state isolation, and partial-power-down safety. The breadboard design therefore uses two fixed-direction transceivers:

- **SN74AHCT245N (5 V):** A port is Pico D0-D7, B port is the 5 V bus, and DIR pin 1 is tied HIGH. Its TTL-compatible A inputs accept 3.3 V and its B outputs provide full 5 V CMOS levels.
- **SN74LVC245AN (3.3 V):** A port is Pico D0-D7, B port is the 5 V bus, and DIR pin 1 is tied LOW. Its B inputs tolerate 5.5 V, A outputs stay in the Pico domain, and I_{off} prevents back-powering while 3.3 V is absent.

Bus bit	Pico GPIO / both A ports	5 V bus / both B ports
D0	GP10 / A1 pin 2	B1 pin 18
D1	GP11 / A2 pin 3	B2 pin 17
D2	GP12 / A3 pin 4	B3 pin 16
D3	GP13 / A4 pin 5	B4 pin 15
D4	GP14 / A5 pin 6	B5 pin 14
D5	GP15 / A6 pin 7	B6 pin 13
D6	GP16 / A7 pin 8	B7 pin 12
D7	GP17 / A8 pin 9	B8 pin 11

Connect both pin 10s to GND. Connect AHCT pin 20 to 5 V and LVC pin 20 to 3.3 V. Each device needs its own local 100 nF capacitor.

The existing ATF22V10 provides the hardware interlock using spare pins and product terms. Connect GP7 DATA_ENABLE to GAL pin 9 and GP6 DATA_DIR to pin 11. GAL pin 17 drives AHCT245 OE# pin 19; pin 18 drives LVC245 OE# pin 19. The two equations are documented in Section 1.2. No 5 V output drives GP6/GP7: they are GAL inputs with 10 kOhm pull-downs. During Pico power-off both inputs read LOW, so both GAL OE# outputs are HIGH. The LVC245's I_{off} specification protects its 3.3 V-powered side while GAL pin 18 remains at a 5 V-domain HIGH.

System state	DATA_ENABLE GP7	DATA_DIR GP6	AHCT OE#	LVC OE#	Result
Isolated / run	0	X	1	1	Both Pico data paths high-impedance
DMA write	1	1	0	1	Pico → AHCT245 → 5 V bus
Readback / OUT trap	1	0	1	0	5 V bus → LVC245 → Pico

Firmware always drives DATA_ENABLE LOW, waits, changes DATA_DIR, waits, and only then drives DATA_ENABLE HIGH. The GAL truth table also makes simultaneous enables impossible for every static GP6/GP7 state. Do not substitute TXS0108E, TXB0108, BSS138, or resistor-divider breakouts: their automatic/pass-device behavior and loading assumptions are not equivalent to this controlled, multi-load push-pull bus.

5.2 Direct MCP23S17 Address-Bus Modes

System state	ADDR_ENABLE GP9	MCP RESET#	IODIRA/B	Functional role
DMA injection	1	1	0x00/0x00 after OLAT preload	MCP drives A0-A15
Trap address read	1	1	0xFF/0xFF	MCP samples the frozen Z80 address
Active execution / isolated	0	0	Reset default 0xFF/0xFF	MCP pins are inputs; Z80 owns A0-A15

Always assert ADDR_ENABLE LOW before releasing the CPU. Releasing MCP reset is not itself permission to drive: firmware must preload OLAT and hold Z80 RESET# or BUSACK# before writing IODIR outputs.

5.3 SN74LVC244AN 5 V-to-3.3 V Input Buffer

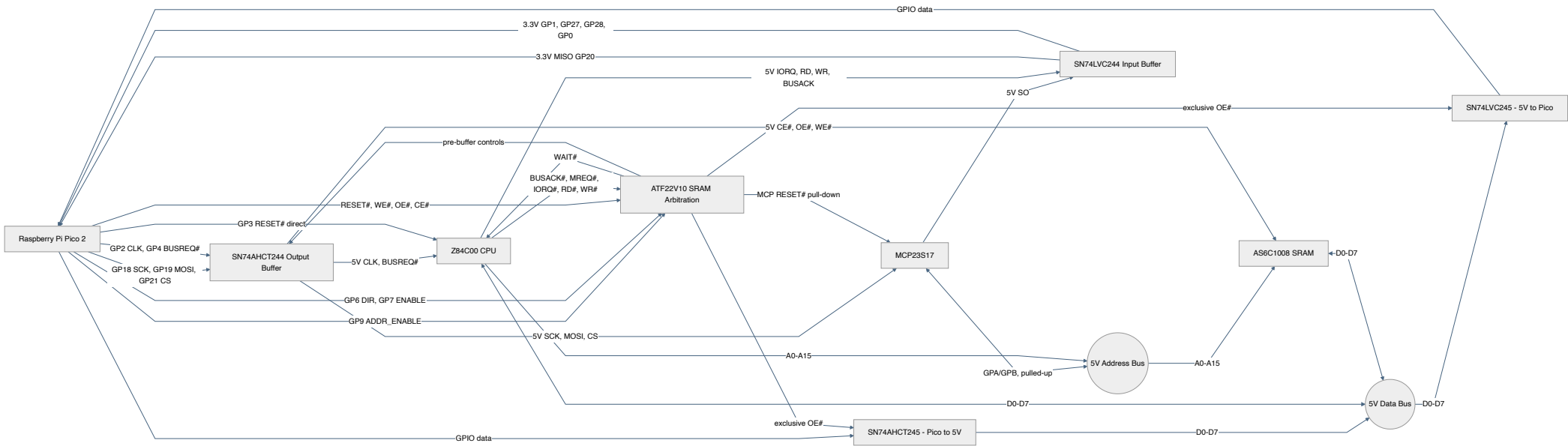
The RP2350's GP0-GP25 are 5 V-tolerant FT pads, but the 5.5 V rating applies only while IOVDD is powered at 3.3 V; with IOVDD at 0 V their absolute maximum is 3.63 V. GP26-GP29 are the QFN-60 package's ADC-capable pads and are not FT at all. Buffering all five incoming 5 V signals therefore avoids a power-sequencing constraint and keeps every Pico GPIO within its normal 3.3 V domain. Power the SN74LVC244AN from the Pico 3.3 V rail. Its inputs accept up to 5.5 V and its Ioff specification protects both sides when VCC is 0 V. Tie both output enables (pins 1 and 19) to GND: every buffered signal -- BUSACK#, IORQ#, RD#, WR#, and MCP SO -- must always be readable, and none of them share a Pico GPIO with any other driver, so neither output enable needs gating. Wire the channels as follows; tie every unused input to GND and leave unused outputs open.

5 V source	LVC244 input	3.3 V output	Pico destination
Z80 BUSACK# pin 23	1A1 pin 2	1Y1 pin 18	GP0
Z80 IORQ# pin 20	1A2 pin 4	1Y2 pin 16	GP1
Z80 RD# pin 21	1A3 pin 6	1Y3 pin 14	GP27
Z80 WR# pin 22	1A4 pin 8	1Y4 pin 12	GP28
MCP23S17 SO pin 14	2A1 pin 11	2Y1 pin 9	GP20
GND	2A2/2A3/2A4 pins 13/15/17	2Y2/2Y3/2Y4 pins 7/5/3 open	Unused

VCC (pin 20) connects to 3.3 V and GND (pin 10) to common ground. The 5 V-side pull-ups in Section 0.3 keep every input defined when its source is absent or high-impedance.

5.4 Bus and Supervisor Signal Interconnections

5.4 Bus and Supervisor Signal Interconnections



6. Architectural Operational Boundaries

6.1 Hardware-WAIT-Assisted Clock-Stop Trap Protocol

When the Z80 executes an I/O instruction, IORQ# LOW reaches ATF22V10 pin 13 and drives WAIT# LOW through GAL pin 20 while DATA_ENABLE remains LOW. In parallel, the buffered IORQ# falling edge trips a Pico interrupt. The handler disables the hardware PWM clock slice after GPIO synchronization and interrupt-entry latency. Because the Z84C00 is fully static, the clock can then remain stopped indefinitely in either a HIGH or LOW state.

After resolving RD#/WR#, the handler configures the appropriate data path. The data-bus helper raises DATA_ENABLE only after direction and data are stable; the GAL then releases WAIT#. The handler resumes CLK, waits until both IORQ# and the active RD#/WR# control are HIGH, and only then lowers DATA_ENABLE to isolate the bus and re-arm WAIT# for the next cycle. Hardware WAIT# removes dependence on interrupt latency alone, but the maximum qualified frequency remains measured because GAL delay, Z80 WAIT setup/hold, clock phase, and breadboard signal integrity still need logic-analyzer evidence.

PIO is deliberately not placed in the WAIT# assertion path. Raw IORQ# already reaches the GAL directly, so WAIT# is asserted after the GAL's combinational propagation delay without GPIO synchronization, PIO sampling, or processor interrupt latency. A PIO state machine could notify firmware or gate a PIO-generated clock, but it would not make this existing assertion path faster and C would still perform the MCP23S17 and data-bus service. The design therefore keeps the GAL as the timing-critical interlock and uses the RP2350 interrupt only after the Z80 has been held safely.

6.2 Terminal I/O over Pico WebSocket

The final terminal is a virtual I/O peripheral implemented by the Pico, not a Z80-side UART. Z80 software performs ordinary IN and OUT instructions against a small terminal port pair; the Pico intercepts those cycles through the Section 6.1 clock-stop trap, then resumes the CPU after sampling or supplying the data byte. A browser connects to the Pico over Wi-Fi and receives the terminal stream through a WebSocket server running on the Pico 2 W.

Wi-Fi/WebSocket is the intended final and day-one operating terminal. Before networking is introduced, Phase 8 uses USB CDC over the Pico's existing USB connector as an intermediate terminal transport. USB CDC consumes no GPIO and appears as /dev/cu.usbmodem... on macOS. It lets the same 0x00/0x01 virtual-port contract and bounded queues be tested without lwIP timing in the loop. Keep diagnostics framed or on a separate CDC interface so debug text cannot become CP/M input. Phase 10 replaces the USB transport endpoint with Wi-Fi/WebSocket; it does not remove USB diagnostics or make USB the final user interface.

The WebSocket server must run on the Pico's other core so Wi-Fi, lwIP, HTTP serving, and WebSocket polling cannot interfere with the timing of the Z80-facing supervisor path. Core 0 owns GPIO, MCP23S17 SPI, clock stop/resume, DMA, and the I/O trap. Core 1 owns Wi-Fi association, the embedded terminal page, WebSocket accept/send/receive, and network polling. The cores share terminal byte queues, one immutable disk-write queue, a small request/result pair for Z80 bus ownership, and atomic terminal/disk status words; they share no raw bus GPIO ownership.

Use the pico-altair-8800 WebSocket console as the software pattern: one embedded HTML terminal page, one WebSocket server, and byte queues between the CPU-facing side and the network-facing side. The trap hook must never call lwIP, print, sleep, wait for a browser, or block on a queue. It may only enqueue Z80 output bytes, dequeue already-buffered browser input bytes, and report terminal status. Core 1 may drop, drain, or back-pressure network data; it must not take locks needed by the trap or touch any Z80 bus GPIO.

The initial terminal decode is intentionally small:

Port	Direction	Function
0x00	Z80 OUT	Terminal transmit byte from Z80 to browser
0x00	Z80 IN	Terminal receive byte from browser to Z80; returns 0x00 if empty
0x01	Z80 IN	Terminal status: bit 0 = receive byte available, bit 1 = transmit queue has room, bit 7 = WebSocket client connected

This is enough for ROM monitors, BASIC, CP/M console glue, and small diagnostics. If later software needs modem-control style signals, add more virtual status bits before adding hardware.

6.3 Onboard Flash CP/M Disk Storage

Boot images and CP/M disk images live in a reserved region of the Pico's own onboard flash, not on any external card or bus. This needs zero extra GPIO and zero extra parts: the same physical flash chip that already holds the Pico's own firmware is memory-mapped and directly readable by both cores at any time, so no SPI bus, socket, or level-shifted chip-select is involved. GP27 and GP28 keep their original job monitoring Z80 RD# and WR# (Section 5.3); nothing needs reclaiming for storage.

Capacity budget. The Pico 2 W's onboard flash is 4 MiB. Reserve the top 1.4375 MiB for a power-fail journal, a distinct Z80 boot package, and four complete 320 KiB disks, leaving 2.5625 MiB for Pico firmware. That still comfortably covers this project's code plus the Wi-Fi firmware/CLM blob embedded by `pico_cyw43_arch`.

Region	Flash offset (from flash base)	Size	Contents
Journal	0x290000	64 KiB	Eight rotating metadata/data sector pairs
Boot	0x2A0000	128 KiB	Manifest plus a maximum 64 KiB Z80 RAM image
Drive A	0x2C0000	320 KiB	CP/M system disk
Drive B	0x310000	320 KiB	CP/M data disk
Drive C	0x360000	320 KiB	CP/M data disk
Drive D	0x3B0000	320 KiB	CP/M data disk

Each 320 KiB slot is exactly eighty 4 KiB flash erase sectors with no partial sector left over, so every slot boundary is also a sector boundary.

Disk-image contract. Here, "320 KiB" means exactly 327,680 bytes: 2,560 linear 128-byte CP/M records. The default BIOS presents those as 80 logical tracks of 32 records, numbered 1-32, with no skew. This is a project-specific logical geometry, not the IBM 3740 8-inch SSSD format (77 tracks x 26 records x 128 bytes = 256,256 bytes); existing images in another 8-inch format must be converted rather than copied into a slot unchanged. The matching CP/M 2.2 DPB is SPT=32, BSH=4, BLM=15, EXM=0, DSM=155, DRM=63, AL0=0x80, AL1=0x00, CKS=0, and OFF=2. Keep these values in the Z80 BIOS and host image builder from one shared generated definition.

Reservation mechanism. Set the CMake variable `PICO_FLASH_SIZE_BYTES` to `0x290000` before `pico_sdk_init()`. In Pico SDK 2.2 this value sizes the generated linker's `FLASH` region, while the `PICO_FLASH_SIZE_BYTES` C macro still comes from `pico2_w.h` and must remain the physical 4 MiB. That deliberate same-name split lets the linker reject code or const data that grows into storage while `hardware_flash` still accepts physical offsets through `0x3FFFFFF`. Do not pass `-DPICO_FLASH_SIZE_BYTES=0x290000` to the compiler: its flash erase/program bounds would then reject every storage write. Add `static_assert(PICO_FLASH_SIZE_BYTES == 4u * 1024u * 1024u, "Pico 2 W flash size changed")` to the storage module and inspect the link map in CI to require `__flash_binary_end <= 0x10290000`.

The build must also link `pico_flash`, `hardware_flash`, `hardware_watchdog`, `pico_multicore`, and `pico_util`. Define `PICO_FLASH_ASSUME_CORE1_SAFE=1`: core 0 uses `flash_safe_execute()` only for journal recovery before core 1 is launched, while later core-1 writes still lock out core 0 normally. Core 0 must never erase/program flash after core 1 starts. The essential CMake ordering is:

```
set(PICO_FLASH_SIZE_BYTES 0x290000)
pico_sdk_init()

target_compile_definitions(z80_supervisor PRIVATE
    PICO_FLASH_ASSUME_CORE1_SAFE=1)
target_link_libraries(z80_supervisor PRIVATE
    pico_flash hardware_flash hardware_watchdog pico_multicore pico_util)
```

Provisioning. Build each disk as a flat, exactly-320-KiB binary and build the boot package described below, then write them outside the running firmware. Use `-v` so `picotool` verifies every load:

```
picotool load -v -o 0x102A0000 -t bin z80boot.pkg
picotool load -v -o 0x102C0000 -t bin drive-a.img
```

```
picotool load -v -o 0x10310000 -t bin drive-b.img
picotool load -v -o 0x10360000 -t bin drive-c.img
picotool load -v -o 0x103B0000 -t bin drive-d.img
```

(0x10000000 is the Pico's XIP flash base address, so these absolute addresses are the flash-offset column above plus that base.) Reject any disk file whose host-side size is not exactly 327,680 bytes. If an RP2350 partition table is later embedded, declare these as data partitions and load them by partition ID; do not casually add `--ignore-partitions`. Firmware-only UF2 updates normally leave these addresses untouched, but `flash_nuke.uf2`, mass erase, or a replacement partition table destroys them, so keep host backups. There is no runtime bulk-image upload path in this design.

`z80boot.pkg` is little-endian. Its first 20 bytes are, in order, 32-bit magic 0x5442385A ("Z8BT"), 16-bit version 1, 16-bit header length 20, 32-bit image length, 32-bit IEEE CRC32 of the image, and 32-bit IEEE CRC32 of the preceding 16 header bytes. Pad the remainder of the first 4 KiB with 0xFF; the image begins at package offset 0x1000 and must be 1-65,536 bytes. The Z80 reset vector is at image offset zero. Generate this package and the BIOS DPB constants from one host tool so geometry and integrity metadata cannot drift.

Read path. A disk read copies from a single 4 KiB SRAM cache when it matches the selected drive and flash block; otherwise it copies directly from the memory-mapped flash region. There is no SPI transaction, DMA request, or cross-core handshake. The READY status uses release/acquire ordering, so core 0 cannot inspect the cache while core 1 is changing it.

Write and recovery path. CP/M still transfers 128-byte records, but the Pico coalesces them in one 4 KiB erase-block cache. The BIOS passes the standard CP/M WRITE classification from register C: normal writes and the first record of a newly allocated CP/M block may remain dirty in SRAM; directory writes flush immediately. Selecting another flash block flushes the previous block first, 250 ms without another changed record triggers an idle flush, and warm boot issues an explicit flush before it reloads CCP/BDOS. Thus CP/M cannot persist directory metadata ahead of its referenced data, a completed overwrite cannot remain indefinitely only in SRAM, and sequential writes to one track need at most one journaled flash update instead of as many as 32. An unchanged record does not dirty the cache.

The trap copies each complete write request into `disk_write_queue` and reports BUSY; it never erases flash itself. When a flush is required, core 1 asks core 0 to acquire BUSACK# while trapping remains enabled, and core 0 disables the trap only after BUSACK# is LOW. Core 1 then uses a bounded `flash_safe_execute()` call, which parks the registered core-0 victim and disables core-1 interrupts.

Each update rotates through one of eight 8 KiB journal pairs:

1. Erase the pair, then program the replacement 4 KiB data block.
2. Program a one-page header containing sequence, target offset, and CRC32. Until this valid header exists, the target remains untouched.
3. Erase/program the target block and verify all 4 KiB through XIP.
4. Erase the journal header only after verification succeeds.

At cold boot, core 0 scans valid journal headers before loading the Z80 image and restores them in sequence order. Therefore power loss before step 2 leaves the old target intact; power loss during or after step 3 leaves a valid replacement block from which boot recovery can finish. After a write, core 0 arms the trap while BUSACK# is still LOW and only then releases BUSREQ#, eliminating any interval in which the Z80 can run without I/O trapping. A bus-acquisition failure leaves the trap enabled; a release failure asserts RESET# and disables the trap. IORQ#, RD#, and WR# all tri-state whenever BUSACK# is asserted (Z8400 datasheet), so `io_trap_handler()` itself first confirms BUSACK# is still HIGH before touching anything; a stray edge while the Pico holds the bus disables the IORQ# interrupt and is ignored. The fitted 5 V-side 10 kOhm pull-ups in Section 0.3 keep the LVC244 inputs defined throughout the grant; Pico-side pulls cannot bias an input on the other side of the buffer. A successful write or recovery requires both a `PIC0_0K` safe-execute result and callback verification of the flash contents. A runtime safe-execute failure asserts RESET#, isolates the buses, stops CLK, and forces a watchdog reboot before attempting another core-0 request: SDK lockout-exit failure can leave core 0 parked, so continuing to its release queue would deadlock instead of recovering.

Flash write endurance is finite and chip-specific; confirm the Pico 2 W's actual onboard flash part's rated erase-cycle endurance before relying on this for a write-heavy workload. The erase-block cache substantially reduces wear for sequential CP/M writes, but this partition remains a poor fit for a disk used as constant scratch/swap space.

Core assignment. Reads happen inline in `io_trap_handler()` on core 0. Writes and flushes are deferred to core 1, the same task that owns the WebSocket terminal (Section 6.2). A cache-only write returns READY without a flash operation; after 250 ms of write inactivity, core 1 flushes it. Every required flush asks core 0's foreground loop to freeze the Z80, performs the journaled update, clears the cache's dirty state, and only then releases the Z80. A successful idle flush does not modify protocol status, so it cannot consume or overwrite a foreground command state. Wi-Fi association is a bounded polling state machine, so storage remains available with no access point. Flash access never touches SPI0, the MCP23S17, or any Z80 bus GPIO.

6.4 System Performance Envelope & Constraints

- **I/O Decode Width:** Strictly limited to **8-bit** decoding (monitoring address lines A0-A7 via the lower expander port).
- **Trap Latency Profile:** GAL-generated WAIT# covers the interval from IORQ# falling until DATA_ENABLE reports a configured data path. The Pico still stops the static CPU clock for unrestricted SPI servicing. Measure IORQ#-to-WAIT#, WAIT# setup/hold, the final PWM edge, and DATA_ENABLE-to-WAIT# release at every claimed rate. The design remains suitable for low-rate virtual peripherals, not high-speed line tracing.
- **Clock Validation Targets:**
 - *Bring-Up Target:* **1 MHz**, qualified only after a logic-analyzer capture proves every I/O cycle stops before the Z80 samples or releases its data.
 - *Qualification Range:* **2 MHz – 6 MHz**, tested in the increments in Section 8.12. No rate in this range is guaranteed in advance.
 - *Experimental Range:* **6.5 MHz – 8 MHz** may be attempted in 500 kHz steps only after 6 MHz passes. These rates are exploratory, not design claims, because the 55 ns SRAM and breadboard margin dominate despite the faster buffer.
 - *Failure Boundary:* Any WAIT setup/hold failure, SRAM setup failure, malformed clock edge, or repeatable memory/I/O error ends qualification at the preceding passing step.
 - *20 MHz CPU Rating:* The Z84C0020PEC rating applies to the CPU, not this no-wait-state breadboard system. At 20 MHz a clock period is 50 ns, shorter than the conservative 79.5 ns component-delay sum for the worst-case GAL + AHCT244 + SRAM select-to-data path. That sum excludes Z80 setup and breadboard delay. Reaching 20 MHz requires a redesigned control path, hardware-generated memory and I/O wait states (or deterministic clock gating), and a PCB-level signal- integrity review; changing the Pico PWM frequency is insufficient.
 - *Future PCB:* A PCB should make 6-8 MHz more credible by reducing stubs, contact resistance, loop area, and uncontrolled return paths. It cannot remove the 55 ns SRAM or GAL delays. A true zero-wait 20 MHz PCB needs roughly 10-15 ns SRAM plus faster decode/control logic; alternatively it can apply hardware WAIT# to memory cycles.

7. Reference Firmware Implementations

The maintained, buildable firmware is the canonical implementation: [browse the source tree](#) or use the [complete source index](#). The corresponding phase sections in Section 8 provide design-level excerpts for the safety invariants and integration order: variable-frequency clock generation, bus acquisition, synchronous I/O trapping, flash image loading, and terminal integration. Do not copy those excerpts in place of the maintained source.

7.1 Firmware and CP/M Build

The repository implements the ten cumulative firmware stages under `src/`. Stage 10 includes the journaled flash disks, CP/M boot loader, and WebSocket terminal. Its host build also assembles the board-native CP/M 2.2 BIOS, reconstructs pristine CCP/BDOS bytes from the preserved Altair source image, and emits all provisionable images.

Install CMake, Ninja, Python 3, `picotool`, and `z80asm`. The firmware needs a complete Arm GNU bare-metal toolchain with Newlib; the Homebrew compiler alone does not provide the required runtime. A known working setup is:

```
brew install cmake ninja picotool z80asm
export PATH="$HOME/.local/share/arm-gnu-toolchain-15.3.rel1/bin:$PATH"
cmake -S . -B build -G Ninja \
  -DPICO_BOARD=pico2_w \
  -DZ80_WIFI_SSID='your-network' \
  -DZ80_WIFI_PASSWORD='your-password'
cmake --build build --target z80_cpm_images -j
```

Wi-Fi credentials are written only to the generated build tree. An empty SSID leaves networking disabled while flash-disk service continues to operate. The `z80_cpm_images` target builds Stage 10 and writes these files to `build/cpm/`:

Artifact	Purpose
z80boot.pkg	Manifest, CRCs, and the reset-ready 64 KiB Z80 image
drive_a_cpm63k-z80.img	Native CP/M system disk with the Z80 SBC BIOS
drive_b_bdsc.img through drive_d_blank.img	Converted 320 KiB disks
z80romless-flash.bin	Complete 4 MiB initial-provisioning image
manifest.json	Geometry, addresses, sizes, and SHA-256 values

Run the host regression checks with:

```
python3 -m unittest discover -s src/cpm -p 'test_*.py' -v
```

7.2 Flash Provisioning

For a new or disposable flash, put the Pico in BOOTSEL mode and write the complete image. This erases existing CP/M disk contents and the journal:

```
picotool load -v build/cpm/z80romless-flash.bin -t bin -o 0x10000000
picotool reboot
```

For normal updates, load only the firmware and selected storage regions. These commands preserve regions not named by the input file:

```
picotool load -v build/src/stage10_websocket_terminal/z80_stage10_websocket_terminal.uf2
picotool load -v build/cpm/z80boot.pkg -t bin -o 0x102A0000
picotool load -v build/cpm/drive_a_cpm63k-z80.img -t bin -o 0x102C0000
picotool load -v build/cpm/drive_b_bdsc.img -t bin -o 0x10310000
```

```
picotool load -v build/cpm/drive_c_escape.img -t bin -o 0x10360000  
picotool load -v build/cpm/drive_d_blank.img -t bin -o 0x103B0000  
picotool reboot
```

After association, browse to <http://<pico-dhcp-address>:8088/>. The USB serial console reports Stage 10 startup and accepts `s` to print terminal queue/drop counts and disk/fatal status. CP/M disk writes are persistent, so retain host backups before full reprovisioning.

8. Progressive Build and Bring-Up Plan

Build and test one functional block at a time. Do not install the next chip until the current phase passes. Use sockets for all DIP devices, place a 100 nF ceramic capacitor directly across each IC's supply pins, and fit at least one 22 uF bulk capacitor per breadboard. Use a current-limited 5 V supply, multimeter, oscilloscope, and preferably a logic analyzer. Start each first power-up at a 100 mA current limit and remove power immediately if a rail falls by more than 5%, current rises unexpectedly, or a device becomes warm.

Fit the pull-ups and pull-downs in Section 0.3 so every signal has its specified state before firmware starts. Use temporary 1 kOhm series resistors when first connecting two potentially driven nodes. Record idle current after every phase. Unless stated otherwise, keep all chips from later phases out of their sockets.

The phase-specific Pico SDK fragments below show the safety-critical core of each test and finish with the required two-core `main()` integration order. Board-specific Wi-Fi/WebSocket hooks and the optional nonblocking USB command parser remain external. Every diagnostic command must print PASS or a detailed failure and call `isolate_buses()` before returning. `PIN_BUSACK_N` is GP0, driven from Z80 `BUSACK#` (pin 23) through the SN74LVC244AN input buffer (Section 5.3); `IORQ#`, `RD#`, and `MCP SO` use the same buffer, so no 5 V output reaches the Pico directly. `PIN_SRAM_CE_N` is fixed at GP5 (ATF22V10 pin 7; Section 1.2). GP23 on the Pico 2 W is dedicated to the CYW43439 wireless module's control interface (shared with GP24/GP25/GP29) and must never be repurposed. `PIN_DATA_0-` `PIN_DATA_7` occupy GP10-GP17. There is no `PIN_SRAM_SOURCE_SELECT` GPIO: the ATF22V10 combines `RESET#` and `BUSACK#` inside each programmed SRAM-control equation (Section 1.2), so ownership still switches automatically with no extra Pico pin. `PIN_WR_N` is GP28, buffered through the same SN74LVC244AN (Section 5.3); `io_trap_handler()` reads both `PIN_RD_N` and `PIN_WR_N` to resolve cycle intent.

8.1 Phase 0 - Empty Sockets and Power Distribution

Install: Breadboards, sockets, decoupling capacitors, pull-ups, power wiring, and signal wiring. Install no active device, including the Pico 2.

Implementation: [Phase 0 power checklist](#).

Test plan:

1. With every IC still removed, verify each socket's occupied rows, notch direction, pin-1 corner, and width against Section 3.1. Mark pin 1 on the breadboard and socket with a paint pen, and photograph the empty-board orientation before wiring over the socket outlines.
2. With power disconnected, check resistance from each supply rail to ground. Investigate readings below 1 kOhm after capacitors charge.
3. **GAL removed:** Check every address, data, and control net end-to-end, then verify no continuity between neighboring bus lines. At the empty GAL socket, verify each signal has no unintended short to GND, 5 V, or an adjacent pin. Do not attempt to verify GAL output levels while the GAL is removed.
4. Apply 5 V and measure every 5 V-powered DIP socket supply pin. Require 4.75 V to 5.25 V at VCC and less than 50 mV at each ground pin. The AHCT245 and GAL VCC socket contacts must read 5 V, while the LVC245 and LVC244 VCC contacts must remain at 0 V because their 3.3 V source, the absent Pico, is not yet installed. With the GAL removed, do not test GAL output levels. Verify the GAL socket and associated nets have no unintended continuity to GND, 5 V, or adjacent signals. GAL output-level verification is performed in Phase 2 after the GAL is installed.
5. If using the photographed plug-in supply, confirm that its body obscures no more than Core Board rows 1-3. Load its 5 V output to at least 500 mA, require 4.75 V to 5.25 V at the farthest board, and confirm no regulator becomes too hot to touch. Leave its 3.3 V output disconnected.
6. Verify the external +5 V rail reaches Pico `VSYS` only through the 1N5819 and does not reach Pico `VBUS`, the 3.3 V rail, or any GPIO contact. With external power applied, `VSYS` must be one Schottky drop below the +5 V rail. Verify each 5 V-side active-low control is pulled HIGH. With power removed, measure approximately 10 kOhm from every Pico-side pull-up contact to the unpowered 3.3 V rail and from every pull-down contact to GND, as listed in Section 0.3; powered Pico-side logic levels are checked in Phase 1.

Measure approximately 10 kOhm from each GP10-GP17 contact to GND through the data SIP network. Specifically require approximately 10 kOhm from Z80 WAIT# pin 24 / GAL pin 20 to +5 V, continuity from Z80 IORQ# pin 20 to GAL pin 13, and no WAIT# short to IORQ#, GND, or an adjacent GAL pin.

Pass gate: No shorts or crossed nets, correct supply voltage at every socket, and negligible current with all devices removed.

8.2 Phase 1 - Raspberry Pi Pico 2 Supervisor

Install: Pico 2 only.

Firmware feature: A diagnostic image must establish safe output levels before enabling any GPIO output: GP7 and GP9 LOW to isolate the data path and hold MCP RESET# asserted; GP3 LOW to assert Z80 RESET#; GP4, GP5, GP21, GP22, and GP26 HIGH to deassert BUSREQ#, SRAM CE#, SPI CS#, SRAM WE#, and SRAM OE#; GP2 LOW to stop the clock; and GP6 LOW for the inactive data direction. GP8 remains an input. It must also provide a slow walking-one GPIO test selected through the USB serial console.

Implementation: [Phase 1 application](#), backed by the shared [supervisor module](#).

Safe Startup and Walking Output (Phases 1-2)

Maintained source: [pins.h](#), [supervisor.h](#), and [supervisor.c](#).

Preload each output latch while the pin is still an input, then enable the output driver. This prevents a brief LOW pulse on active-low lines.

```
#include <stdint.h>
#include <stdio.h>
#include "pico/stdlib.h"

enum {
    PIN_IORQ_N = 1, PIN_CLK = 2, PIN_RESET_N = 3,
    PIN_BUSREQ_N = 4, PIN_BUSACK_N = 0,
    PIN_DATA_DIR = 6, PIN_DATA_ENABLE = 7,
    PIN_UNUSED_8 = 8, PIN_ADDR_ENABLE = 9,
    PIN_DATA_0 = 10, PIN_DATA_1 = 11, PIN_DATA_2 = 12, PIN_DATA_3 = 13,
    PIN_DATA_4 = 14, PIN_DATA_5 = 15, PIN_DATA_6 = 16, PIN_DATA_7 = 17,
    PIN_SPI_SCK = 18, PIN_SPI_MOSI = 19, PIN_SPI_MISO = 20,
    PIN_SPI_CS_N = 21,
    PIN_SRAM_WE_N = 22, PIN_SRAM_CE_N = 5, PIN_SRAM_OE_N = 26,
    PIN_RD_N = 27, PIN_WR_N = 28
};

static void output_with_initial_level(uint pin, bool level) {
    gpio_init(pin);
    gpio_put(pin, level); // Preload SIO output latch.
    gpio_set_dir(pin, GPIO_OUT);
}

static void input_with_no_pull(uint pin) {
    gpio_init(pin);
    gpio_set_dir(pin, GPIO_IN);
    gpio_disable_pulls(pin);
}

static void diagnostic_safe_startup(void) {
    output_with_initial_level(PIN_DATA_ENABLE, 0);
    output_with_initial_level(PIN_ADDR_ENABLE, 0);
    output_with_initial_level(PIN_RESET_N, 0); // Hold CPU reset.
    output_with_initial_level(PIN_BUSREQ_N, 1);
    output_with_initial_level(PIN_SRAM_WE_N, 1);
    output_with_initial_level(PIN_SRAM_CE_N, 1);
    output_with_initial_level(PIN_SRAM_OE_N, 1);
    output_with_initial_level(PIN_SPI_CS_N, 1);
    output_with_initial_level(PIN_CLK, 0);
    output_with_initial_level(PIN_DATA_DIR, 0);
    input_with_no_pull(PIN_BUSACK_N);
    input_with_no_pull(PIN_IORQ_N); // Section 0.3 pulls up the LVC244 input.
    input_with_no_pull(PIN_RD_N);
    input_with_no_pull(PIN_WR_N);
}

static void walking_output_test(const uint *pins, size_t count) {
```



```

for (size_t active = 0; active < count; ++active) {
    for (size_t index = 0; index < count; ++index)
        gpio_put(pins[index], index == active);
    sleep_ms(250);           // Probe or logic-analyzer capture point.
}
diagnostic_safe_startup();
}

```

Run `walking_output_test()` only in Phases 1-2 while the destination driver/bus chips are absent. It deliberately changes raw pin levels and is not safe as an in-system diagnostic after Phase 3.

Test plan:

1. With the Section 0.4 1N5819 fitted, USB and external power may be connected together. Confirm neither source back-powers the other, then require 3.20 V to 3.40 V on the Pico 3.3 V rail, at the still-absent SN74LVC245AN and SN74LVC244AN VCC contacts.
2. Scope GP7 and GP9 through reset and startup; GP7 must remain LOW and GP9 must remain LOW so both bus interfaces stay isolated.
3. Verify GP3 and the directly connected Z80 RESET# socket pin are LOW, and all other Pico control pins have the inactive levels listed above before, during, and after startup. Verify the RESET# node never exceeds the Pico 3.3 V rail; no 5 V pull-up is permitted on it.
4. Before configuring GP10-GP17 as outputs, require all eight to read LOW from the external SIP network. Drive each HIGH in turn and verify 3.20-3.40 V while the other seven remain LOW.
5. Run the walking-one test and probe each destination socket. Require one-to-one routing, 0 V/3.3 V levels, and no change on neighboring pins. Restore safe levels when the test ends or the USB link drops.

Pass gate: Stable 3.3 V, safe startup levels, and correct routing for every Pico signal.

8.3 Phase 2 - ATF22V10 Arbitration and SN74AHCT244 Buffer

Install: Program and verify the ATF22V10 outside the circuit, then install it with the SN74AHCT244 and SRAM still removed. After the GAL truth-table tests pass, install the AHCT244. Keep the Z80, MCP23S17, and SRAM removed. Connect GAL pins 9/10/11 to GP7/GP9/GP6 with their fitted pull-downs, connect raw IORQ# to GAL pin 13, connect GAL pin 20 to the pulled-up Z80 WAIT# node, and tie both AHCT244 output-enable pins LOW. Require 4.75 V to 5.25 V at GAL VCC before testing any pulled-up GAL output.

Firmware feature: Add commands to toggle each supervisor output at 10 Hz and generate selectable 1 kHz, 100 kHz, and 1 MHz 50% duty-cycle clocks on GP2.

Implementation: [Phase 2 application](#), using the shared [clock module](#).

Variable-Frequency Clock Generation (Phase 2, Phases 7-8 Run Modes)

Maintained source: [clock.h](#) and [clock.c](#).

Configure PIN_CLK as a PWM output once during Phase 2 bring-up. Reuse the same slice for the selectable 10 Hz/1 kHz/100 kHz/1 MHz run modes in Phases 7-8, and to freeze the clock during the Phase 8 I/O trap.

```

#include "hardware/pwm.h"
#include "hardware/clocks.h"

static bool set_z80_clock_hz(uint32_t hz) {
    if (hz < 10 || hz > 8000000)
        return false;

    uint slice_num = pwm_gpio_to_slice_num(PIN_CLK);
    pwm_set_enabled(slice_num, false);
    uint32_t sys_clk = clock_get_hz(clk_sys);
    uint64_t sys_clk16 = (uint64_t)sys_clk * 16u;
    uint32_t best_divider16 = 0;
    uint32_t best_count = 0;
    uint64_t best_error = UINT64_MAX;

    // Search all legal 8.4 dividers; this runs only when frequency changes.
    for (uint32_t divider16 = 16; divider16 <= 4095; ++divider16) {
        uint64_t denominator = (uint64_t)hz * divider16;
        uint64_t count = (sys_clk16 + denominator / 2u) / denominator;
    }
}

```



```

if (count < 1) count = 1;
if (count > 65536) count = 65536;
uint64_t product = (uint64_t)divider16 * count;
uint64_t target_product = (uint64_t)hz * product;
uint64_t error = sys_clk16 > target_product ?
    sys_clk16 - target_product : target_product - sys_clk16;
uint64_t best_product = (uint64_t)best_divider16 * best_count;
if (best_divider16 == 0 ||
    error * best_product < best_error * product) {
    best_divider16 = divider16;
    best_count = (uint32_t)count;
    best_error = error;
}
}

uint16_t wrap = (uint16_t)(best_count - 1u);
pwm_set_clkdiv(slice_num, (float)best_divider16 / 16.0f);
pwm_set_wrap(slice_num, wrap);
pwm_set_chan_level(slice_num, PWM_CHAN_A,
    (uint16_t)(best_count / 2u)); // Exact 50% when count is even.
pwm_set_counter(slice_num, 0);
gpio_set_function(PIN_CLK, GPIO_FUNC_PWM);
pwm_set_enabled(slice_num, true);
return true;
}

static void stop_z80_clock(void) {
    pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), false); // May freeze HIGH or LOW.
}

static void resume_z80_clock(void) {
    pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), true);
}

```

Test plan:

1. Require a successful programmer readback/verify of the exact src/pld/sram_control.pld JEDEC image before inserting the GAL.
2. With RESET# LOW, toggle Pico CE#/OE#/WE# one at a time and require the corresponding GAL pin 16/15/14 to follow while the other two remain HIGH. Then set RESET# HIGH and manually exercise pulled-up BUSACK#/MREQ#/RD#/WR# through 1 kOhm; verify all three CPU-side truth table paths and the BUSACK# LOW override.
3. With each Pico and Z80 candidate control held HIGH, toggle RESET# and BUSACK# separately while observing GAL pins 14-16 on the scope. The consensus terms must hold every output continuously HIGH; any active-low pulse fails the programmed image.
4. **GAL installed; data transceivers removed:** Hold DATA_ENABLE pin 9 LOW and toggle DATA_DIR pin 11; GAL pins 17 and 18 must both remain HIGH. Drive DATA_ENABLE HIGH: pin 17 must be LOW only when DATA_DIR is HIGH, and pin 18 must be LOW only when DATA_DIR is LOW. Scope both outputs while changing DATA_DIR with DATA_ENABLE LOW; neither may pulse LOW.
5. Hold DATA_ENABLE LOW. Drive the pulled-up pin-13 IORQ# test node LOW through 1 kOhm and require GAL pin 20 / WAIT# LOW. Raise DATA_ENABLE and require WAIT# HIGH; release IORQ# and require WAIT# to remain HIGH for either DATA_ENABLE state. Scope IORQ#-to-WAIT# assertion and DATA_ENABLE-to-WAIT# release; any glitch or inverted case fails.
6. Install the AHCT244. Use the Stage 2 walking command to toggle its eight functional input paths independently; RESET# was already tested in Phase 1 and is not an AHCT244 input. At the selected output require LOW below 0.3 V, HIGH at or above 4.4 V, correct polarity, and no activity on adjacent outputs.
7. Test AHCT244 channel 1A1 at each clock frequency. Require 45% to 55% duty cycle and clean transitions at Z80 socket pin 6.
8. After the Pico 3.3 V rail reaches 3.20 V, verify RESET# remains LOW while BUSREQ#, SRAM CE#/WE#/OE#, and SPI CS# remain HIGH for at least 100 ms. Before 3.3 V is valid, RESET# must remain LOW but SRAM control levels are not used as a retention guarantee.
9. Power-cycle ten times while monitoring these signals. Any active-low transition after 3.3 V becomes valid fails the phase.

Pass gate: Programmer verification and every GAL truth-table case pass, all eight AHCT244 outputs have valid 5 V levels and correct polarity, the 1 MHz clock is clean, and startup creates no active-low glitch.

8.4 Phase 3 - MCP23S17 SPI Address Generator

Install: The 3.3 V-powered SN74LVC244AN first, then Q1 and the MCP23S17-E/SP. The already-tested AHCT244 supplies all three SPI inputs. Keep Z80 and SRAM removed; their empty sockets expose the pulled-up shared address bus for probing.

Electrical hold point: Fit level translation on all SPI inputs. The MCP23S17 datasheet specifies $V_{IH} \geq 0.8V_{DD}$ for CS#, SCK, and SI, which is 4.0 V with a 5 V supply; a Pico 3.3 V HIGH is therefore not compliant. Translate Pico CS#, SCK, and MOSI through SN74AHCT244 channels 3-5 (Section 4). Buffer MCP SO/MISO down through SN74LVC244AN channel 2A1/2Y1 (Section 5.3); do not connect it directly to GP20. Tie the MCP23S17 A0/A1/A2 hardware address pins to GND (Section 2). Confirm this wiring against the schematic before proceeding.

Firmware feature: Add byte-level SPI register read/write, a write-then-read register test, and 16-bit walking-one/walking-zero tests that can configure both MCP ports as either inputs or outputs.

Implementation: [Phase 3 application](#), using the shared [MCP23S17 driver](#).

MCP23S17 Register and Port Test (Phases 3-4)

Maintained source: [mcp23s17.h](#) and [mcp23s17.c](#).

The SPI translator sits between these Pico pins and the 5 V MCP23S17; MISO/SO returns through the SN74LVC244AN. The register test proves communication before either address transceiver is enabled.

```
#include "hardware/spi.h"

enum { MCP_WRITE = 0x40, MCP_READ = 0x41 };
enum { IODIRA = 0x00, IODIRB = 0x01, GPIOA = 0x12, GPIOB = 0x13,
      OLATA = 0x14, OLATB = 0x15 };

static void mcp_spi_init(void) {
    output_with_initial_level(PIN_SPI_CS_N, 1);
    spi_init(spi0, 4000000);
    gpio_set_function(PIN_SPI_SCK, GPIO_FUNC_SPI);
    gpio_set_function(PIN_SPI_MOSI, GPIO_FUNC_SPI);
    gpio_set_function(PIN_SPI_MISO, GPIO_FUNC_SPI);
}

static void mcp_write(uint8_t reg, uint8_t value) {
    uint8_t frame[] = { MCP_WRITE, reg, value };
    gpio_put(PIN_SPI_CS_N, 0);
    spi_write_blocking(spi0, frame, sizeof frame);
    gpio_put(PIN_SPI_CS_N, 1);
}

static uint8_t mcp_read(uint8_t reg) {
    uint8_t tx[] = { MCP_READ, reg, 0x00 };
    uint8_t rx[sizeof tx];
    gpio_put(PIN_SPI_CS_N, 0);
    spi_write_read_blocking(spi0, tx, rx, sizeof tx);
    gpio_put(PIN_SPI_CS_N, 1);
    return rx[2];
}

static bool mcp_register_test(void) {
    const uint8_t patterns[] = { 0x55, 0xAA };
    mcp_write(IODIRA, 0x00);
    mcp_write(IODIRB, 0x00);
    for (size_t i = 0; i < sizeof patterns; ++i) {
        mcp_write(OLATA, patterns[i]);
        mcp_write(OLATB, (uint8_t)~patterns[i]);
        if (mcp_read(OLATA) != patterns[i] ||
            mcp_read(OLATB) != (uint8_t)~patterns[i])
            return false;
    }
    return true;
}
```

Test plan:

1. Before fitting the MCP, pull each used LVC244 input LOW through 1 kOhm, then release it HIGH through its fitted 10 kOhm pull-up. Verify the matching Pico input reads LOW and HIGH at 3.3 V levels and unused channels do not change.
2. With GP9 LOW, verify Q1 holds MCP RESET# LOW and all A0-A15 nodes read HIGH through the two SIP networks. Drive GP9 HIGH and require a clean 5 V RESET# release.
3. With compliant translation fitted, write and read back 0x55 and 0xAA in IODIRA, IODIRB, OLATA, and OLATB.
4. Configure outputs and probe walking-one and walking-zero patterns at the empty Z80 and SRAM sockets.
5. Configure inputs, apply 0 V or 5 V through 10 kOhm to each pin, and verify only the corresponding GPIO register bit changes.
6. Run 10,000 alternating register writes and reads with zero errors.

Pass gate: Compliant SPI levels, error-free register access, and correct operation of all 16 port bits in both directions.

8.5 Phase 4 - Direct Address Bus and Reset Isolation

Install: No additional IC. Keep Z80 and SRAM removed.

Firmware feature: Address helpers must assert ADDR_ENABLE LOW, release MCP reset, wait, preload OLAT, and only then set IODIR outputs. On every exit they must assert reset again.

Implementation: [Phase 4 application](#), using the shared [bus module](#).

Contention-Safe Bus Isolation (Phases 4-6)

Maintained source: [bus.h](#) and [bus.c](#).

ADDR_ENABLE and DATA_ENABLE must both be LOW for isolation. DATA_ENABLE must remain LOW before DATA_DIR changes.

```
static void isolate_buses(void) {
    gpio_put(PIN_ADDR_ENABLE, 0);
    gpio_put(PIN_DATA_ENABLE, 0);
}
```

Data and Address Bus GPIO Helpers (Phases 4-6)

Maintained source: [bus.h](#) and [bus.c](#).

PIN_DATA_0-PIN_DATA_7 (GP10-GP17) connect to both fixed-direction data transceivers. PIN_DATA_DIR and PIN_DATA_ENABLE feed the GAL, which enables exactly one selected path. Address bytes reach the shared bus through the MCP23S17's own output latches, already exercised in the Phase 3-4 register test. Call these helpers only while RESET# is asserted or request_cpu_bus() has returned true, and keep SRAM CE# HIGH while changing address or data direction.

```
static const uint DATA_PINS[8] = {
    PIN_DATA_0, PIN_DATA_1, PIN_DATA_2, PIN_DATA_3,
    PIN_DATA_4, PIN_DATA_5, PIN_DATA_6, PIN_DATA_7
};

static void data_bus_drive(uint8_t value) {
    gpio_put(PIN_DATA_ENABLE, 0);
    for (size_t i = 0; i < 8; ++i) {
        gpio_put(DATA_PINS[i], (value >> i) & 1); // Preload before enabling output.
        gpio_set_dir(DATA_PINS[i], GPIO_OUT);
    }
    gpio_put(PIN_DATA_DIR, 1); // Select AHCT Pico-to-bus path.
    busy_wait_us_32(1);
    gpio_put(PIN_DATA_ENABLE, 1);
}

static void data_bus_prepare_input(void) {
    gpio_put(PIN_DATA_ENABLE, 0);
    for (size_t i = 0; i < 8; ++i)
```

```

    gpio_set_dir(DATA_PINS[i], GPIO_IN);
    gpio_put(PIN_DATA_DIR, 0); // Select LVC bus-to-Pico path.
    busy_wait_us_32(1);
    gpio_put(PIN_DATA_ENABLE, 1);
}

static uint8_t data_bus_sample(void) {
    uint8_t value = 0;
    for (size_t i = 0; i < 8; ++i)
        value |= gpio_get(DATA_PINS[i]) << i;
    return value;
}

static void address_bus_drive(uint16_t address) {
    gpio_put(PIN_ADDR_ENABLE, 0);
    busy_wait_us_32(1); // MCP RESET# low-pulse minimum.
    gpio_put(PIN_ADDR_ENABLE, 1); // Release MCP reset; ports default to inputs.
    busy_wait_us_32(1);
    mcp_write(OLATA, (uint8_t)address);
    mcp_write(OLATB, (uint8_t)(address >> 8));
    mcp_write(IODIRA, 0x00);
    mcp_write(IODIRB, 0x00);
}

```

Test plan:

1. With ADDR_ENABLE LOW, verify RESET# LOW and all 16 MCP pins high-impedance; each bus line must sit HIGH through its 10 kOhm pull-up.
2. Release reset, select MCP output direction, and test 0x55, 0xAA, walking-one, and walking-zero patterns on every shared address line.
3. Assert reset and use a 1 kOhm test pull-down to prove each bus line moves independently while MCP is isolated.
4. Release reset with IODIR inputs. Drive each bus line through 1 kOhm and verify the MCP reads it without driving back.
5. Test 0x0000, 0xFFFF, 0x5555, 0xAAAA, and a walking one across A0-A15, repeat with a walking zero, then run 1,000 release/configure/reset cycles while checking current. Scope A0, A7, A8, and A15 at their SRAM socket contacts during 0x0000, 0xFFFF, 0x5555, and 0xAAAA; require valid levels without double-clocking, sustained mid-rail plateaus, or ringing that crosses the MCP/SRAM input thresholds.

Pass gate: Every address bit passes in both directions and MCP reset reliably returns every port to high-impedance input mode.

8.6 Phase 5 - All-PDIP Data Transceivers and Interlock

Install sequence: Start with the verified GAL installed and both data transceivers removed. Perform test 1, power off, then install the SN74AHCT245N and SN74LVC245AN in that order. Keep Z80 and SRAM removed. Tie AHCT DIR HIGH and LVC DIR LOW before insertion.

Firmware feature: Add an 8-bit data-bus test using the same disable-change-enable sequence and fixed, walking-one, and walking-zero patterns on the Pico data GPIOs.

Implementation: [Phase 5 application](#), using the shared [bus module](#).

Test plan:

1. **GAL installed; data transceivers removed:** With GP6/GP7 LOW, verify GAL pins 17 and 18 are both HIGH. Toggle GP7 HIGH and GP6 LOW/HIGH; require exactly one GAL OE# output to go LOW and verify the other remains HIGH.
2. **GAL and data transceivers installed:** With GP7 LOW, verify GAL pins 17/18 and both transceiver OE# pin 19 contacts are HIGH. Confirm both A and B buses remain high-impedance and neither supply current changes abnormally. This is the end-to-end safety-property check: DATA_ENABLE LOW must disable both physical data drivers. Verify the AHCT A1-A8 pins remain below 0.8 V while Pico data GPIOs are inputs.
3. Select Pico-to-bus direction and test 0x00, 0xFF, 0x55, 0xAA, walking-one, and walking-zero. Verify levels and bit order.
4. Drive GP7 LOW, change GP6 LOW, and drive GP7 HIGH. Drive each bus input with 0 V and 5 V through 1 kOhm and verify the Pico reading.
5. Run 1,000 disable-change-enable cycles while checking readback and supply current.

Pass gate: All eight bits pass both ways, isolation works, and no direction change causes contention or unexpected current.

8.7 Phase 6 - AS6C1008 SRAM and DMA Path

Install: The AS6C1008-55PCN only after the programmed ATF22V10, AHCT244 channels 2A2-2A4, and final SRAM-side pull-ups have passed Phase 2. Keep the Z80 removed.

Electrical hold point: During DMA the GAL must select only Pico CE#/OE#/WE#; during execution it must select only Z80 MREQ#/RD#/WR#. CE# must not be tied LOW, and no Pico output is joined directly to a Z80 output. With the Z80 removed, BUSACK# floats HIGH via its pull-up; hold RESET# LOW so the programmed equations select the Pico side regardless. Do not install SRAM until all three GAL outputs and all three corresponding AHCT244 outputs have passed static truth-table, continuity, and voltage-level tests.

Firmware feature: Add single-byte DMA read/write primitives, a walking address/data test, a two-pass full-memory pattern test, and a March C- or equivalent RAM test. Every failure must report its address, expected byte, and actual byte over USB serial.

Implementation: Phase 6 application, using the shared [SRAM DMA module](#).

SRAM DMA Access and Pattern Test (Phase 6)

Maintained source: [sram.h](#) and [sram.c](#).

The address and data helper functions above represent the already tested MCP23S17 and GPIO bus operations. The control-source selector must grant exclusive SRAM control to the Pico before this code runs.

```
static void dma_write_byte(uint16_t address, uint8_t value) {
    gpio_put(PIN_SRAM_CE_N, 1);
    gpio_put(PIN_SRAM_OE_N, 1);
    gpio_put(PIN_SRAM_WE_N, 1);
    address_bus_drive(address);           // MCP directly drives A0-A15.
    data_bus_drive(value);               // Pico -> AHCT245 -> D0-D7.
    gpio_put(PIN_SRAM_CE_N, 0);
    busy_wait_us_32(1);
    gpio_put(PIN_SRAM_WE_N, 0);
    busy_wait_us_32(1);
    gpio_put(PIN_SRAM_WE_N, 1);
    gpio_put(PIN_SRAM_CE_N, 1);
}

static uint8_t dma_read_byte(uint16_t address) {
    uint8_t value;
    gpio_put(PIN_SRAM_CE_N, 1);
    gpio_put(PIN_SRAM_WE_N, 1);
    address_bus_drive(address);
    data_bus_prepare_input();           // Disable, reverse, then enable.
    gpio_put(PIN_SRAM_CE_N, 0);
    gpio_put(PIN_SRAM_OE_N, 0);
    busy_wait_us_32(1);
    value = data_bus_sample();
    gpio_put(PIN_SRAM_OE_N, 1);
    gpio_put(PIN_SRAM_CE_N, 1);
    gpio_put(PIN_DATA_ENABLE, 0);
    return value;
}

static bool sram_pattern_test(bool complement) {
    for (uint32_t address = 0; address < 65536; ++address) {
        uint8_t expected = (uint8_t)address ^ (uint8_t)(address >> 8);
        dma_write_byte((uint16_t)address,
            complement ? (uint8_t)~expected : expected);
    }
    for (uint32_t address = 0; address < 65536; ++address) {
        uint8_t expected = (uint8_t)address ^ (uint8_t)(address >> 8);
        if (complement)
            expected = (uint8_t)~expected;
        uint8_t actual = dma_read_byte((uint16_t)address);
        if (actual != expected) {
            printf("FAIL %04lx expected=%02x actual=%02x\n",
                (unsigned long)address, expected, actual);
            isolate_buses();
        }
    }
}
```

```

    return false;
}
}
isolate_buses();
return true;
}

```

Test plan:

1. With transceivers disabled, require A16 LOW, CE2 HIGH, and CE#, OE#, and WE# HIGH directly at the SRAM.
2. **GAL -> AHCT244 -> SRAM end-to-end:** Keep the data transceivers disabled. With RESET# LOW, toggle one Pico CE#/OE#/WE# candidate at a time and probe the corresponding GAL pre-buffer pin, AHCT244 output, and final SRAM pin. Keep CE# HIGH while testing OE#/WE#; test CE# only with OE#/WE# HIGH. Then set RESET# and pulled-up BUSACK# HIGH and repeat by pulling MREQ#/RD#/WR# LOW individually through 1 kOhm. Require the selected path to reach below 0.3 V at the SRAM, return to at least 4.4 V, preserve polarity, and leave the other two SRAM controls HIGH.
3. In DMA mode, write and read one byte. Require WE# HIGH before address or data changes, and disable the Pico data driver before asserting OE# for readback.
4. Write unique values at 0x0000 and each power-of-two address from 0x0001 through 0x8000. Verify every value remains independent.
5. At several addresses test 0x00, 0xFF, 0x55, 0xAA, walking-one, and walking-zero data.
6. Fill all 65,536 bytes with the XOR of the address bytes, verify it, then repeat with the complement. Run the March test afterward.
7. Repeat after ten power cycles and with the intended 1 MHz timing.

Pass gate: Every end-to-end control path reaches the correct SRAM pin with no adjacent-control activity, zero address/data/full-range pattern or March-test errors, and no overlap between CPU and Pico SRAM control sources.

8.8 Phase 7 - Z84C0020PEC CPU, Installed Last

Install: Z84C0020PEC. Preload SRAM first. Disable both bus transceivers, drive the Pico-side SRAM CE#/OE#/WE# controls inactive, hold RESET# LOW and BUSREQ# HIGH, and stop the clock LOW before insertion and power-up. RESET# LOW deliberately keeps the programmed logic on the inactive Pico side; the GAL equations switch to CPU controls automatically only when RESET# is released with BUSACK# HIGH.

Firmware feature: Add clock single-step and selectable 10 Hz, 1 kHz, 100 kHz, and 1 MHz run modes; reset pulse control; timed BUSREQ#/BUSACK# acquisition; and a command that preloads and verifies a small test program before releasing reset.

Implementation: [Phase 7 application](#), using the shared [CPU ownership module](#).

Z80 Single-Step and Timed Bus Request (Phases 7-8)

Maintained source: [cpu.h](#) and [cpu.c](#).

Single-step uses SIO rather than PWM. `request_cpu_bus()` and `release_cpu_bus()` each take an explicit `timeout_us` bound so a wiring fault or a missing Z80 cannot hang the supervisor waiting on BUSACK#. The clock must be running while requesting or releasing BUSREQ#; after using `clock_one_cycle()`, call `set_z80_clock_hz()` to restore the CLK pin's PWM function before either handshake.

```

static void clock_one_cycle(uint32_t half_period_us) {
    pwm_set_enabled(pwm_gpio_to_slice_num(PIN_CLK), false);
    gpio_set_function(PIN_CLK, GPIO_FUNC_SIO);
    gpio_set_dir(PIN_CLK, GPIO_OUT);
    gpio_put(PIN_CLK, 0);
    busy_wait_us_32(half_period_us);
    gpio_put(PIN_CLK, 1);
    busy_wait_us_32(half_period_us);
    gpio_put(PIN_CLK, 0);
}

static bool request_cpu_bus(uint32_t timeout_us) {
    absolute_time_t deadline = make_timeout_time_us(timeout_us);
    isolate_buses();

```



```

gpio_put(PIN_BUSREQ_N, 0);
while (gpio_get(PIN_BUSACK_N) != 0) {
    if (time_reached(deadline)) {
        gpio_put(PIN_BUSREQ_N, 1);
        return false;
    }
    tight_loop_contents();
}
return true;           // RESET# HIGH: BUSACK# LOW selects DMA controls.
}

static bool release_cpu_bus(uint32_t timeout_us) {
    absolute_time_t deadline = make_timeout_time_us(timeout_us);
    gpio_put(PIN_SRAM_CE_N, 1);
    gpio_put(PIN_SRAM_OE_N, 1);
    gpio_put(PIN_SRAM_WE_N, 1);
    isolate_buses();
    gpio_put(PIN_BUSREQ_N, 1);
    while (gpio_get(PIN_BUSACK_N) == 0) {
        if (time_reached(deadline)) {
            gpio_put(PIN_RESET_N, 0);
            return false;
        }
        tight_loop_contents();
    }
    return true;        // BUSACK# HIGH restores CPU controls.
}

```

Test plan:

1. Verify RESET# LOW, BUSREQ# HIGH, both transceiver OE# signals HIGH, and SRAM write inactive immediately after power-up.
2. Clock at 10 Hz with RESET# asserted for at least three full cycles. Release reset and verify the first opcode fetch at 0x0000, including the expected M1#, MREQ#, and RD# sequence.
3. Preload NOP; NOP; JP 0000h. Single-step and verify address progression and the repeating jump while Pico bus drivers stay high-impedance.
4. Repeat at 1 kHz, 100 kHz, and 1 MHz with no unexpected SRAM writes.
5. Assert BUSREQ# while clocking. Require BUSACK# LOW after the current machine cycle and verify CPU bus outputs are high-impedance. Release BUSREQ# and require execution to resume.
6. Assert RESET# for at least three clocks during execution, release it, and verify a new fetch from 0x0000.

Pass gate: Correct reset fetch and execution through 1 MHz, valid BUSREQ#/BUSACK# transfer, and no bus contention.

8.9 Phase 8 - Integrated Virtual-ROM Boot and I/O Trap

Install: No further chips. Load final supervisor firmware.

Firmware feature: Combine safe startup, timed bus acquisition, image injection and readback, run control, and the synchronous IN/OUT trap. Maintain counters for boots, DMA failures, readback mismatches, trap timeouts, and unexpected RD#/WR# control states.

Implementation: [Phase 8 application](#), using the shared [I/O trap](#), [CPU](#), and [SRAM](#) modules.

Synchronous I/O Trap Handler (Phase 8)

Maintained source: [io_trap.h](#) and [io_trap.c](#).

A falling-edge IRQ on PIN_IORQ_N freezes the clock, reverses the address transceiver with the same contention-safe helper used elsewhere, reads the trapped port from the lower MCP23S17 port (Section 6.4's 8-bit decode limit), and reuses the already-tested data bus helpers from the SRAM DMA code to sample or drive the data byte. The handler first confirms PIN_BUSACK_N is still HIGH; IORQ# tri-states along with RD#/WR# whenever BUSACK# is asserted (Section 6.3), so a falling edge seen while the Pico already owns the bus cannot be a real Z80 cycle and is ignored before any bus or SPI0 state is touched. The GAL has already asserted WAIT# before the handler runs. Both MCP ports must be forced to inputs before the two transceivers' shared OE# is enabled, even though only GPIOA is read; otherwise the still-output GPB port fights Z80 A8-A15. For IN, the data byte must stay driven

until the Z80 samples it, so the clock resumes and RD# is polled before the data bus is isolated. DATA_ENABLE is also the GAL's WAIT-ready input, so the selected data path must remain enabled until IORQ# and RD#/WR# have both returned HIGH.

process_virtual_io_read and process_virtual_io_write are the only application-supplied hooks. They must use the signatures below, finish without sleeping, printing, waiting on USB, or taking a lock held by main code, and must not start another bus operation. Move expensive work to the main loop through a lock-free queue.

```
#include "hardware/watchdog.h"

uint8_t process_virtual_io_read(uint8_t port);
void process_virtual_io_write(uint8_t port, uint8_t value);

enum { TRAP_RELEASE_TIMEOUT_US = 500000 }; // Covers the 10 Hz test mode.
static volatile uint32_t trap_timeout_count;
static volatile uint32_t unexpected_control_count;

static _Noreturn void reset_after_trap_fault(void) {
    isolate_buses();
    gpio_put(PIN_RESET_N, 0);
    stop_z80_clock();
    for (unsigned int cycle = 0; cycle < 3; ++cycle)
        clock_one_cycle(1); // RESET# setup and each half-cycle exceed Z80 minima.
    watchdog_reboot(0, 0, 0); // Fail-closed: same recovery path as flash faults.
    while (true)
        tight_loop_contents(); // watchdog_reboot() takes effect asynchronously.
}

static void resume_and_wait_for_release(uint control_pin) {
    absolute_time_t deadline = make_timeout_time_us(TRAP_RELEASE_TIMEOUT_US);
    resume_z80_clock();
    while (!gpio_get(PIN_IORQ_N) || !gpio_get(control_pin)) {
        if (time_reached(deadline)) {
            ++trap_timeout_count;
            reset_after_trap_fault();
        }
        tight_loop_contents();
    }
}

static void io_trap_handler(uint gpio, uint32_t events) {
    (void)events;
    if (gpio != PIN_IORQ_N)
        return; // Only IORQ# is ever armed, but never trust a shared callback.
    if (!gpio_get(PIN_BUSACK_N)) {
        gpio_set_irq_enabled(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL, false);
        return; // The foreground release path re-arms this interrupt.
    }
    stop_z80_clock();
    gpio_put(PIN_ADDR_ENABLE, 1); // Release MCP reset.
    busy_wait_us_32(1);
    mcp_write(IODIRA, 0xFF);
    mcp_write(IODIRB, 0xFF);
    uint8_t port = mcp_read(GPIOA); // 8-bit I/O decode.
    gpio_put(PIN_ADDR_ENABLE, 0); // Reset makes all ports inputs.

    bool is_read = !gpio_get(PIN_RD_N);
    bool is_write = !gpio_get(PIN_WR_N);
    if (is_read == is_write) { // Neither or both asserted: hardware fault.
        ++unexpected_control_count;
        reset_after_trap_fault();
    }

    if (is_write) {
        data_bus_prepare_input(); // Bus -> Pico.
        uint8_t value = data_bus_sample();
        process_virtual_io_write(port, value);
        resume_and_wait_for_release(PIN_WR_N);
        isolate_buses();
    } else {
        data_bus_drive(process_virtual_io_read(port)); // Pico -> Bus.
        resume_and_wait_for_release(PIN_RD_N);
        isolate_buses();
    }
}

static void enable_io_trap(void) {
    input_with_no_pull(PIN_IORQ_N); // Pull-up is on the LVC244's 5 V input.
    gpio_acknowledge_irq(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL);
    gpio_set_irq_enabled_with_callback(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL,
```



```

    true, &io_trap_handler);
}

static void disable_io_trap(void) {
    gpio_set_irq_enabled(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL, false);
    gpio_acknowledge_irq(PIN_IORQ_N, GPIO_IRQ_EDGE_FALL);
}

```

Test plan:

1. Acquire ownership by one of two complete procedures. Either use the timed BUSREQ#/BUSACK# handshake, or drive the Pico SRAM controls inactive, assert RESET#, continue the clock for at least three full cycles, and then stop it. In either case, verify the Z80 address and data buses are high-impedance before enabling a transceiver. The Section 1.2 AND gate then grants the Pico exclusive SRAM control; inject a known 64 KB image, read back every byte, and compare it before CPU release.
2. Run a RAM increment loop for one hour at 1 MHz without corruption or supervisor timeout.
3. Execute OUT on ports 0x00, 0x01, 0x55, 0xAA, and 0xFF with matching data. Verify correct clock stop, address/data capture, driver disable, and clock resume.
4. Execute matching IN tests and verify each Pico-generated byte is stored correctly in SRAM.
5. Bind the same terminal ports to a bounded USB CDC queue. On macOS, connect through /dev/cu.usbmodem...; verify bidirectional bytes, disconnect/reconnect, full-queue counters, and no blocking call in the trap. This is an intermediate transport test, not the final terminal.
6. With a logic analyzer, prove IORQ# asserts WAIT# before the Z80 WAIT sampling edge, DATA_ENABLE releases WAIT# only after direction/data are stable, and WAIT# remains released until IORQ# and RD#/WR# are HIGH.
7. With a logic analyzer, prove both transceiver OE# signals are HIGH during CPU cycles, the CPU bus is high-impedance before DMA enable, both DIR controls change only while their OE# is HIGH, and no overlap occurs between SRAM control sources.
8. Repeat cold boot, injection, execution, and I/O tests 100 times with zero logged failures.

Pass gate: Zero image or boot failures, correct IN/OUT behavior, one-hour stable execution, and contention-free ownership transitions.

8.10 Phase 9 - Flash Disk Image Loader and CP/M Storage

Install: No hardware rework. Set PICO_FLASH_SIZE_BYTES to 0x290000 in CMakeLists.txt, define PICO_FLASH_ASSUME_CORE1_SAFE=1, and link the libraries listed in Section 6.3. Provision the manifest-backed boot package and all four 320 KiB disk slots with the verified picotool commands there.

Firmware feature: With RESET# held LOW, recover any valid journal, validate the boot manifest and CRC32, DMA-write its payload to SRAM, and compare every byte before RESET# release. Do not wait for BUSACK# during this cold-boot path: RESET# itself selects the Pico's SRAM controls through Section 1.2's GAL equations. Once running, ports 0x10-0x14 provide command/status, drive, 16-bit LBA, and 128-byte data transfers. Reads are synchronous XIP copies; writes use the journaled core-1 service and BUSY/READY/ERROR status defined in Section 8.9.

Implementation: [Phase 9 application](#), with the shared [disk device](#), [flash backend](#), and [flash layout](#).

Flash Disk Image Loader (Final Phase 9 Integration)

Maintained source: [flash_disk.h](#), [flash_layout.h](#), [disk_device.c](#), and [flash_backend.c](#).

Loading a boot image is now a synchronous, core-0-only operation: a flash read is an ordinary memory access through the XIP-mapped pointer (Section 6.3), so bringing the initial image into SRAM needs no filesystem, blocking I/O call, or core 1 task. Only CP/M's live disk-sector *writes* still need to run on core 1 and cross back to core 0's foreground loop, because only they need to freeze the Z80 around a flash erase/program cycle (Section 6.3).

```

#include <stdint.h>
#include <stdbool.h>
#include <stddef.h>
#include <stdio.h>
#include <string.h>

```

```

#include "hardware/flash.h"
#include "hardware/watchdog.h"
#include "pico/flash.h"
#include "pico/util/queue.h"

enum {
    FLASH_JOURNAL_BASE_OFFSET = 0x290000u,
    FLASH_JOURNAL_BYTES = 0x10000u,
    FLASH_JOURNAL_PAIR_BYTES = 2u * FLASH_SECTOR_SIZE,
    FLASH_JOURNAL_PAIR_COUNT = FLASH_JOURNAL_BYTES / FLASH_JOURNAL_PAIR_BYTES,
    FLASH_BOOT_BASE_OFFSET = 0x2A0000u,
    FLASH_BOOT_REGION_BYTES = 0x20000u,
    FLASH_BOOT_PAYLOAD_OFFSET = FLASH_BOOT_BASE_OFFSET + FLASH_SECTOR_SIZE,
    FLASH_DISK_BASE_OFFSET = 0x2C0000u,    // Section 6.3 partition table.
    FLASH_DISK_SLOT_BYTES = 0x50000u,    // 320 KiB per drive.
    FLASH_DISK_SLOT_COUNT = 4u,
    FLASH_DISK_RECORD_BYTES = 128u,
    FLASH_DISK_RECORD_COUNT = 2560u,
    SRAM_SIZE_BYTES = 65536
};

_Static_assert(PICO_FLASH_SIZE_BYTES == 4u * 1024u * 1024u,
    "Pico 2 W physical flash size changed");
_Static_assert(FLASH_DISK_SLOT_BYTES ==
    FLASH_DISK_RECORD_BYTES * FLASH_DISK_RECORD_COUNT,
    "disk geometry does not fill its slot");
_Static_assert(FLASH_BOOT_PAYLOAD_OFFSET + SRAM_SIZE_BYTES <=
    FLASH_BOOT_BASE_OFFSET + FLASH_BOOT_REGION_BYTES,
    "boot payload exceeds its reserved region");
_Static_assert(FLASH_DISK_BASE_OFFSET +
    FLASH_DISK_SLOT_COUNT * FLASH_DISK_SLOT_BYTES == PICO_FLASH_SIZE_BYTES,
    "disk slots must end at physical flash boundary");
_Static_assert(FLASH_JOURNAL_PAIR_COUNT == 8,
    "journal layout no longer matches the partition table");
_Static_assert(PICO_FLASH_ASSUME_CORE1_SAFE,
    "core 0 journal recovery runs only before core 1 is launched");

static const uint8_t *flash_disk_slot_ptr(unsigned drive) {
    uint32_t offset = FLASH_DISK_BASE_OFFSET + drive * FLASH_DISK_SLOT_BYTES;
    return (const uint8_t *) (XIP_BASE + offset);
}

static uint32_t crc32_bytes(const uint8_t *data, size_t length) {
    uint32_t crc = UINT32_MAX;
    for (size_t i = 0; i < length; ++i) {
        crc ^= data[i];
        for (unsigned int bit = 0; bit < 8; ++bit)
            crc = (crc >> 1) ^ (0xEDB88320u & (uint32_t)~(int32_t)(crc & 1u));
    }
    return ~crc;
}

enum { Z80_BOOT_MAGIC = 0x5442385Au, Z80_BOOT_VERSION = 1u }; // "Z8BT".

typedef struct {
    uint32_t magic;
    uint16_t version;
    uint16_t header_bytes;
    uint32_t image_bytes;
    uint32_t image_crc32;
    uint32_t header_crc32;
} z80_boot_manifest_t;

_Static_assert(sizeof(z80_boot_manifest_t) == 20,
    "boot manifest layout must match the host packer");

// Core 0 only, entirely synchronous: a flash read needs no filesystem,
// queue, or core 1 task.
static bool prepare_reset_held_dma(void) {
    isolate_buses();
    gpio_put(PIN_RESET_N, 0);
    for (unsigned int cycle = 0; cycle < 3; ++cycle)
        clock_one_cycle(1); // Bit-banged SIO pulses; set_z80_clock_hz() runs later.
    stop_z80_clock();

    if (!gpio_get(PIN_BUSACK_N) || !gpio_get(PIN_IORQ_N) ||
        !gpio_get(PIN_RD_N) || !gpio_get(PIN_WR_N))
        return false;

    return true; // RESET# LOW selects the Pico SRAM controls (Section 1.2).
}

```

```

static bool boot_image_from_flash(void) {
    const z80_boot_manifest_t *manifest =
        (const z80_boot_manifest_t *) (XIP_BASE + FLASH_BOOT_BASE_OFFSET);
    const uint8_t *source =
        (const uint8_t *) (XIP_BASE + FLASH_BOOT_PAYLOAD_OFFSET);

    if (manifest->magic != Z80_BOOT_MAGIC ||
        manifest->version != Z80_BOOT_VERSION ||
        manifest->header_bytes != sizeof(*manifest) ||
        manifest->image_bytes == 0 || manifest->image_bytes > SRAM_SIZE_BYTES)
        return false;

    uint32_t header_crc = crc32_bytes((const uint8_t *) manifest,
        offsetof(z80_boot_manifest_t, header_crc32));
    if (header_crc != manifest->header_crc32 ||
        crc32_bytes(source, manifest->image_bytes) != manifest->image_crc32)
        return false;

    if (!prepare_reset_held_dma())
        return false;

    for (uint32_t i = 0; i < manifest->image_bytes; ++i)
        dma_write_byte((uint16_t) i, source[i]);

    bool ok = true;
    for (uint32_t i = 0; i < manifest->image_bytes; ++i) {
        if (dma_read_byte((uint16_t) i) != source[i]) {
            printf("boot verify failed near %04lx\n", (unsigned long) i);
            ok = false;
            break;
        }
    }

    if (ok)
        printf("loaded boot bytes=%lu crc32=%08lx\n",
            (unsigned long) manifest->image_bytes,
            (unsigned long) manifest->image_crc32);
    isolate_buses();
    // RESET# remains asserted; the caller releases it only after success.
    return ok;
}

static bool flash_recover_journal(void);

static bool boot_cpm_from_flash(void) {
    return flash_recover_journal() && boot_image_from_flash();
}

enum { FLASH_SERVICE_ACQUIRE_BUS, FLASH_SERVICE_RELEASE_BUS };

typedef struct {
    uint8_t operation;
} flash_service_request_t;

static queue_t flash_service_request_queue; // Core 1 -> Core 0.
static queue_t flash_service_result_queue; // Core 0 -> Core 1.

static void flash_service_queues_init(void) {
    queue_init(&flash_service_request_queue, sizeof(flash_service_request_t), 4);
    queue_init(&flash_service_result_queue, sizeof(bool), 4);
}

// Core 0's foreground loop only; never called from io_trap_handler().
static void core0_service_flash_requests(void) {
    flash_service_request_t request;
    if (!queue_try_remove(&flash_service_request_queue, &request))
        return;

    bool ok;
    switch (request.operation) {
        case FLASH_SERVICE_ACQUIRE_BUS:
            ok = request_cpu_bus(500000);
            if (ok)
                disable_io_trap(); // BUSACK# LOW: no new cycle can start.
            break;
        case FLASH_SERVICE_RELEASE_BUS:
            enable_io_trap(); // Arm before BUSACK# lets the Z80 run.
            ok = release_cpu_bus(500000);
            if (!ok)
                disable_io_trap(); // release_cpu_bus() asserted RESET#.
            break;
        default:
            ok = false;
    }
}

```

```

        break;
    }
    queue_add_blocking(&flash_service_result_queue, &ok);
}

// Core 1 only; blocks its caller, never core 0's foreground loop.
static bool core0_request(uint8_t operation) {
    flash_service_request_t request = { operation };
    queue_add_blocking(&flash_service_request_queue, &request);
    bool ok = false;
    queue_remove_blocking(&flash_service_result_queue, &ok);
    return ok;
}

enum { FLASH_JOURNAL_MAGIC = 0x314C4E4Au }; // "JNL1".

typedef struct {
    uint32_t magic;
    uint32_t sequence;
    uint32_t target_offset;
    uint32_t data_crc32;
    uint32_t header_crc32;
    uint8_t reserved[FLASH_PAGE_SIZE - 20u];
} flash_journal_header_t;

_Static_assert(sizeof(flash_journal_header_t) == FLASH_PAGE_SIZE,
    "journal header must occupy one flash page");

typedef struct {
    uint32_t header_offset;
    uint32_t data_offset;
    uint32_t target_offset;
    uint8_t *data;
    flash_journal_header_t header;
    bool committed;
} flash_write_params_t;

static bool flash_disk_block_offset_valid(uint32_t offset) {
    uint32_t disk_end = FLASH_DISK_BASE_OFFSET +
        FLASH_DISK_SLOT_COUNT * FLASH_DISK_SLOT_BYTES;
    return offset >= FLASH_DISK_BASE_OFFSET &&
        offset <= disk_end - FLASH_SECTOR_SIZE &&
        (offset & (FLASH_SECTOR_SIZE - 1u)) == 0;
}

static _Noreturn void reboot_after_flash_safe_failure(void) {
    gpio_put(PIN_RESET_N, 0);
    isolate_buses();
    stop_z80_clock();
    watchdog_reboot(0, 0, 0);
    while (true)
        tight_loop_contents();
}

static void flash_write_callback(void *param) {
    flash_write_params_t *p = (flash_write_params_t *)param;
    flash_range_erase(p->header_offset, FLASH_JOURNAL_PAIR_BYTES);
    flash_range_program(p->data_offset, p->data, FLASH_SECTOR_SIZE);
    flash_range_program(p->header_offset, (const uint8_t *)&p->header,
        FLASH_PAGE_SIZE);
    flash_range_erase(p->target_offset, FLASH_SECTOR_SIZE);
    flash_range_program(p->target_offset, p->data, FLASH_SECTOR_SIZE);

    p->committed = memcmp((const void *) (XIP_BASE + p->target_offset),
        p->data, FLASH_SECTOR_SIZE) == 0;
    if (p->committed) {
        flash_range_erase(p->header_offset, FLASH_SECTOR_SIZE);
        p->committed = *(const uint32_t *) (XIP_BASE + p->header_offset) ==
            UINT32_MAX;
    }
}

typedef struct {
    uint32_t header_offset;
    uint32_t target_offset;
    uint8_t *data;
    bool restored;
} flash_restore_params_t;

static void flash_restore_callback(void *param) {
    flash_restore_params_t *p = (flash_restore_params_t *)param;
    flash_range_erase(p->target_offset, FLASH_SECTOR_SIZE);
    flash_range_program(p->target_offset, p->data, FLASH_SECTOR_SIZE);

```

```

p->restored = memcmp((const void *) (XIP_BASE + p->target_offset),
p->data, FLASH_SECTOR_SIZE) == 0;
if (p->restored) {
    flash_range_erase(p->header_offset, FLASH_SECTOR_SIZE);
    p->restored = *(const uint32_t *) (XIP_BASE + p->header_offset) ==
        UINT32_MAX;
}
}

// Core 0 only, before core 1 launch and before RESET# release.
static bool flash_recover_journal(void) {
    static flash_journal_header_t headers[FLASH_JOURNAL_PAIR_COUNT];
    static bool valid[FLASH_JOURNAL_PAIR_COUNT];
    static uint8_t recovery_block[FLASH_SECTOR_SIZE];

    for (unsigned pair = 0; pair < FLASH_JOURNAL_PAIR_COUNT; ++pair) {
        uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
            pair * FLASH_JOURNAL_PAIR_BYTES;
        memcpy(&headers[pair], (const void *) (XIP_BASE + header_offset),
            sizeof headers[pair]);
        valid[pair] = headers[pair].magic == FLASH_JOURNAL_MAGIC &&
            headers[pair].header_crc32 == crc32_bytes((const uint8_t *)&headers[pair],
                offsetof(flash_journal_header_t, header_crc32)) &&
            flash_disk_block_offset_valid(headers[pair].target_offset);
    }

    for (unsigned recovered = 0; recovered < FLASH_JOURNAL_PAIR_COUNT;
        ++recovered) {
        int selected = -1;
        for (unsigned pair = 0; pair < FLASH_JOURNAL_PAIR_COUNT; ++pair) {
            if (valid[pair] && (selected < 0 || headers[pair].sequence <
                headers[(unsigned)selected].sequence))
                selected = (int)pair;
        }
        if (selected < 0)
            break;

        unsigned pair = (unsigned)selected;
        uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
            pair * FLASH_JOURNAL_PAIR_BYTES;
        uint32_t data_offset = header_offset + FLASH_SECTOR_SIZE;
        memcpy(recovery_block, (const void *) (XIP_BASE + data_offset),
            sizeof recovery_block);
        if (crc32_bytes(recovery_block, sizeof recovery_block) !=
            headers[pair].data_crc32)
            return false;

        flash_restore_params_t params = {
            header_offset, headers[pair].target_offset, recovery_block, false
        };
        int rc = flash_safe_execute(flash_restore_callback, &params, 1000);
        if (rc != PICO_OK || !params.restored)
            return false;
        valid[pair] = false;
    }
    return true;
}

typedef struct {
    bool valid;
    bool dirty;
    unsigned drive;
    uint32_t block_offset;
    uint8_t data[FLASH_SECTOR_SIZE];
} flash_disk_cache_t;

static flash_disk_cache_t disk_cache;
static absolute_time_t disk_cache_flush_deadline;

// Core 1 only. Commits the cached erase block, freezing the Z80 while
// flash_safe_execute() pauses XIP fetches on both cores (Section 6.3).
static bool flash_disk_flush_cache(void) {
    if (!disk_cache.dirty)
        return true;

    uint32_t flash_offset = FLASH_DISK_BASE_OFFSET +
        disk_cache.drive * FLASH_DISK_SLOT_BYTES + disk_cache.block_offset;

    if (!core0_request(FLASH_SERVICE_ACQUIRE_BUS))
        return false;

    static uint32_t sequence;
    static unsigned next_pair;

```

```

unsigned pair = next_pair++ % FLASH_JOURNAL_PAIR_COUNT;
uint32_t header_offset = FLASH_JOURNAL_BASE_OFFSET +
    pair * FLASH_JOURNAL_PAIR_BYTES;

flash_write_params_t params;
memset(&params, 0, sizeof params);
params.header_offset = header_offset;
params.data_offset = header_offset + FLASH_SECTOR_SIZE;
params.target_offset = flash_offset;
params.data = disk_cache.data;
memset(&params.header, 0xFF, sizeof params.header);
params.header.magic = FLASH_JOURNAL_MAGIC;
params.header.sequence = ++sequence;
params.header.target_offset = flash_offset;
params.header.data_crc32 = crc32_bytes(disk_cache.data,
    sizeof disk_cache.data);
params.header.header_crc32 = crc32_bytes((const uint8_t *)&params.header,
    offsetof(flash_journal_header_t, header_crc32));

int rc = flash_safe_execute(flash_write_callback, &params, 1000);
if (rc != PIC0_OK)
    reboot_after_flash_safe_failure();
if (params.committed)
    disk_cache.dirty = false;
bool released = core0_request(FLASH_SERVICE_RELEASE_BUS);
return released && params.committed;
}

static bool flash_disk_select_cache(unsigned drive, uint16_t lba) {
    uint32_t sector_offset = (uint32_t)lba * FLASH_DISK_RECORD_BYTES;
    uint32_t block_offset = sector_offset & ~(FLASH_SECTOR_SIZE - 1u);
    if (disk_cache.valid && disk_cache.drive == drive &&
        disk_cache.block_offset == block_offset)
        return true;
    if (!flash_disk_flush_cache())
        return false;

    uint32_t flash_offset = FLASH_DISK_BASE_OFFSET +
        drive * FLASH_DISK_SLOT_BYTES + block_offset;
    memcpy(disk_cache.data, (const void *)(XIP_BASE + flash_offset),
        sizeof disk_cache.data);
    disk_cache.drive = drive;
    disk_cache.block_offset = block_offset;
    disk_cache.valid = true;
    return true;
}

static bool flash_disk_write_record(unsigned drive, uint16_t lba,
    const uint8_t *data, uint8_t write_type) {
    if (!flash_disk_select_cache(drive, lba))
        return false;
    size_t offset = ((size_t)lba * FLASH_DISK_RECORD_BYTES) &
        (FLASH_SECTOR_SIZE - 1u);
    if (memcmp(disk_cache.data + offset, data, FLASH_DISK_RECORD_BYTES)) {
        memcpy(disk_cache.data + offset, data, FLASH_DISK_RECORD_BYTES);
        disk_cache.dirty = true;
        disk_cache_flush_deadline = make_timeout_time_ms(250);
    }
    return write_type == 1 ? flash_disk_flush_cache() : true;
}

static bool flash_disk_read_record(unsigned drive, uint16_t lba,
    uint8_t *data) {
    uint32_t sector_offset = (uint32_t)lba * FLASH_DISK_RECORD_BYTES;
    uint32_t block_offset = sector_offset & ~(FLASH_SECTOR_SIZE - 1u);
    uint32_t within_block = sector_offset - block_offset;
    if (disk_cache.valid && disk_cache.drive == drive &&
        disk_cache.block_offset == block_offset) {
        memcpy(data, disk_cache.data + within_block, FLASH_DISK_RECORD_BYTES);
    } else {
        uint32_t flash_offset = FLASH_DISK_BASE_OFFSET +
            drive * FLASH_DISK_SLOT_BYTES + sector_offset;
        memcpy(data, (const void *)(XIP_BASE + flash_offset),
            FLASH_DISK_RECORD_BYTES);
    }
    return true;
}

enum {
    DISK_COMMAND_STATUS_PORT = 0x10,
    DISK_DRIVE_PORT = 0x11,
    DISK_LBA_LOW_PORT = 0x12,
    DISK_LBA_HIGH_PORT = 0x13,

```

```

DISK_DATA_PORT = 0x14,
DISK_COMMAND_CLEAR = 0,
DISK_COMMAND_READ = 1,
DISK_COMMAND_WRITE_NORMAL = 2,
DISK_COMMAND_WRITE_DIRECTORY = 3,
DISK_COMMAND_WRITE_UNALLOCATED = 4,
DISK_COMMAND_FLUSH = 5,
DISK_STATUS_READY = 1u << 0,
DISK_STATUS_DATA_READY = 1u << 1,
DISK_STATUS_DATA_ROOM = 1u << 2,
DISK_STATUS_BUSY = 1u << 3,
DISK_STATUS_ERROR = 1u << 7,
DISK_WRITE_QUEUE_DEPTH = 2
};

typedef struct {
    uint8_t command;
    uint8_t drive;
    uint16_t lba;
    uint8_t data[FLASH_DISK_RECORD_BYTES];
} disk_request_t;

static queue_t disk_write_queue;           // Z80/Core 0 -> Core 1.
static uint32_t disk_status = DISK_STATUS_READY;
static uint32_t disk_fatal_error;
static uint8_t disk_drive;
static uint16_t disk_lba;
static uint8_t disk_write_drive; // Snapshot of drive/lba at WRITE issue,
static uint16_t disk_write_lba;  // immune to changes during data transfer.
static uint8_t disk_write_type;
static uint16_t disk_data_index;
static uint8_t disk_data[FLASH_DISK_RECORD_BYTES];

static uint32_t disk_status_load(void) {
    return __atomic_load_n(&disk_status, __ATOMIC_ACQUIRE);
}

static void disk_status_store(uint32_t status) {
    __atomic_store_n(&disk_status, status, __ATOMIC_RELEASE);
}

static bool disk_address_valid(void) {
    return disk_drive < FLASH_DISK_SLOT_COUNT &&
        disk_lba < FLASH_DISK_RECORD_COUNT;
}

static void disk_service_init(void) {
    queue_init(&disk_write_queue, sizeof(disk_request_t),
        DISK_WRITE_QUEUE_DEPTH);
    __atomic_store_n(&disk_fatal_error, 0, __ATOMIC_RELEASE);
    disk_status_store(DISK_STATUS_READY);
}

static void disk_start_command(uint8_t command) {
    if (disk_status_load() & DISK_STATUS_BUSY)
        return;

    bool fatal = __atomic_load_n(&disk_fatal_error, __ATOMIC_ACQUIRE);
    if (command == DISK_COMMAND_CLEAR) {
        disk_data_index = 0;
        disk_status_store(DISK_STATUS_READY | (fatal ? DISK_STATUS_ERROR : 0));
        return;
    }
    if (fatal) {
        disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
        return;
    }

    if (command == DISK_COMMAND_FLUSH) {
        disk_request_t request = { .command = command };
        if (queue_try_add(&disk_write_queue, &request))
            disk_status_store(DISK_STATUS_BUSY);
        else
            disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
        return;
    }
    if (!disk_address_valid()) {
        disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
        return;
    }

    disk_data_index = 0;
    if (command == DISK_COMMAND_READ) {

```

```

    bool ok = flash_disk_read_record(disk_drive, disk_lba, disk_data);
    disk_status_store(DISK_STATUS_READY |
        (ok ? DISK_STATUS_DATA_READY : DISK_STATUS_ERROR));
} else if (command >= DISK_COMMAND_WRITE_NORMAL &&
    command <= DISK_COMMAND_WRITE_UNALLOCATED) {
    disk_write_drive = disk_drive;
    disk_write_lba = disk_lba;
    disk_write_type = command - DISK_COMMAND_WRITE_NORMAL;
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_DATA_ROOM);
} else {
    disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
}
}

static uint8_t disk_virtual_io_read(uint8_t port) {
    if (port == DISK_COMMAND_STATUS_PORT)
        return (uint8_t)disk_status_load();
    if (port != DISK_DATA_PORT ||
        !(disk_status_load() & DISK_STATUS_DATA_READY))
        return 0;

    uint8_t value = disk_data[disk_data_index++];
    if (disk_data_index == sizeof disk_data)
        disk_status_store(DISK_STATUS_READY);
    return value;
}

static void disk_virtual_io_write(uint8_t port, uint8_t value) {
    uint32_t status = disk_status_load();
    if (port == DISK_COMMAND_STATUS_PORT) {
        disk_start_command(value);
    } else if (status & DISK_STATUS_BUSY) {
        return;
    } else if (port == DISK_DRIVE_PORT) {
        disk_drive = value;
    } else if (port == DISK_LBA_LOW_PORT) {
        disk_lba = (uint16_t)((disk_lba & 0xFF00u) | value);
    } else if (port == DISK_LBA_HIGH_PORT) {
        disk_lba = (uint16_t)((disk_lba & 0x00FFu) | ((uint16_t)value << 8));
    } else if (port == DISK_DATA_PORT &&
        (status & DISK_STATUS_DATA_ROOM)) {
        disk_data[disk_data_index++] = value;
        if (disk_data_index == sizeof disk_data) {
            disk_request_t request = {
                .command = DISK_COMMAND_WRITE_NORMAL + disk_write_type,
                .drive = disk_write_drive,
                .lba = disk_write_lba
            };
            memcpy(request.data, disk_data, sizeof request.data);
            if (queue_try_add(&disk_write_queue, &request))
                disk_status_store(DISK_STATUS_BUSY);
            else
                disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
        }
    }
}

// Core 1 only; call on every loop iteration, including while Wi-Fi is down.
static void core1_service_disk_request(void) {
    disk_request_t request;
    if (!queue_try_remove(&disk_write_queue, &request)) {
        if (!disk_cache.dirty || !time_reached(disk_cache_flush_deadline))
            return;
        bool ok = flash_disk_flush_cache();
        if (!ok)
            __atomic_store_n(&disk_fatal_error, 1, __ATOMIC_RELEASE);
        if (!ok)
            disk_status_store(DISK_STATUS_READY | DISK_STATUS_ERROR);
        return;
    }

    bool ok = request.command == DISK_COMMAND_FLUSH
        ? flash_disk_flush_cache()
        : flash_disk_write_record(request.drive, request.lba, request.data,
            request.command - DISK_COMMAND_WRITE_NORMAL);
    if (!ok)
        __atomic_store_n(&disk_fatal_error, 1, __ATOMIC_RELEASE);
    disk_status_store(DISK_STATUS_READY | (ok ? 0 : DISK_STATUS_ERROR));
}

```


The Z80 BIOS sets drive and 16-bit LBA, then writes command 1 for a read. Write commands 2, 3, and 4 mean normal, directory, and first record of a newly allocated CP/M block respectively; command 5 explicitly flushes the cache. A read returns 128 bytes from the data port. A write accepts exactly 128 bytes there, then changes status to BUSY until core 1 has cached it and completed any required journaled flush. Command 0 clears a transient protocol/queue error when not busy; a flash or journal failure remains latched until reboot and recovery. The BIOS polls READY/DATA_READY/DATA_ROOM/BUSY instead of assuming Pico timing, but reads each status only once per poll and uses Z80 INIR/OTIR for payload transfer. It also waits for READY before writing a command's drive/LBA registers. Idle flushes are serialized separately by holding BUSACK# from before the flash operation until dirty state is cleared. No erase, program, blocking queue, or flash-safe call runs inside `io_trap_handler()`.

Test plan:

1. With the Z80 socket populated and Phase 8 passing, confirm `io_trap_handler()` still passes every Phase 8 IN/OUT test unchanged; Phase 9 adds no new pins or rework to disturb it.
2. Confirm the link fails if code crosses `0x10290000`, while a runtime print/static assertion still reports the physical C macro as 4 MiB.
3. Provision the boot package and four exact-size disk images with `picotool -v`; read every region's first and last page back and compare host CRC32 values. Give every image a distinct known directory sentinel (DUMP.COM on A, CC2.COM on B, ATTNC11.COM on C, and DOCTOR.COM on D) so a valid but aliased drive mapping cannot pass.
4. Cold-boot with RESET# held LOW. Require journal recovery and SRAM verification to finish without waiting for BUSACK#, then release RESET# only after success. Corrupt the manifest, payload, and SRAM readback separately and require each case to remain fail-closed.
5. Exercise the decoded port path with real Z80 IN/OUT instructions, not direct calls to the C handlers. Prove that `0x10` selects only command/status, `0x11` only the drive, `0x12/0x13` the complete LBA, and `0x14` only record data; also prove that terminal ports `0x00/0x01` cannot intercept or alias any disk access.
6. Exercise all 2,560 LBAs on every drive through ports `0x10-0x14`, alternating the selected drive between commands. Verify exact 128-byte transfers, distinct per-drive sentinel records, no cross-drive aliasing, invalid drive/LBA rejection, command while BUSY rejection, and test-injected queue-full/error clearing behavior. After an injected flash or journal failure, require every READ/WRITE command to retain READY|ERROR until reboot recovery.
7. Start a write, then change the live drive and LBA registers before core 1 services its queue. Verify the completed write uses the drive, LBA, write type, and 128-byte payload captured when the command began and changes no other record.
8. Boot the packaged image and exercise the filesystem through CP/M itself: run DIR on A-D and require each sentinel on only its expected drive; run LS through natural transient exit and BIOS warm boot; run multi-extent C:ATTNC11; and use PIP to copy files across drives. Reboot and compare the affected records and directories byte-for-byte with the expected host images.
9. Probe BUSREQ#, BUSACK#, IORQ#, and CLK during a write. Require the trap to remain armed until BUSACK# is LOW and to be armed again before BUSACK# returns HIGH, with no untrapped I/O edge in either interval. Inject an IORQ# falling edge while BUSACK# is LOW and require the handler to disable the IRQ without touching CLK, SPI0, or either bus.
10. Add test-only power-cut hooks after journal-data program, header program, target erase, partial target program, target verification, and header clear. Reboot after every hook and require recovery to produce either the complete old block (before valid header) or the complete new block (after valid header), never a mixture.
11. Repeat reads and writes with Wi-Fi absent, associating, connected, and reconnecting. Disk completion must not depend on network state, and WebSocket queue overflow must remain counted rather than block.
12. Rewrite hot directory blocks repeatedly while tracking journal-pair rotation and the flash part's rated erase endurance. Treat this as a smoke test, not proof of lifetime.
13. Inject safe-execute entry and exit failures. Require RESET# LOW, isolated buses, stopped CLK, and a watchdog reboot without waiting on the core-0 release queue; recovery must retain the verified old or new disk block.

Pass gate: Boot and all four disks match host CRC32 values; A-D expose their distinct expected sentinel files with no aliasing; real CP/M DIR, transient execution, warm boot, multi-extent loading, and cross-drive copy complete through the BIOS; queued writes retain their command-time drive/LBA snapshot; every fault-injection reboot recovers an intact old or new block; all bounds and manifest failures remain fail-closed; disk service works without Wi-Fi; the linker protects the storage boundary; and logic-analyzer captures show no untrapped Z80 cycle around a flash write.

8.11 Phase 10 - WebSocket Terminal Console

Install: No further bus hardware. Use a Pico 2 W for the final networked terminal build. Wi-Fi/WebSocket is the required final user terminal; the Phase 8 USB CDC transport remains available for bring-up and diagnostics but is not the target operating interface. A non-W Pico 2 may compile the same hooks as stubs only for host-side development.

Firmware feature: Start the WebSocket console service on core 1 after core 0 has completed safe GPIO startup, queue initialization, and the Phase 9 boot-image load (which finishes entirely on core 0 before core 1 is launched). Core 0 continues to own the Z80 clock, bus transceivers, MCP23S17, SRAM DMA, and I/O trap. Core 1 owns Wi-Fi connection management, the embedded HTTP terminal page, WebSocket client state, network polling, and the Section 6.3 flash disk-write service -- all in the same `core1_main()` task, since `multicore_launch_core1()` only accepts one entry point. The two cores exchange only terminal bytes, flash disk-write requests, and status through nonblocking queues, following the `pico-altair-8800` console bridge pattern.

Implementation: [Phase 10 application](#), with the shared [terminal bridge](#), [network service](#), and [browser terminal](#).

WebSocket Terminal I/O Bridge (Final Phase 10 Integration)

Maintained source: [terminal.h](#), [terminal_bridge.c](#), and [terminal_network.cpp](#).

The terminal bridge follows the `pico-altair-8800` model: initialize the queues on core 0, load the boot image from flash (Section 6.3, entirely on core 0), then launch core 1 to own Wi-Fi and WebSocket work -- and, per Section 6.3, the flash disk-write service, in the same task. The exact HTTP/WebSocket library can be `pico-ws-server` as in that project, or another lwIP-based server with the same callback shape. Only the queue functions are visible to the Z80 trap.

```
#include "pico/time.h"
#include "pico/multicore.h"
#include "pico/util/queue.h"

enum {
    TERM_DATA_PORT = 0x00,
    TERM_STATUS_PORT = 0x01,
    TERM_RX_DEPTH = 128,
    TERM_TX_DEPTH = 512,
    TERM_STATUS_RX_READY = 1u << 0,
    TERM_STATUS_TX_ROOM = 1u << 1,
    TERM_STATUS_CLIENT = 1u << 7
};

static queue_t terminal_rx_queue; // Browser/Core 1 -> Z80/Core 0.
static queue_t terminal_tx_queue; // Z80/Core 0 -> Browser/Core 1.
static uint32_t terminal_client_connected;
static uint32_t terminal_rx_drop_count;
static uint32_t terminal_tx_drop_count;

static void terminal_queues_init(void) {
    queue_init(&terminal_rx_queue, sizeof(uint8_t), TERM_RX_DEPTH);
    queue_init(&terminal_tx_queue, sizeof(uint8_t), TERM_TX_DEPTH);
}

uint8_t process_virtual_io_read(uint8_t port) {
    if (port >= DISK_COMMAND_STATUS_PORT && port <= DISK_DATA_PORT)
        return disk_virtual_io_read(port);

    if (port == TERM_DATA_PORT) {
        uint8_t value = 0;
        queue_try_remove(&terminal_rx_queue, &value);
        return value;
    }

    if (port == TERM_STATUS_PORT) {
        uint8_t status = 0;
        if (queue_get_level(&terminal_rx_queue) != 0)
            status |= TERM_STATUS_RX_READY;
        if (queue_get_level(&terminal_tx_queue) < TERM_TX_DEPTH)
            status |= TERM_STATUS_TX_ROOM;
        if (__atomic_load_n(&terminal_client_connected, __ATOMIC_ACQUIRE))
            status |= TERM_STATUS_CLIENT;
        return status;
    }
}
```

```

    return 0xFF;
}

void process_virtual_io_write(uint8_t port, uint8_t value) {
    if (port >= DISK_COMMAND_STATUS_PORT && port <= DISK_DATA_PORT) {
        disk_virtual_io_write(port, value);
        return;
    }
    if (port != TERM_DATA_PORT)
        return;
    if (!queue_try_add(&terminal_tx_queue, &value))
        __atomic_fetch_add(&terminal_tx_drop_count, 1, __ATOMIC_RELAXED);
}

// Called by the WebSocket server on core 1 when browser bytes arrive.
static bool terminal_ws_receive(const uint8_t *payload, size_t length,
    void *user_data) {
    (void)user_data;
    for (size_t i = 0; i < length; ++i) {
        uint8_t value = payload[i] == '\n' ? '\r' : payload[i];
        if (!queue_try_add(&terminal_rx_queue, &value)) {
            uint8_t discard;
            queue_try_remove(&terminal_rx_queue, &discard);
            if (!queue_try_add(&terminal_rx_queue, &value))
                __atomic_fetch_add(&terminal_rx_drop_count, 1, __ATOMIC_RELAXED);
        }
    }
    return true;
}

// Called by the WebSocket server on core 1 when it can send browser data.
static size_t terminal_ws_supply(uint8_t *buffer, size_t max_length,
    void *user_data) {
    (void)user_data;
    size_t count = 0;
    while (count < max_length && queue_try_remove(&terminal_tx_queue,
        &buffer[count]))
        ++count;
    return count;
}

static void terminal_ws_connected(void *user_data) {
    (void)user_data;
    __atomic_store_n(&terminal_client_connected, 1, __ATOMIC_RELEASE);
}

static void terminal_ws_disconnected(void *user_data) {
    (void)user_data;
    __atomic_store_n(&terminal_client_connected, 0, __ATOMIC_RELEASE);
    uint8_t discard;
    while (queue_try_remove(&terminal_rx_queue, &discard)) {}
    while (queue_try_remove(&terminal_tx_queue, &discard)) {}
}

enum { WS_OUTPUT_TIMER_INTERVAL_MS = 20, WS_INPUT_TIMER_INTERVAL_MS = 10 };

static uint32_t pending_ws_output;
static uint32_t pending_ws_input;
static struct repeating_timer ws_output_timer;
static struct repeating_timer ws_input_timer;

static bool ws_output_timer_callback(struct repeating_timer *t) {
    (void)t;
    __atomic_store_n(&pending_ws_output, 1, __ATOMIC_RELEASE);
    return true;
    // Keep repeating.
}

static bool ws_input_timer_callback(struct repeating_timer *t) {
    (void)t;
    __atomic_store_n(&pending_ws_input, 1, __ATOMIC_RELEASE);
    return true;
}

bool wifi_service_poll(void);
void terminal_websocket_server_start(uint16_t port,
    bool (*receive)(const uint8_t *, size_t, void *),
    size_t (*supply)(uint8_t *, size_t, void *),
    void (*connected)(void *), void (*disconnected)(void *));
void terminal_websocket_server_poll_output(void);
void terminal_websocket_server_poll_input(void);
void supervisor_usb_poll_nonblocking(void);

```

```

// The single core 1 entry point: WebSocket terminal and the Section 6.3
// flash disk-write service share this one task, as
// `multicore_launch_core1()` only accepts one function. The boot image
// is already in SRAM by the time this runs (Section 6.3/8.10).
static void core1_main(void) {
    bool websocket_started = false;

    while (true) {
        core1_service_disk_request();

        bool network_ready = wifi_service_poll();
        if (network_ready && !websocket_started) {
            terminal_websocket_server_start(8088, terminal_ws_receive,
            terminal_ws_supply, terminal_ws_connected, terminal_ws_disconnected);
            websocket_started = true;
        }
        if (network_ready && websocket_started &&
            __atomic_exchange_n(&pending_ws_output, 0, __ATOMIC_ACQ_REL)) {
            terminal_websocket_server_poll_output();
        }
        if (network_ready && websocket_started &&
            __atomic_exchange_n(&pending_ws_input, 0, __ATOMIC_ACQ_REL)) {
            terminal_websocket_server_poll_input();
        }
        tight_loop_contents();
    }
}

static bool start_core1_services(void) {
    terminal_queues_init();           // Core 0 creates queues before launch.
    flash_service_queues_init();
    disk_service_init();
    if (!flash_safe_execute_core_init())
        return false;                // Core 0 registers as lockout victim.
    if (!add_repeating_timer_ms(-WS_OUTPUT_TIMER_INTERVAL_MS,
        ws_output_timer_callback, NULL, &ws_output_timer))
        return false;
    if (!add_repeating_timer_ms(-WS_INPUT_TIMER_INTERVAL_MS,
        ws_input_timer_callback, NULL, &ws_input_timer)) {
        cancel_repeating_timer(&ws_output_timer);
        return false;
    }
    multicore_launch_core1(core1_main);
    return true;
}

static void supervisor_fail_closed(const char *reason) {
    gpio_put(PIN_RESET_N, 0);
    isolate_buses();
    stop_z80_clock();
    printf("supervisor halted: %s\n", reason);
    while (true)
        tight_loop_contents();
}

int main(void) {
    diagnostic_safe_startup();         // First GPIO action; RESET# stays LOW.
    stdio_init_all();
    mcp_spi_init();

    if (!boot_cpm_from_flash())
        supervisor_fail_closed("boot package or journal recovery failed");
    if (!start_core1_services())
        supervisor_fail_closed("flash lockout or timer initialization failed");
    if (!set_z80_clock_hz(1000000))
        supervisor_fail_closed("invalid Z80 clock configuration");

    enable_io_trap();                 // Arm before the Z80 can issue I/O.
    gpio_put(PIN_RESET_N, 1);         // Boot image verified; begin execution.

    while (true) {
        core0_service_flash_requests();
        supervisor_usb_poll_nonblocking();
        tight_loop_contents();
    }
}

```

wifi_service_poll(), terminal_websocket_server_start(), and the two server poll functions stand for the network layer, not new Z80-facing logic. Their implementation belongs entirely to core 1 and should mirror the reference project's core1_io_mgr.c pattern. wifi_service_poll() is an idempotent, bounded lifecycle state machine: it initializes CYW43, enables station mode, and associates without sleeping; after a partial initialization failure it

cleans up and retries with an internal backoff. Core 1 calls it on every loop even after the server starts. It returns false while unavailable, re-enters association after link loss, and returns true once the station link is usable again. This guarantees `core1_service_disk_request()` runs even with Wi-Fi absent or reconnecting. Once associated, disable power-saving with `cyw43_wifi_pm(&cyw43_state, CYW43_NO_POWERSAVE_MODE)` for lower terminal latency. `supervisor_usb_poll_nonblocking()` similarly stands for an optional command parser that must return promptly so core 0 cannot starve flash ownership requests.

Required Integration Order

Maintained source: [Stage 10 main.c](#) and [Stage 10 CMakeLists.txt](#).

The command-loop application must call `diagnostic_safe_startup()` as its first GPIO action. After Phase 3 hardware is fitted, call `mcp_spi_init()` before any MCP access. Keep `enable_io_trap()` disabled during DMA and single-step operation; call `set_z80_clock_hz()` first, then enable the trap immediately before releasing RESET# for a PWM-run test. Before returning to DMA from a running CPU, call `disable_io_trap()` and either request the CPU bus while the clock still runs or assert RESET# and supply at least three further full clocks. Only after the selected ownership procedure completes and the Z80 bus is verified high-impedance may the firmware enable either transceiver. While the trap is enabled, reserve SPI0 for the handler: no other IRQ, core, or main-loop operation may access the MCP23S17. Drain `core0_service_flash_requests()` continuously from core 0's nonblocking foreground loop. For acquisition, leave trapping enabled while asserting BUSREQ# and waiting for BUSACK# LOW, then disable it. For release, enable trapping while BUSACK# is still LOW, then deassert BUSREQ# and wait for BUSACK# HIGH. Core 0 must call and check `flash_safe_execute_core_init()` before launching core 1 and must not use the multicore FIFO for anything else, because the lockout handler owns it. Journal recovery is the only core-0 flash write and occurs before core 1 launch; all runtime writes execute on core 1 while the Z80 is held in BUSACK#.

Test plan:

1. Boot with no browser connected. Verify the Z80 still runs the Phase 8 I/O tests, IN 0x01 reports no client, and terminal output does not accumulate without bound.
2. Connect a browser to `http://<pico-ip>:8088/`, or a WebSocket client to `ws://<pico-ip>:8088/`. Verify IN 0x01 sets the client-connected bit without disturbing the Z80 clock.
3. Run a Z80 program that writes a continuous alphabet pattern to OUT 0x00. Verify the browser receives the stream in order and that queue-full conditions are counted rather than blocking the trap.
4. Type from the browser and verify the Z80 receives each byte through IN 0x00 only after IN 0x01 reports data available. Test single characters, pasted bursts, delayed characters, an empty queue, and queue overflow; RX_READY must never be asserted unless an immediate data read returns a real queued byte.
5. At the CP/M prompt, run DIR, LS, switch through B-D, and repeat the Phase 9 sentinel and cross-drive checks. This proves the terminal and disk port ranges remain independently routed in the final combined firmware.
6. Disconnect and reconnect the browser while the Z80 test program runs. Verify stale input is cleared, output resumes for the new client, and no trap timeout counter increments.
7. Exhaust the default alarm pool before service startup and require the supervisor to remain fail-closed rather than launching core 1 without both WebSocket polling timers.

Pass gate: The WebSocket service remains responsive while the Z80 runs at the Phase 8 qualified 1 MHz setting; terminal status never claims data that cannot be read; interactive CP/M commands, warm boot, distinct A-D directories, and cross-drive copies still pass in the final combined firmware; no network path runs on the core that services Z80 timing; and all terminal queue overflow or client disconnect conditions are visible through counters rather than blocking the CPU trap.

8.12 Frequency Qualification

Begin only after Phase 10 passes at 1 MHz. Test 2 MHz, then increase in 500 kHz steps to 6 MHz. If and only if 6 MHz passes with margin, continue experimentally in 500 kHz steps to 8 MHz. At each step repeat SRAM readback, the one-hour memory loop, and continuous IN/OUT tests while measuring stop latency. At 1, 2, 3, 4, 5, and 6 MHz, plus every experimental step, also apply address patterns 0x0000, 0xFFFF, 0x5555, 0xAAAA, walking one, and walking zero while capturing A0, A7, A8, and A15 at the SRAM pins. Capture CLK, MREQ#, RD#/WR#, SRAM CE#/OE#/WE#, and D0-D7 as well; require valid read data before the Z80 setup deadline and every SRAM write

pulse to meet the 45 ns minimum after propagation through the GAL and AHCT244. For every I/O cycle, also require WAIT# LOW before the Z80 sampling edge, WAIT# HIGH only after DATA_ENABLE and data direction are valid, and no WAIT# reassertion until IORQ# and RD#/WR# are inactive. Use the Section 0.5 DHO814 four-channel groups and repeat the listed alternatives for each analog signal. Use the logic analyzer for the simultaneous digital capture; do not infer whole-bus ordering from four analog channels. The qualified frequency is the highest error-free step at or below 6 MHz for which the logic analyzer proves memory timing and the complete WAIT/clock-stop handshake. Report 6.5-8 MHz separately as experimental even if they pass; do not claim any rate without equivalent timing evidence and repeated cold/runtime tests.

Appendix A: Terms and Abbreviations

This glossary explains the recurring terminology used in this document. Signal names ending in # are **active LOW**: for example, RESET# is asserted (active) at 0 V and deasserted (inactive) at a valid HIGH voltage. An overbar or _N suffix in a datasheet or source file commonly means the same thing.

System and Processor Terms

Term	Meaning	Purpose in this project
SBC	Single-board computer	The complete Z80 computer built across the three breadboards.
Z80	Zilog 8-bit microprocessor family	Executes the monitor, test programs, and CP/M after the Pico loads SRAM.
CPU	Central processing unit	The Z84C0020PEC Z80 that executes program instructions.
Pico / Pico 2 W	Raspberry Pi Pico 2 W microcontroller board	Supervises startup, loads SRAM, traps I/O, provides virtual disks, and hosts the Wi-Fi terminal.
Supervisor	Controller responsible for safe startup and system management	The Pico firmware that owns reset, clock, DMA, trapped I/O, and fault handling.
Bus	Shared group of electrical signals	Carries addresses, data, or control information between the Z80, SRAM, Pico interface, and MCP23S17.
Address bus / A0–A15	Sixteen wires selecting a memory location or I/O port	Addresses 65,536 byte locations; A0 is the least-significant bit and A15 the most-significant bit.
Data bus / D0–D7	Eight bidirectional wires carrying one byte	Transfers instruction, memory, and I/O data between devices.
Control bus	Timing and ownership signals such as MREQ#, RD#, and BUSACK#	Identifies the type and phase of each Z80 transaction.
I/O	Input/output	Communication with peripherals rather than ordinary memory. This design implements most peripherals in Pico firmware.
Port	An I/O address selected by a Z80 IN or OUT instruction	Selects terminal, disk, status, or other virtual-device registers.
Trap	Firmware interception of a Z80 I/O cycle	Stops the static Z80 clock so the Pico can inspect or complete the cycle.
DMA	Direct memory access	Lets the Pico take ownership of the buses and read or write SRAM without Z80 execution.
ROMless	Having no physical Z80 ROM	The Pico writes the boot image into SRAM before releasing the Z80.
Static CPU	CPU whose clock may be stopped without losing internal state	Allows the Pico to freeze the Z80 safely during a trapped I/O cycle.
Clock / CLK	Repeating timing signal	Advances the Z80 state machine; generated by the Pico and translated by the AHCT244.
Machine cycle	Group of clock periods used for one bus operation	Defines when addresses, controls, and data must become valid.
Wait state	Extra clock period inserted into a bus cycle	Not used by this design; operation is limited to frequencies that pass measurement without one.

Memory, Storage, and Software Terms

Term	Meaning	Purpose in this project
RAM	Random-access memory	Read/write working memory; contents are normally lost when power is removed.
SRAM	Static random-access memory	The AS6C1008 provides the Z80's 64 KiB address space and needs no refresh clock.
Flash	Nonvolatile electrically erasable storage	Pico onboard flash holds firmware, the Z80 boot package, CP/M disks, and the write journal.

Term	Meaning	Purpose in this project
XIP	Execute in place	Pico address mapping that permits flash contents to be read like ordinary memory.
CP/M	Control Program for Microcomputers	The disk operating system run by the Z80.
BIOS	Basic Input/Output System	CP/M's board-specific layer for console and disk access. It is unrelated to a modern PC BIOS.
BDOS	Basic Disk Operating System	CP/M service layer implementing files, console functions, and other operating-system calls.
CCP	Console Command Processor	CP/M command interpreter that reads and runs commands.
DPB	Disk Parameter Block	CP/M table describing logical disk geometry and allocation rules.
LBA	Logical block address	Linear number identifying one 128-byte virtual-disk record.
Sector / record	Fixed-size disk transfer unit	CP/M exchanges one 128-byte record; flash updates preserve its containing 4 KiB erase sector.
Journal	Temporary recovery record written before changing disk data	Allows an interrupted flash write to recover the complete old or new block after reboot.
Manifest	Header describing an image and its integrity data	Gives boot-image length, version, and CRC values before SRAM loading.
CRC / CRC32	Cyclic redundancy check, using a 32-bit result here	Detects corruption in boot images, journal entries, and generated artifacts.
SHA-256	256-bit cryptographic hash	Records reproducible host-artifact identities; it is not calculated by the Z80.
Firmware	Software executed by a microcontroller	The staged Pico applications under <code>src/</code> .
Boot loader	Code that prepares and starts another program	Validates the flash package, copies the Z80 image to SRAM, then releases reset.
UF2	USB Flashing Format	Drag-and-drop or <code>picotool</code> firmware image format supported by Pico boot ROM.
BOOTSEL	Pico USB boot-selection mode	Allows firmware or complete flash images to be loaded while the application is not running.

Digital Electronics Terms

Term	Meaning	Purpose in this project
GPIO	General-purpose input/output	Individually programmable Pico pins used for clocks, controls, buses, and status.
SPI	Serial Peripheral Interface	Four-wire link used by the Pico to control the MCP23S17.
SCK / SCLK	SPI serial clock	Times each SPI bit sent to or received from the MCP23S17.
MOSI / SI	Master-out, slave-in	SPI data from Pico to MCP23S17.
MISO / SO	Master-in, slave-out	SPI data from MCP23S17 to Pico through the LVC244.
CS#	Chip select	Active-LOW SPI signal selecting the MCP23S17 for a transaction.
PWM	Pulse-width modulation hardware	Pico peripheral used to generate an accurate, variable-frequency Z80 clock.
SIO	Single-cycle I/O block in the RP2350	Direct Pico GPIO hardware used for static levels and manual clock stepping; not Z80 serial I/O.
IRQ	Interrupt request	Hardware event that diverts Pico execution to the I/O trap handler.
ISR / handler	Interrupt service routine	Time-critical Pico function that freezes and services a trapped Z80 cycle.
PLD	Programmable logic device	General class of programmable logic; the ATF22V10 implements arbitration and interlocks.
GAL	Generic Array Logic	Reprogrammable PLD family; shorthand used here for the ATF22V10.
JEDEC / JED file	Standard programming-data format for a PLD	File written to the ATF22V10 by a compatible programmer.

Term	Meaning	Purpose in this project
Logic equation / product term	Boolean expression implemented inside the GAL	Selects SRAM control ownership and guarantees mutually exclusive data drivers.
Consensus term	Redundant Boolean term that removes a static hazard	Prevents a brief false SRAM-control pulse while ownership inputs change.
Static hazard / glitch	Unwanted short output pulse caused by unequal logic delays	Could accidentally assert SRAM CE#, OE#, or WE#; explicitly prevented and tested.
Buffer	Logic device that strengthens, isolates, or translates a signal	AHCT244 drives 5 V clock/SPI/SRAM controls; LVC244 protects Pico inputs.
Transceiver	Bidirectional or direction-controlled bus buffer	AHCT245 and LVC245 form separate Pico-to-bus and bus-to-Pico data paths.
Tri-state / high-impedance / Hi-Z	Output state that electrically disconnects its driver	Permits several devices to share a bus as long as only the intended owner drives it.
Bus contention	Two outputs driving different levels on one wire	Can corrupt data or damage parts; GAL and firmware interlocks are designed to prevent it.
Level shifting / translation	Converting signals between voltage domains	Safely connects Pico 3.3 V logic to the 5 V Z80, SRAM, GAL, and MCP23S17.
5 V tolerant	Input may safely receive a specified 5 V-domain signal	Required for every path from installed 5 V logic into a 3.3 V-powered device.
Ioff	Partial-power-down input/output protection	Prevents LVC devices from being back-powered through their pins while their supply is off.
Pull-up / pull-down	Resistor weakly biasing a signal HIGH / LOW	Establishes a safe default when no active device drives the signal.
Decoupling / bypass capacitor	Small capacitor beside an IC supply pair	Supplies brief switching current locally and suppresses rail noise.
Bulk capacitor	Larger capacitor on a board or supply entry	Reduces slower supply dips; it does not replace local decoupling.
Power domain / rail	Group of devices sharing a supply voltage	The design has regulated 5 V logic and Pico-derived 3.3 V logic domains with common ground.
GND	Common electrical reference, nominally 0 V	Return path and voltage reference shared by all boards and instruments.
VCC / VDD	Positive IC supply voltage	Usually 5 V for Z80-side logic and 3.3 V for Pico-side LVC devices in this design.
VSYS / VBUS / 3V3	Pico system-input, USB 5 V, and regulated 3.3 V rails	Must be connected exactly as specified to avoid back-powering USB or the Pico regulator.
Schottky diode	Low-forward-voltage diode	The 1N5819 safely ORs external 5 V into Pico VSYS.
NPN transistor	Three-terminal bipolar transistor type	Q1, a 2N3904, pulls MCP RESET# LOW when address access is disabled.

Signal Names and Bus Notation

The descriptions below state the signal's purpose when it is asserted. Because most Z80 controls are active LOW, an inactive signal normally sits HIGH.

Signal or notation	Meaning	Purpose in this project
CLK	Clock	Advances the Z80. The Pico generates it and the AHCT244 translates it to a compliant 5 V clock level.
RESET#	Reset	LOW initializes the Z80 and also forces GAL arbitration to select the Pico's SRAM controls.
BUSREQ#	Bus request	Pico drives it LOW to ask the Z80 to release its address, data, and main control buses for DMA.
BUSACK#	Bus acknowledge	Z80 drives it LOW after granting BUSREQ#; tells the Pico and GAL that DMA ownership is established.
MREQ#	Memory request	Z80 drives it LOW during a memory transaction; becomes the CPU-owned source for SRAM CE#.

Signal or notation	Meaning	Purpose in this project
IORQ#	Input/output request	Z80 drives it LOW during an I/O transaction; its falling edge starts the Pico clock-stop trap.
RD#	Read	Z80 drives it LOW when the selected memory or I/O device must place data onto D0–D7.
WR#	Write	Z80 drives it LOW when the selected memory or I/O device must accept data from D0–D7.
M1#	Machine-cycle-one	Z80 drives it LOW during an opcode fetch and interrupt acknowledgement; used here as a Phase 7 probe point.
RFSH#	Dynamic-memory refresh	Z80 output for refreshing DRAM after an opcode fetch; this SRAM design does not use it.
HALT#	Halt state	Z80 drives it LOW after executing HALT; not connected to the Pico in this design.
WAIT#	Wait request	GAL pin 20 drives it LOW for an I/O cycle until DATA_ENABLE reports that the selected data path is ready; a 10 kOhm pull-up defines it if the GAL is absent.
INT#	Maskable interrupt request	An external device could pull it LOW to request a normal Z80 interrupt; unused and pulled HIGH here.
NMI#	Non-maskable interrupt request	A falling edge requests the Z80's highest-priority interrupt; unused and pulled HIGH here.
CE# / CS#	Chip enable / chip select	LOW selects an IC. CE# selects SRAM; CS# selects the MCP23S17 for SPI.
CE2	Second chip-enable input	Active-HIGH SRAM enable tied permanently to 5 V; CE# remains the switched enable.
OE#	Output enable	LOW allows a memory, buffer, or transceiver to drive its output pins.
WE#	Write enable	LOW commands SRAM to store the current data-bus value.
DIR / DATA_DIR	Direction select	Chooses which fixed-direction data transceiver may be enabled; firmware changes it only while isolated.
DATA_ENABLE	Data-interface master enable	HIGH permits the GAL to enable exactly one data path; LOW makes both paths high-impedance.
ADDR_ENABLE	Address-interface enable	HIGH releases MCP reset for address access; LOW resets it so all address pins return to inputs.
DATA_UP_OE#	Pico-to-5 V data-path output enable	GAL output that enables the AHCT245 only for Pico-to-bus transfers.
DATA_DOWN_OE#	5 V-to-Pico data-path output enable	GAL output that enables the LVC245 only for bus-to-Pico transfers.
SRAM_*_PRE#	Pre-buffer SRAM control	GAL-generated TTL-level CE#, OE#, or WE# signal before the AHCT244 raises it to 5 V CMOS levels.
PIC0_* / Z80_*	Source-name prefix	Distinguishes Pico-generated controls from Z80-generated controls in equations and net names.
A0–A15	Address bits 0 through 15	Shared 16-bit address bus; A0 is least significant and A15 most significant.
A16	SRAM address bit 16	Tied LOW so only the lower 64 KiB of the 128 KiB SRAM is used.
D0–D7	Data bits 0 through 7	Shared 8-bit bidirectional data bus; D0 is least significant.
GPA0–GPA7 / GPB0–GPB7	MCP23S17 port A / B bits	Connect directly to address buses A0–A7 / A8–A15. These are not Pico GP pins.
SCK / SI / SO	Serial clock / serial input / serial output	MCP23S17 datasheet names for SPI clock, MOSI, and MISO respectively.
IN / OUT	Z80 input / output instructions	Read a byte from, or write a byte to, an I/O port serviced by the Pico trap.
#, _N, or overbar	Active-LOW notation	Indicates that LOW means asserted. For example, PIN_RD_N in source code represents hardware RD#.
x in a truth table	Don't-care value	Either LOW or HIGH is allowed because that input cannot alter the stated result.

Component Names and Pin Labels

Name or fragment	Meaning	Purpose in this project
Z84C0020PEC	Zilog CMOS Z80, 20 MHz grade, plastic DIP commercial part	The exact CPU specified for the build; 20 MHz is a part rating, not the qualified system speed.
AS6C1008–55PCN	Alliance Memory 128 KiB x 8 SRAM, 55 ns, plastic DIP	Provides the Z80's lower 64 KiB memory space.
MCP23S17–E/SP	Microchip 16-bit SPI GPIO expander in SPDIP	Drives or samples the 16-bit address bus under Pico control.
ATF22V10B/C	Microchip 22-input/10-output reprogrammable GAL variants	Implements SRAM arbitration, data interlocking, MCP reset control, and I/O WAIT#.
SN74AHCT244N / AHCT244	8-channel non-inverting 5 V buffer with TTL inputs	Translates clock, bus request, SPI, and GAL SRAM controls to 5 V outputs.
SN74AHCT245N / AHCT245	8-bit 5 V bus transceiver with TTL inputs	Implements the fixed Pico-to-5 V data path.
SN74LVC245AN / LVC245	8-bit low-voltage bus transceiver with 5 V-tolerant inputs	Implements the fixed 5 V-to-Pico data path.
SN74LVC244AN / LVC244	8-channel low-voltage input buffer with 5 V-tolerant inputs	Buffers Z80 status/control and MCP S0 into the Pico domain.
2N3904 / Q1	General-purpose NPN transistor / schematic reference for transistor 1	Pulls MCP RESET# LOW under GAL control.
1N5819	Schottky rectifier diode	ORs external 5 V into Pico V _{SYS} without feeding the USB V _{BUS} node.
RN1, RN2, RN3	Resistor-network references	RN1/RN2 pull up address bits; RN3 pulls down Pico data GPIOs.
A1–A8 / B1–B8	The two eight-bit ports of a 245 transceiver	A faces the Pico and B faces the 5 V bus in this design.
1A1, 1Y1, etc.	Input and output labels on a 244/541 buffer bank	A is a buffer input and Y is its corresponding output; the leading digit selects a bank where present.
0E1# / 0E2#	Output-enable inputs for buffer bank 1 / 2	Tied LOW on the always-readable LVC244 and AHCT244 buffers. A 245 transceiver instead has one 0E# for all eight bits.
GP0, GP1, etc.	Pico general-purpose pin number	RP2350 GPIO index used by firmware; it is not the Pico header's physical pin number.
pin 1, notch, dot	Package orientation references	Identify how to insert a device without rotation or mirroring.
N, AN, PCN, PEC, PU, SP suffixes	Manufacturer package/grade codes	Distinguish exact orderable parts; consult that manufacturer's datasheet rather than assuming suffixes are interchangeable.

Pico Pad, Power, and MCP Register Terms

Term	Meaning	Purpose in this project
RP2350	Microcontroller chip used by Pico 2	Executes the supervisor firmware and provides GPIO, PWM, SPI, flash access, and two CPU cores.
GP0–GP29	RP2350 GPIO numbers exposed or used by Pico 2	Firmware-level pin names; map them to physical header pins with Section 1.0.
Pad	Physical input/output cell on the RP2350 die	Determines electrical properties such as 5 V tolerance and ADC capability.
FT pad	Five-volt-tolerant digital pad	GP0–GP25 pad type; tolerance still depends on RP2350 power conditions.
ADC pad	Analog-to-digital-converter-capable pad	GP26–GP29 pad type; not 5 V tolerant, so GP27/GP28 monitors require the LVC244.
ADC	Analog-to-digital converter	Converts voltage to a number; not used for Z80 monitoring despite GP26–GP29 being ADC-capable.
IOVDD	RP2350 input/output supply	Must be present at 3.3 V for the FT-pad 5.5 V absolute-maximum condition to apply.
AGND	Pico analog ground pin	Bonded to common ground because this project does not use a separate analog subsystem.

Term	Meaning	Purpose in this project
ADC_VREF	Pico ADC reference-voltage pin	Left open because the project does not use Pico ADC measurements.
3V3 / 3V3(OUT)	Pico regulated 3.3 V output	Powers LVC244/LVC245 and 3.3 V pull-ups.
3V3_EN	Pico 3.3 V regulator enable	Left open; pulling it LOW would disable the Pico's 3.3 V rail.
RUN	Pico reset input	Left open; pulling it LOW resets the RP2350.
VBUS	USB connector's nominal 5 V node	Must not be tied to the external 5 V rail.
VSYS	Pico system power input	Receives external 5 V only through the 1N5819 and USB power through Pico's onboard diode.
IODIRA / IODIRB	MCP port A / B direction registers	Bit value 1 selects input; 0 selects output. Reset makes both registers 0xFF.
OLATA / OLATB	MCP port A / B output-latch registers	Preload the next address value before changing port direction to outputs.
GPIOA / GPIOB	MCP port A / B pin-value registers	Read the actual logic levels present on the address-bus pins.
IOCON	MCP I/O configuration register	Holds global expander options, including hardware-address enable.
HAEN	MCP hardware-address-enable bit	Enables SPI opcode comparison with A0/A1/A2 pins; those pins are tied LOW here regardless.
SPI0	RP2350 SPI controller instance 0	Dedicated to MCP23S17 access while the I/O trap is armed.
Core 0 / Core 1	The two RP2350 processor cores	Core 0 owns timing and buses; core 1 owns networking and runtime flash writes.
FIFO	First-in, first-out queue	Ordered communication buffer; the multicore FIFO is reserved by Pico flash lockout support.

Software, Network, File, and Test Abbreviations

Term	Meaning	Purpose in this project
SDK	Software development kit	Pico headers, libraries, tools, and build support used by the firmware.
API	Application programming interface	Defined functions through which code or tools access a subsystem.
CLI	Command-line interface	Text-based invocation of tools such as CMake, picotool, and test scripts.
CI	Continuous integration	Automated build/test environment that can reject boundary or artifact failures.
USB	Universal Serial Bus	Programs the Pico, carries diagnostics, and connects test instruments to the Mac.
UART	Universal asynchronous receiver/transmitter	Common serial protocol supported by decoders; the Z80 terminal itself is virtual, not a UART chip.
I2C / I ² C	Inter-Integrated Circuit	Two-wire serial protocol supported by tools but not used for the MCP23S17 in this design.
TCP/IP	Transmission Control Protocol / Internet Protocol	Networking foundation used by HTTP and WebSocket traffic.
HTTP	Hypertext Transfer Protocol	Serves the browser terminal page from Pico port 8088.
HTML	Hypertext Markup Language	Source format of the embedded browser terminal interface.
JSON	JavaScript Object Notation	Structured text format used for generated manifests.
CSV	Comma-separated values	Plain-text tabular export format for captures, measurements, and generated data.
SVG	Scalable Vector Graphics	Vector image format used for diagrams and schematic exports.
PDF	Portable Document Format	Generated printable edition of this specification and schematic.
FFT	Fast Fourier transform	Converts sampled time-domain data into a frequency spectrum.
DVM	Digital voltmeter	Instrument function that reports numeric voltage measurements.
RLE	Run-length encoding	Compression that stores repeated identical digital samples efficiently.
IEEE CRC32	Standard 32-bit CRC convention	Exact CRC algorithm used by boot and journal formats.
XIP_BASE	Base address of Pico's memory-mapped external flash	Added to a flash offset to obtain a readable pointer.

Term	Meaning	Purpose in this project
PICO_OK	Pico SDK success result	Required return from <code>flash_safe_execute()</code> before a flash operation is accepted.
watchdog	Hardware recovery timer	Reboots into journal recovery if runtime flash coordination cannot continue safely.
atomic	Operation observed as indivisible between cores	Protects shared status words without a lock needed by the trap.
nonblocking	Returns immediately rather than waiting	Required for trap hooks and foreground services that must not stall another critical task.
CRC, hash, verify	Different integrity checks	CRC detects transfer/storage corruption, SHA-256 identifies host artifacts, and verify compares programmed/read-back contents.

Units, Numeric Notation, and Data Sizes

Notation	Meaning	How to read it here
0x prefix	Hexadecimal (base-16) number	0xFF is decimal 255; convenient for bytes, addresses, masks, and register values.
bit	One binary digit	Has value 0 or 1. Eight bits form one byte.
byte	Eight bits	One Z80 data-bus transfer and one addressable SRAM location.
K, M, G in rates	Decimal kilo, mega, giga	Multipliers of 1,000; for example, 1 MHz is 1,000,000 cycles per second.
KiB / MiB	Kibibyte / mebibyte	Binary sizes of 1,024 bytes / 1,048,576 bytes.
KB / MB	Kilobyte / megabyte	Sometimes used informally in prose or vendor material; exact project image sizes are stated in bytes or KiB.
Hz, kHz, MHz	Hertz, kilohertz, megahertz	Cycles per second; used for clock, signal, and bandwidth values.
S/s, MS/s, GSa/s	Samples per second, mega-samples/s, giga-samples/s	Logic-analyzer or oscilloscope sampling rate, not signal frequency.
s, ms, us, ns	Second, millisecond, microsecond, nanosecond	Time units: each step is one thousand times smaller than the previous. Source identifiers use <code>us</code> for microseconds.
V, mV	Volt, millivolt	Electrical potential; 1 V equals 1,000 mV.
A, mA	Ampere, milliampere	Electrical current; 1 A equals 1,000 mA.
Ohm, kOhm	Ohm, kilo-ohm	Resistance; 10 kOhm means 10,000 ohms.
pF, nF, uF	Picofarad, nanofarad, microfarad	Capacitance units; adjacent steps differ by 1,000. <code>uF</code> is ASCII notation for μ F.
Mpts, samples	Million points / recorded measurements	Capture-memory depth; available duration depends on sample rate and active channels.
x or ×	Multiplication or count	128K × 8 means 128K locations each eight bits wide; 8×10 kOhm means eight resistors.
1x / 10x	Probe attenuation	Probe delivers full voltage / one tenth voltage to the scope; software must use the matching setting.

Logic Families, Levels, and Timing

Term	Meaning	Purpose in this project
TTL	Transistor-transistor logic levels	A TTL-compatible input accepts a lower HIGH voltage than a 5 V CMOS input.
CMOS	Complementary metal-oxide-semiconductor logic	Technology used by the static Z80, SRAM, Pico, and most interface devices.
HCT	High-speed CMOS with TTL-compatible inputs	Slower family considered for U4; the final design uses AHCT instead.
AHCT	Faster HCT-family logic	AHCT244 buffers clock/control/SPI and AHCT245 drives the 5 V data bus from Pico 3.3 V outputs.
LVC	Low-voltage CMOS logic family	LVC244/LVC245 run at 3.3 V while accepting protected 5 V inputs.
HIGH / LOW	The two digital logic states	Voltage ranges, not exact voltages; valid limits are defined by each datasheet.

Term	Meaning	Purpose in this project
Assert / deassert	Make a control active / inactive	Independent of voltage wording; asserting RESET# means driving it LOW.
V_{IH} / V_{IL}	Minimum input-HIGH / maximum input-LOW voltage	Determines whether a receiving input is guaranteed to recognize a signal.
V_{OH} / V_{OL}	Guaranteed output-HIGH / output-LOW voltage	Determines whether a driver meets the receiving device's input thresholds under load.
Propagation delay	Time from an input transition to its output response	Delays through GAL, AHCT244, and SRAM constrain the maximum clock rate.
Setup time	Time data must be stable before a sampling clock edge	SRAM read data must arrive early enough for the Z80 to sample it correctly.
Hold time	Time data must remain stable after a sampling edge	Drivers must not release or change the bus too soon.
Rise / fall time	Time for a signal edge to change between LOW and HIGH	Fast edges, rather than only clock frequency, cause breadboard ringing and crosstalk.
Duty cycle	Percentage of a period that a signal is HIGH	The Z80 clock is targeted at 50%, with a stated acceptance range.
Ringing / overshoot	Oscillation or voltage excursion following an edge	Can cross logic thresholds or exceed ratings; checked with short-ground scope probing.
Crosstalk	Unwanted coupling from one signal into another	Reduced by short grouped wiring and nearby ground returns.
Timing margin	Remaining safe time or voltage beyond a required limit	Measured during qualification rather than inferred from part speed ratings alone.

Package, Construction, and Schematic Terms

Term	Meaning	Purpose in this project
IC	Integrated circuit	A packaged electronic device such as the Z80, SRAM, or logic buffer.
DIP / PDIP	Dual in-line package / plastic dual in-line package	Through-hole IC package with two parallel pin rows, suitable for sockets and breadboards.
SPDIP	Shrink plastic dual in-line package	Narrower-pin-pitch DIP variant used by the MCP23S17.
SIP	Single in-line package	Resistor network with one common pin and eight resistor pins.
TO-92	Small three-lead transistor package	Package used by Q1; its lead order must be checked against the purchased datasheet.
Pin 1 / notch / dot	Package orientation marks	Prevent an IC from being inserted rotated or mirrored.
Breadboard ravine	Insulating center gap of a solderless breadboard	Separates the two IC pin rows so opposite pins are not shorted together.
Net	All electrically connected points sharing one signal name	Fundamental connection represented and checked in the KiCad schematic.
Junction	Explicit connection where wires meet	Distinguishes an intended shared node from crossing but unconnected wires.
ERC	Electrical rules check	KiCad check for incompatible, missing, or improperly driven schematic connections.
Netlist	Machine-readable list of components, pins, and nets	Independently compared with the generated manifest to validate schematic connectivity.
BOM	Bill of materials	List and quantity of components needed to build the system.

Measurement and Build Terms

Term	Meaning	Purpose in this project
Oscilloscope / scope	Instrument displaying voltage versus time	Measures analog levels, edge shape, pulse width, ringing, and timing between two signals.

Term	Meaning	Purpose in this project
Logic analyzer	Instrument recording many digital HIGH/LOW signals	Correlates bus values and control ordering across many channels simultaneously.
MSO	Mixed-signal oscilloscope	Instrument combining analog scope channels with synchronized digital inputs; the DHO814 has four analog channels but no integrated digital bus inputs.
Probe	Instrument lead designed to minimize loading	A compensated 10x scope probe is used for analog measurements; logic probes capture digital states.
1x / 10x probe	Probe attenuation ratio	10x mode usually loads fast logic less and provides better bandwidth than 1x mode.
Ground spring	Very short probe-ground attachment	Prevents the probe's own ground-loop inductance from creating apparent ringing.
Bandwidth	Highest sine-wave frequency passed within a stated attenuation	The DHO814's 100 MHz front end has an approximate 3.5 ns calculated rise-time limit; an edge displayed near that limit may be scope-limited.
Sample rate	Number of measurements recorded per second	Determines time resolution; it is distinct from analog bandwidth.
Real-time sampling	Samples used for one displayed acquisition are collected from the same trigger event	The DHO814 uses real-time sampling, so Single mode can preserve a one-off transition for analysis.
Memory depth	Number of samples retained in one acquisition	The DHO814 provides up to 25 Mpts, allowing substantial pre-trigger and post-trigger context while retaining a useful sample rate.
AWG	Arbitrary waveform generator	Produces programmable test waveforms; available in many USB scopes but not required for Z80 operation.
SCPI	Standard Commands for Programmable Instruments	Text-command interface for automating compatible bench instruments.
KiCad	Electronic schematic and PCB design application	Holds the native electrical schematic and runs ERC.
CMake / Ninja	Build-system generator / build executor	Configure and compile the Pico firmware and generated CP/M artifacts.
Toolchain	Compiler, assembler, linker, libraries, and utilities	Converts source code into executable Pico and Z80 images.
Bare metal	Software running without a hosted desktop operating system	The Pico firmware links its own embedded runtime rather than macOS libraries.
lwIP	Lightweight IP networking stack	Supplies TCP/IP networking used by the Pico WebSocket terminal.
WebSocket	Persistent bidirectional browser/server connection	Carries terminal bytes between the browser and Pico with low overhead.

Appendix B: Datasheet References

- [Z84C00 CPU datasheet](#)
- [AS6C1008 SRAM datasheet](#)
- [MCP23017/MCP23S17 datasheet](#)
- [ATF22V10C datasheet](#)
- [SN74AHCT244 datasheet](#)
- [SN74AHCT245 datasheet](#)
- [SN74LVC245A datasheet](#)
- [SN74LVC244A datasheet](#)
- [1N5817/1N5818/1N5819 Schottky diode datasheet](#)
- [RP2350 datasheet](#)
- [Raspberry Pi Pico 2 board datasheet](#)
- [BusBoard BB830 breadboard datasheet](#)
- [RIGOL DH0800 Series user guide](#)

Appendix C: Source Code Index

Canonical repository: github.com/gloveboxes/Z80ROMlessSBC. All phase applications are cumulative: each stage links the shared modules proven by earlier stages, while remaining independently buildable for hardware bring-up.

Project Build Files

File	Purpose
CMakeLists.txt	Pico SDK import, Pico 2 W target, project languages, and protected firmware flash linker boundary
src/CMakeLists.txt	Shared firmware libraries, stage registration, and the z80_cpm_images artifact target
.gitignore	Generated build, Python cache, and assembler-output exclusions
package.json, package-lock.json	Locked Mermaid and browser-automation dependencies plus the npm run pdf command
scripts/build-readme-pdf.mjs	Pandoc, Mermaid, and Chromium PDF renderer with full-page diagram layout

Cumulative Stage Applications

Phase	Application	Target definition	Responsibility
0	Power checklist	Hardware-only	Empty-socket wiring, rail, resistance, and startup-state checks
1	main.c	CMakeLists.txt	Safe supervisor startup and GPIO walk
2	main.c	CMakeLists.txt	Buffered controls and variable-frequency clock
3	main.c	CMakeLists.txt	MCP23S17 register and port diagnostics
4	main.c	CMakeLists.txt	16-bit address-bus drive, sample, and isolation tests
5	main.c	CMakeLists.txt	8-bit data-bus drive, sample, and isolation tests
6	main.c	CMakeLists.txt	SRAM DMA and full-memory validation
7	main.c	CMakeLists.txt	Z80 reset, execution, clock, and BUSREQ/BUSACK tests
8	main.c	CMakeLists.txt	Virtual-ROM preload and synchronous I/O trapping
9	main.c	CMakeLists.txt	Manifest boot, journal recovery, and persistent flash disks
10	main.c	CMakeLists.txt	Final CP/M, flash-disk, Wi-Fi, and WebSocket integration

Shared Firmware Modules

Module	Public interface	Implementation	Responsibility
Pin map	pins.h	Header-only	Authoritative Pico GPIO assignments
Supervisor	supervisor.h	supervisor.c	Safe GPIO startup and fail-safe isolation defaults
Clock	clock.h	clock.c	PWM frequency selection, stop/resume, and single-cycle clocking
MCP23S17	mcp23s17.h	mcp23s17.c	SPI register access and 16-bit address expansion
Buses	bus.h	bus.c	Contention-safe address/data direction, isolation, drive, and sample operations
SRAM	sram.h	sram.c	DMA byte access, image load/verify, and RAM diagnostics
CPU	cpu.h	cpu.c	Reset-held DMA, run control, bus ownership, and fail-closed state
I/O trap	io_trap.h	io_trap.c	Synchronous Z80 IN/OUT interception and fault counters
Flash disk	flash_disk.h	disk_device.c	Z80-facing command/status, drive, LBA, and 128-byte data ports
Flash layout	flash_layout.h	Header-only	Firmware, journal, boot-package, and disk-slot offsets

Module	Public interface	Implementation	Responsibility
Flash backend	flash_backend.h	flash_backend.c	Manifest validation, SRAM boot load, journal recovery, and core-1 commits
Terminal	terminal.h	terminal_bridge.c, terminal_network.cpp	Queue-backed terminal ports, Wi-Fi lifecycle, HTTP, and WebSocket service

CP/M, Disk, and Web Tooling

Area	Files	Purpose
Native CP/M	README, z80_bios.asm, build_images.py, test_build_images.py	64K CCP/BDOS image, native BIOS, boot package, full-flash composition, and host regression tests
Dynamic debug I/O	adapter README, adapter entry point, port dispatcher	Project-hosted SBC disk adapter, complete Altair host port drivers, interrupt service, and ANSI terminal-input pipeline
Disk geometry and conversion	README, geometry.py, convert_altair_disks.py	Shared CP/M geometry and deterministic Altair-media conversion
Generated disk artifacts	generated directory, manifest.json	Four exact 320 KiB intermediate disk slots and source/output hashes
Preserved source media	source-altair directory, altair_88dskrom.h, altair_disk_loader.h	Original framed disks, Altair loader references, and upstream license
Browser terminal	terminal.html, embed_html.py	Embedded Stage 10 terminal client and build-time HTML conversion
Network configuration	lwipopts.h, wifi_config.h.in	lwIP settings and build-time Wi-Fi credential template

Appendix D: CP/M BIOS and dcc Compatibility

This system runs CP/M 2.2 with a custom BIOS designed specifically for this board. The image builder packages the 64K CP/M CCP and BDOS with the native BIOS assembled from [src/cpm/z80_bios.asm](#). The BIOS is the layer that translates standard CP/M console and disk operations into this board's Pico-serviced virtual I/O ports. Applications do not need to know that the terminal is a WebSocket or that disk records reside in Pico flash.

[dcc](#) targets CP/M 2.2 on the Z80 and emits ordinary .COM programs linked with its Z80 runtime. That execution model is compatible with this system: the dcc runtime enters CP/M through the standard page-zero vectors, CP/M BDOS performs console and file-system policy, and the custom BIOS performs the final hardware-dependent transfer.

D.1 CP/M Image Provenance and Machine Boundary

The resident CCP and BDOS come from the Burcon CP/M 2.2 distribution for the MITS Altair 88-DCDD controller. Its BIOS reservation is 0x0700 bytes, so a 64K generation places the CCP at 0xE300, BDOS at 0xEB00, and BIOS at 0xF900. The checked-in `src/cpm/cpm64_system.bin` is the 5,632-byte CCP/BDOS RAM region captured after the matching Burcon MOVCPM 64 and SYSGEN tools produced a system that cold-booted successfully in the Altair emulator. Its SHA-256 fingerprint is enforced by the image builder.

CP/M generators are machine-specific. To reproduce this artifact, use the [Burcon CP/M 2.2 utilities](#): the archive's MOVCPM.COM and SYSGEN.COM, following its [Using MOVCPM guide](#). The matching generator reports SAVE 38 after MOVCPM 64. Preserve the Burcon 0x0700 BIOS reservation and verify the relocated entry vectors before accepting any regenerated CCP/BDOS image.

The fingerprinted binary has also been recursively disassembled as Intel 8080 source in `src/cpm/cpm64_i8080.asm`; its control-flow, opcode-compatibility, and optimization findings are recorded in `src/cpm/cpm64_i8080_audit.md`. The boot-proven binary remains immutable. The image builder assembles the separate `src/cpm/cpm64_z80.asm` port, whose unoptimized generation mode reproduces the reference byte-for-byte. The active port uses relative branches, ten guarded DJNZ sites, LDIR, native SBC HL,DE subtraction, SUB 2 fusion, a branch inversion, branch threading, tail transfers, and register-use rewrites. It reclaims 263 bytes while preserving the fixed section sizes and relocates them upward by 0x0400. Recursive discovery includes the BDOS cleanup block reached through a synthetic stack return. Host tests independently verify relocation and native instruction encodings, while `src/cpm/test_z80_debug_host.py` boots the image and completes DIR, transient loading, and a BIOS warm reload through a model of the SBC terminal and disk ports.

Only the CCP/BDOS and their memory ABI are reused. The Altair disk BIOS and cold loader use ports 0x08-0x0A and are reference material, not executable code for this board. At image-build time they are replaced by the SBC-native BIOS assembled from `src/cpm/z80_bios.asm`, which uses the Pico-serviced terminal and flash-disk ports described below. It intentionally targets the Z80 rather than the 8080-compatible subset, using relative branches, DJNZ, BIT, rotate/carry status tests, carry-to-mask SBC A,A, INIR/OTIR block I/O, and EXX alternate-register banking in `read_record/write_record` (safe because this system never enables interrupts, so the shadow register set is otherwise unused). Build it with `z80asm`.

D.2 CP/M Memory Map and Entry Points

The reset-ready SRAM image uses the following upper-memory layout:

Region	Address range	Role
Page zero	0x0000-0x00FF	CP/M vectors, default FCBs, DMA buffer, and command tail
Transient Program Area (TPA)	0x0100-0xE6FF	.COM program, dcc runtime, static data, heap, and stack
CCP	0xE700-0xEEFF	Console Command Processor, reloaded after a warm boot
BDOS	0xEF00-0xFCFF	CP/M console, file, and disk service layer; entry at 0xEF06
BIOS	0xFD00-0xFFC5	710-byte Z80-optimized boot, console, disk, and translation routines

The TPA contains 0xE600 bytes, or 58,880 bytes, before the CCP boundary. A dcc program and every runtime block selected for it must fit in this space together with its stack and heap. The dcc runtime reads the installed JP 0xEF06 vector at 0x0005 to locate BDOS; application memory must still remain below the CCP boundary at 0xE700. Required CCP/BDOS workspace, the BIOS directory buffer and allocation vectors, disk state, and internal stacks remain inside their relocated resident regions.

The BIOS cold boot installs JP instructions at 0x0000 and 0x0005 for warm boot and BDOS respectively. A dcc program normally starts at 0x0100 and returns to CP/M through the warm-boot vector. Because a transient program may overwrite the resident CCP while using the TPA, the custom warm boot reloads the 44 CCP/BDOS system records before returning to the command prompt.

D.3 Custom BIOS Responsibilities

The BIOS supplies the standard CP/M 2.2 jump table expected by BDOS and by dcc's optional direct-BIOS functions. Its principal mappings are:

CP/M BIOS operation	Board implementation
BOOT / WB00T	Initialize page zero or reload CCP/BDOS from Drive A
CONST	Read terminal status port 0x01; return 0xFF when bit 0 reports input ready
CONIN	Wait for receive-ready, then read terminal data port 0x00
CONOUT	Wait for transmit-room bit 1, then write the character to port 0x00
READER	Return CP/M text EOF (0x1A); no reader device is implemented
LIST / PUNCH	Discard output; printer and punch devices are intentionally not implemented
LISTST	Return not ready because no list device exists
SELDISK	Validate drives A-D and return the corresponding disk parameter header
SETTRK / SETSEC / SETDMA	Record the logical CP/M transfer location and SRAM DMA address
READ / WRITE	Convert track and sector to a linear 128-byte LBA, transfer through ports 0x10-0x14, and preserve CP/M write-type semantics for flash caching
SECTRAN	Return the sector unchanged because native disk images use linear sector order

CP/M BDOS remains responsible for filenames, FCBs, directory searches, allocation, sequential/random record selection, and text-file conventions. The BIOS sees only drive selection and 128-byte logical records. Consequently, dcc file APIs do not require a board-specific runtime backend: their BDOS file calls eventually reach the custom READ and WRITE entries.

Console ports occupy the aligned 0x00 group and disk ports the aligned 0x10 group. Moving them would not reduce Pico trap work because the same 8-bit port decoder handles every I/O cycle, so these established direct-I/O addresses remain stable.

D.4 dcc Console and File-I/O Paths

The normal dcc runtime paths are compatible without recompiling the runtime for these port numbers:

dcc application operation	Runtime and operating-system path
printf, puts, putchar, and stdout/stderr writes	dcc console buffering -> BDOS functions 9 and 2 -> BIOS CONOUT -> terminal ports
getchar, scanf, and stdin reads	BDOS function 1 -> BIOS CONST/CONIN -> terminal ports
kbhit and getch	BDOS direct-console function 6 -> BIOS console entries -> terminal ports
fopen, fread, fwrite, fclose, and related file calls	CP/M FCB operations in BDOS -> BIOS READ/WRITE -> flash-disk ports
Program exit or abort	Page-zero warm boot -> BIOS WB00T -> system reload and CCP prompt

dcc's console runtime performs CP/M-oriented character handling above the BIOS. In particular, normal text input treats Ctrl-Z as end-of-file, converts an entered carriage return to C \n, and supplies the accompanying line-feed echo. Console output converts C \n to CR/LF. The BIOS deliberately transports bytes and reports readiness; it does not duplicate those runtime policies.

Buffered dcc output may not reach the browser immediately. A newline, an explicit fflush, a full buffer, an input operation, or program termination flushes it. Separately, the Pico terminal transmit queue is bounded. With no WebSocket client, limited output can queue for later delivery, but an output-heavy program eventually waits in BIOS CONOUT when status bit 1 reports no room. This wait occurs in the Z80 program, not in the Pico's I/O trap, and a connected client allows transmission to resume.

D.5 Direct CP/M, BIOS, and Port Access

dcc also exposes non-C89 target extensions for software that intentionally bypasses part of the normal runtime:

API	Compatibility on this system
bdos() / bdoshl()	Compatible for implemented CP/M 2.2 BDOS functions through CALL 0x0005
bios() / bioshl() / biosreg()	Compatible with the standard 17-entry BIOS jump table installed at 0xFD00
inp(port)	Executes an 8-bit Z80 IN; suitable for reading the virtual terminal or disk ports
outp(port, value)	Executes an 8-bit Z80 OUT; suitable for writing the virtual terminal or disk ports

Only the low eight bits of a dcc inp or outp port argument are significant, which matches the supervisor's eight-bit I/O decode. Direct terminal code must poll receive-ready before reading port 0x00 and transmit-room before writing it. Direct disk code must follow the complete command/status and exact 128-byte-transfer protocol in Section 6.3. Using BDOS is preferred for ordinary console and file access because it preserves CP/M buffering, FCB, error, and warm-boot behavior.

A normal dcc .COM file linked with the dcc runtime is not a standalone bare-metal binary even if it uses inp and outp: startup, memory discovery, file services, and exit still depend on CP/M page zero, BDOS, and BIOS. A truly bare-metal dcc program would need a different startup/runtime arrangement and is outside this compatibility claim.

D.6 Compatibility Boundaries

- Programs must fit within the 0x0100-0xE6FF TPA after runtime, globals, heap, and stack requirements are included.
- The installed operating system is CP/M 2.2. Optional CP/M 3 or emulator-only BDOS extensions are not supplied by this BIOS/BDOS image. In particular, software should treat dcc's BDOS-105-backed clock functions as unavailable unless separately demonstrated on this target.
- CP/M filenames and disks retain CP/M 2.2 semantics: 8.3 names, FCB-based access, 128-byte logical records, and Ctrl-Z text-file conventions where the dcc API documents them.
- The terminal data port returns 0x00 if read while empty. Correct software must test status bit 0 first; BDOS and the custom BIOS already do so.
- Status bit 7 reports WebSocket client connection state to direct-port software, but CP/M and dcc do not require it for their standard console ABI.
- dcc's own emulator and physical-Z80 test history establishes general CP/M runtime portability, but it does not by itself qualify this board's Pico trap, custom BIOS, flash backend, or electrical timing.

D.7 Validation Status and Required Qualification

The host regression suite assembles the custom BIOS and verifies its placement, the CCP/BDOS fingerprint, page-zero reset target, boot-package headers and CRCs, native disk construction, CP/M geometry, and complete 4 MiB flash layout. These are structural and reproducibility checks; they do not execute dcc programs through this BIOS or measure a physical I/O cycle.

Separately, the source Burcon system has been cold-booted in the Altair host emulator as 64K CP/M, reached the A> prompt, and completed DIR. That test validates the relocated CCP/BDOS artifact and Burcon memory ABI, but it executes the Altair BIOS rather than this board's custom BIOS. It therefore does not validate the Pico I/O trap, virtual terminal and disk ports, flash persistence, or Z80 bus timing.

The design remains unvalidated in hardware as stated in the Overview. Before claiming dcc compatibility on the completed machine, perform the Phase 8-10 tests and additionally run representative dcc .COM programs that cover:

1. printf/puts, literal \$, CR/LF conversion, buffering, and fflush.
2. getchar, scanf, kbhit, getch, Ctrl-Z, and browser reconnect behavior.
3. File create, close, reopen, sequential read/write, random access, rename, delete, directory updates, disk-full handling, and warm boot.
4. Transfers on drives A-D, including repeated writes followed by power-cycle recovery and byte-for-byte host comparison.
5. A dcc program near the TPA limit, confirming that stack/heap growth does not cross 0xE700 and that exit reliably reloads the CCP.

Passing those application tests, the existing all-LBA and fault-injection disk plan, and the logic-analyzer requirements is the point at which console and file-I/O compatibility may be described as proven on this hardware.

D.8 End-to-End Z80 Optimization, Image Build, and Flash Test

The active bootable system is Z80-optimized throughout its resident software: the CCP and BDOS are emitted as cpm64_z80.asm, and the board BIOS is implemented by z80_bios.asm. The immutable cpm64_system.bin is retained as the fingerprinted Burcon reference and as the input to the translation audit; it is not copied directly into the deployed image. The optimized CCP/BDOS port uses Z80 relative branches, guarded DJNZ, block moves, native arithmetic, tail transfers, and safe register-use rewrites. The BIOS uses the Z80 instruction set for its jump table, cold/warm boot paths, console polling, and disk transfers, including INIR/OTIR, SBC, relative branches, and EXX. The resulting image keeps the CP/M ABI and fixed resident boundaries while leaving the transient program area at 0x0100-0xE6FF.

Build the complete CP/M artifact set from the repository root:

```
python3 src/cpm/build_images.py --output-dir build/cpm
```

The builder assembles the optimized CCP/BDOS and BIOS, constructs the reset-ready 64 KiB z80boot.img, wraps it as z80boot.pkg with a manifest and CRC, and replaces only Drive A's reserved system tracks in drive_a_cpm63k-z80.img. Drives B-D remain the converted native 320 KiB images. With a Stage 10 firmware binary, the same builder can compose z80romless-flash.bin, a complete 4 MiB image:

```
python3 src/cpm/build_images.py --output-dir build/cpm \
  --firmware build/src/stage10_websocket_terminal/z80_stage10_websocket_terminal.bin
```

The BIOS and flash backend share one deliberately narrow contract. CP/M passes drive, track, sector, DMA address, and write type to the BIOS; the BIOS converts the request to one 128-byte linear-record transfer through ports 0x10-0x14. The Pico maps those records into the four exact 320 KiB flash slots at 0x2C0000, 0x310000, 0x360000, and 0x3B0000. Normal writes can coalesce in a 4 KiB cache, while directory and warm-boot writes are journaled; the write type is preserved so the Pico can choose the safe persistence path. Cold boot validates the package and journal, copies the image into SRAM while RESET# is held, verifies it, and installs the page-zero warm-boot and BDOS vectors. BIOS warm boot then reloads the 44 resident CP/M records from native Drive A. This avoids a second storage bus while keeping Z80 disk semantics and Pico flash erase/program operations separate.

Run the reproducibility and structure tests with:

```
python3 -m unittest discover -s src/cpm -p 'test_*.py' -v
```

These tests assemble both optimized Z80 components and verify the CP/M 2.2 version byte, resident placement, BIOS jump-table ABI, boot-package header and CRCs, exact 64 KiB and 320 KiB geometries, Drive A system-track replacement, and every flash-region boundary. The end-to-end DCC debug-host test additionally boots the optimized image, observes the CP/M banner and A> prompt, runs DIR, executes LS.COM to its natural exit, confirms BIOS warm reload, selects another drive, loads a multi-extent transient, and copies files across drives. After building the project-owned z80-debug-host, run the end-to-end test with:

```
python3 src/cpm/test_z80_debug_host.py \
--host build/cpm/z80_debug_host/z80-debug-host
```

This is host/emulator evidence, not a claim of completed electrical qualification. Physical proof still requires the Phase 8-10 bus, flash fault-injection, power-cycle recovery, and logic-analyzer tests described above.

D.9 Interactive End-to-End CP/M User Test

The repository includes an interactive launcher for manually exercising the same optimized CP/M image, board BIOS, virtual-disk protocol, and native disk images used by the automated DCC test.

Prerequisites are CMake, a C/C++ compiler, Python 3, and z80asm. In VS Code:

- 1. Open **Run and Debug**.
- 2. Select **Interactive CP/M emulator**.
- 3. Press **Start**.
- 4. Wait for the banner and A> prompt in the **Interactive CP/M emulator** terminal.
- 5. Type CP/M commands normally. Press **Ctrl+]** to detach and stop the session.

The pre-launch task builds the DCC-derived z80-debug-host, the complete project-owned adapter in src/cpm/dcc_io_adapter, and the CP/M artifacts in build/cpm. The **Interactive CP/M emulator** debug configuration always passes that built adapter to z80-debug-host; no optional selection or external Altair checkout is required. The launcher extracts the 64 KiB SRAM payload from z80boot.pkg, starts it through the board BIOS, and attaches Drives A-D from the generated 320 KiB images. A successful start displays:

```
64K CP/M 2.2 - Burcon Z80 Edition

A>
```

The following manual sequence exercises progressively more of the deployed software path:

```
A>DIR
A>LS
A>C:
C>ATTNC11
C>A:PIP D:ATTNC11.COM=C:ATTNC11.COM
C>A:PIP D:ATTN.WTS=C:ATTN.WTS
C>D:
D>DIR
```

- DIR proves CCP command handling, BDOS directory access, BIOS record reads, the 0x10-0x14 disk-port protocol, and Drive A filesystem interpretation.
- LS loads and executes a transient program, exits through page zero, and exercises the BIOS warm-boot reload before returning to A>.
- C: and ATTNC11 exercise drive selection and multi-extent program loading.
- The PIP commands exercise cross-drive reads, writes, directory updates, and persistence in the emulator's generated Drive D image.

The dynamic adapter preserves the SBC ports 0x10-0x15 and also mirrors all active ports in the ESP32 Altair port_drivers/io_ports.c dispatcher:

Ports	Host-emulated function
24-31, 37-39, 41-44	Time and timer requests
24-30	Time and timer input

Ports	Host-emulated function
45, 48, 49, 70	Utility strings and host information
46-47	OpenWeatherMap field request and status
52	DCC-host interrupt timer
60-61	Host file transfer
71-72	Text-file-backed environment variables
120-124	OpenAI-compatible chat request and response
200	Shared response-buffer input

The host adapter intentionally does not add the `altair_local` Sense HAT extensions because those ports are absent from the firmware source-of-truth dispatcher. Weather and chat use libcurl when available and explicitly report an unavailable/error state when network support or configuration is absent. No API keys are stored in the repository.

Terminal input uses the same policy as `altair_local`: Up/Down/Right/Left map to Ctrl-E/Ctrl-X/Ctrl-D/Ctrl-S, Insert/Delete map to Ctrl-O/Ctrl-G, Page Up/Page Down map to Ctrl-R/Ctrl-V, and Backspace or Delete maps to Ctrl-H. A standalone Escape is emitted after a 30 ms grace period, allowing ANSI cursor sequences to be recognized without losing Escape as a CP/M command key.

This is a realistic end-to-end **software-path** test: it executes the optimized CCP/BDOS and `z80_bios.asm` with the SRAM payload extracted from the packaged boot artifact, while the BIOS accesses the same native disk-image format that is provisioned into Pico flash. Disk writes are made to the generated files under `build/cpm`; rerunning the preparation task regenerates those artifacts, so copy out any test result that must be retained.

The launcher intentionally substitutes host files for the physical Pico flash backend. It therefore does not validate Pico package/CRC handling, the physical flash cache and journal implementation, power-loss recovery, the Pico-to-SRAM DMA transfer, WAIT#/bus timing, voltage translation, or the assembled board. Those remain Phase 8-10 hardware and firmware qualification items.

Generate the PDF

The PDF renderer requires Node.js, Pandoc, and Microsoft Edge, Google Chrome, or Chromium. On macOS, install the command-line prerequisites with:

```
brew install node pandoc
```

From the repository root, install the locked dependencies and generate README.pdf:

```
npm ci  
npm run pdf
```

The renderer preserves tables, code highlighting, local images, links, and MathML. Mermaid diagrams are emitted as vector graphics on dedicated pages; wide diagrams automatically use A4 landscape orientation.

Set BROWSER_EXECUTABLE if the browser is installed in a nonstandard location:

```
BROWSER_EXECUTABLE="/path/to/chrome" npm run pdf
```

To render another Markdown file or choose another output path, pass both paths after --:

```
npm run pdf -- README.md README.pdf
```